MES – Technical Manual

Madison Precision Products

- 2008 _ VTS_VCMTraceabilitySystem

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1.0 Introduction

1.1 Purpose

The “Machine Integration – Technical Manual” document contains detailed configuration information of all machine integration devices (software, hardware, communication, etc.) for the implementation of VTS Traceability System. This document is intended to compliment the “Machine Integration – User Manual” document.

1.2 Definitions, Acronyms and Abbreviations

Item	Definition
2D	Two Dimensional
Cognex	Manufacture of camera and imaging software
CPU	Central Processing Unit
CSQ	Computer Systems Quality
DHCP	Dynamic Host Configuration Protocol
Flexware	Publisher of Manufacturing Director software
GTIN	Global Trade Identification Number
I/O	Inputs and Outputs
LAN	Local Area Network
Manufacturing Director	Software package consisting of operator interface and configuration tools, used to store product information and to download parameters to and from camera and printer.
MD	Manufacturing Director
OCV	Optical Character Verification
OLE	Object Linking and Embedding
OPC	OLE for Process Control
PC	Personal Computer
PLC	Programmable Logic Controller
URL	Universal Resource Locator

2.0 Machine Integration Overview

2.1 System Connectivity

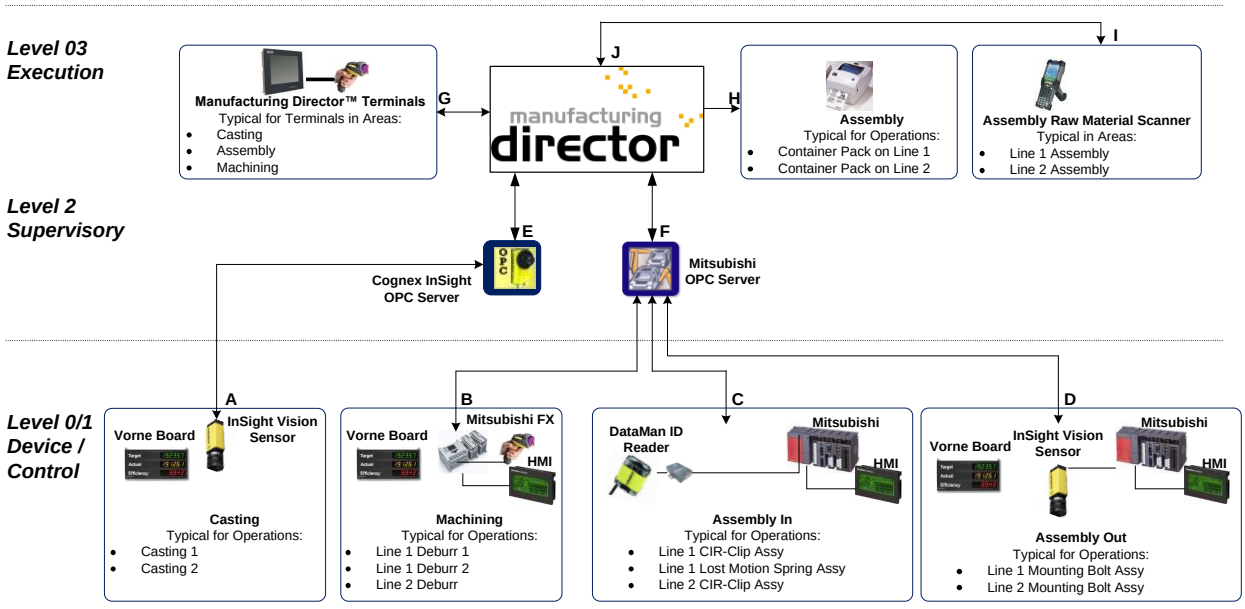
The dataflow through the system can be separated into three levels.

The first level is the "Device/Control" layer. This is where data is collected from the shop floor from devices such as Cognex In-Sight Sensors and DataMan ID Readers. The data is then stored in PLC memory in a format that can be accessed by OPC Servers. The first level also houses the HMI displays and Vorne Boards that are used to provide visual feedback of the MES transaction process to the operators.

The second level is the "Supervisory" layer. This layer houses the two OPC servers used in the system; Cognex In-Sight OPC Server and Mitsubishi OPC Server. The Cognex server communicates directly to the job file in the In-Sight Sensors located in Casting lines 1 and 2 and the Mitsubishi server communicates to the Mitsubishi PLCs located in Machining and Assembly. The OPC server acts as the communication bridge between the shop floor devices and Manufacturing Director. The devices send/receive data to the servers and MD also send/receive data to the servers.

The third level is the "Execution" layer. This is where the data is collected from level 2 and stored into local databases. Manufacturing Director then takes this data and begins the part transaction process. Once the transaction is complete, MD sends new information down to level 2 where it is distributed among the shop floor devices. Level 3 also maintains the production process and is used to print labels to the Zebra Label Printers and communicate to the Manufacturing Director Terminals located on the shop floor.

The following is a diagram showing the three levels of operation and the process dataflow.



2.1.1 Casting (A)

Task	Operation
1	Cognex In-Sight sensor acquires image of barcode.
2	Logic in In-Sight sensor (job file) verifies condition of barcode and assigns a part state. Decode Error = a state of 0, Successful Decode = a state of 1
3	Data Ready bit is turned on in In-Sight sensor to let MES know there is a new part.
4	MD creates and completes part at the casting operation.
5	Vorne Display Boards: displays updated production data based on discrete signals from the Casting control panel.

2.1.2 Machining (B)

Task	Operation
1	Cognex DataMan ID Reader decodes serial number from the 2D Data Matrix bar code and sends to the Mitsubishi PLC.
2	Mitsubishi PLC stores part serial number data.
3	Data Ready is set in the Mitsubishi PLC upon receiving the part serial number.

4	Data Ready triggers MD:
	• MD checks if the part exists, if not then the part is created and it is recorded that the part was started and completed at the operation.
	• If the part exists then MD checks the Part Interlock logic set for this part:
	  o Part must have a valid SN format
	  o Overall Disposition of the part must Good and must not be Scrap or Suspect
	  Note: The part could have a Casting Part Status of Invalid but this would not produce a part Invalid at this operation if the overall part disposition is Good.
	• If the interlock logic determines the part status is Invalid the HMI displays “Bad Part”. This part is not to be run through the process but placed in a tote for supervision to review.
	• If the interlock logic determines the part status is Valid the HMI will display “Good Part” and the part is to be placed on the in-feed conveyor.
	• MD will also record that the part was started and completed at the machining operation upon a part status of Valid.

5	Vorne Display Boards: displays updated production data based on discrete signals from the OP40 machines at each line.

2.1.3 Assembly In – Line 2 Cir-Clip Assembly (C)

Task	Operation
1	Cognex DataMan ID Reader decodes serial number from the 2D Data Matrix bar code and sends to the Mitsubishi PLC.
2	Mitsubishi PLC stores part serial number data.

3	Data Ready is set in the Mitsubishi PLC upon receiving the part serial number.

4	Data Ready triggers MD:
	• MD checks if the part exists, if not then the part is created and it is recorded that the part was started and completed at the operation.
	• If the part exists then MD checks the Part Interlock logic set for this part:
	  o Part must have a valid SN format
	  o Overall Disposition of the part must Good and must not be Scrap or Suspect
	  Note: The part could have a Part Status of Invalid relating to a previous operation (Casting or Machining) but this would not produce a part Invalid if the overall disposition of the part is Good.
	• If the interlock logic determines the part status is Invalid the HMI displays "MES ERROR". This part is not to be run through the process but placed in a tote for supervision to review.
	• If the interlock logic determines the part status is Valid the HMI will display "PART VALID" and the part will be allowed to run into the process.
	• MD will also record that the part was started and completed at the machining operation upon a part status of Valid.

2.1.4 Assembly In – Line 1 Cir-Clip Assembly (C)

Task	Operation
1	Cognex DataMan ID Reader decodes serial number from the 2D Data Matrix bar code and sends to the Mitsubishi PLC.
2	Mitsubishi PLC stores part serial number data.
3	Data Ready is set in the Mitsubishi PLC upon receiving the part serial number.

4	Data Ready triggers MD:
	• MD checks if the part exists, if not then the part is created and it is recorded that the part was started and completed at the operation.
	• If the part exists then MD checks the Part Interlock logic set for this part:
	  o Part must have a valid SN format
	  o Overall Disposition of the part must be "Good" and must not be "Scrap" or "Suspect"
	  Note: The part could have a Part Status of Invalid relating to a previous operation (Casting or Machining) but this would not produce a part Invalid if the overall disposition of the part is Good.
	• If Cir-Clip machine process produces a clip insert error then the part status of the Cir-Clip operation is set to Invalid.
	• If the interlock logic determines that the part status is Invalid the part is to be placed in a tote for supervision to review.
	• If the interlock logic determines the part status is Valid the part is to be queued for the next operation.

	• MD will also record that the part was started and completed at the machining operation upon a part status of Valid.

2.1.5 Assembly In – Line 1 Lost Motion (C)

Task	Operation
1	Cognex DataMan ID Reader decodes serial number from the 2D Data Matrix bar code and sends to the Mitsubishi PLC.
2	Mitsubishi PLC stores part serial number data.
3	Data Ready is set in the Mitsubishi PLC upon receiving the part serial number.

4	Data Ready triggers MD:
	• The following MD interlock logic is checked at this station:
	  o The part must have been successfully processed at Cir-Clip, Cir-Clip part status must be Valid.
	  o Overall Disposition of the part must be "Good" and must not be "Scrap" or "Suspect"
	  Note: The part could have a Part Status of Invalid relating to a previous operation (Casting or Machining) but this would not produce a part Invalid if the overall disposition of the part is Good.
	• If the interlock logic determines that the part status is Invalid the part is to be placed in a tote for supervision to review.
	• If the interlock logic determines the part status is Valid the part will be accepted into the assembly line for processing.
	• MD will also record that the part was started and completed at this operation upon a part status of Valid.

2.1.6 Assembly Out (Line 1/2) (D)

Task	Operation
1	Cognex In-Sight Sensor decodes serial number from the 2D Data Matrix bar code and sends to the Mitsubishi PLC.
2	Mitsubishi PLC stores part serial number data.
3	Part Start is set in the Mitsubishi PLC upon receiving the part serial number.

4	Part Start triggers MD:
	• MD checks if the part exists, if not then the part is created and it is recorded that the part was started this operation.
	• The following MD interlock logic is checked at this station:
	  o Part must have a valid SN format
	  o Overall Disposition of the part must be "Good" and must not be "Scrap" or "Suspect"
	  Note: The part could have a Part Status of Invalid relating to a previous operation (Casting or Machining) but this would not produce a part Invalid if the overall disposition of the part is Good.
	• If the interlock logic determines that the part status is Invalid the

	machine will not process the part and the operator is to remove the part from the system and placed in a tote for supervision to review. • If the interlock logic determines the part status is Valid the part will be accepted into machine for processing.
5	Data Ready triggers MD: • MD checks the status of this Part Status bit in the PLC to determineif the part is a good part. • If the Part Status bit is off this means it is a bad part and the Part Status is Invalid and the machine is interlocked and the operator is to remove the part and place it in a tote for supervision review. • If the Part Status bit is on this means it is a good part and the Part Status is Valid. MD will record the completion of the part processing and the machine will place the part on the out-feed conveyor by the machine.
5	Vorne Display displays updated production data.

2.1.7 Cognex In-Sight OPC Server (E)

Task	Operation
1	Data is received from In-Sight Sensors in Casting Lines 1 and 2.
2	Data is sent to MD.
3	Transaction status data is received by MD.
4	Transaction status data is sent to Casting Lines 1 and 2.

2.1.8 Mitsubishi OPC Server (F)

Task	Operation
1	Data is received from PLCs in Machining Lines and Assembly Lines
2	Data is sent to MD.
3	Transaction status data is received by MD.
4	Transaction status data is sent to PLCs in Machining Lines and Assembly Lines

2.1.9 Manufacturing Director Terminals (G)

Task	Operation
1	MES data is retrieved from the MD database.
2	MES data is displayed to the Operator.
3	Operators can track and modify parts and containers.

2.1.10 Assembly Printers (H)

Step #	Task
1	MD populates the label with MES data and sends label to the printer in ZPL format.

2	Zebra Label Printer prints the container labels for completed containers in Assembly Out lines 1 and 2.

2.1.11 Assembly Raw Material Scanner (I)

Step #	Task
1	Data is scanned from container labels.
2	Data is sent to MD.
3	MES data is received from MD.

3.0 Manufacturing Director™ - System Components

3.1 Manufacturing Director™ - Modeling Client

The Modeling Client allows for the flexibility of the Manufacturing Director™ system to form to the facilities production "model". Flexware has base models defined that fit the typical MES track and trace solution but these models will be modified for each implementation.

> A production system's model typically does not changed after an implementation is in place and should not be altered unless Flexware Innovation, Inc. is consulted prior to any changes being made.

3.2 Manufacturing Director™ - Configuration Client

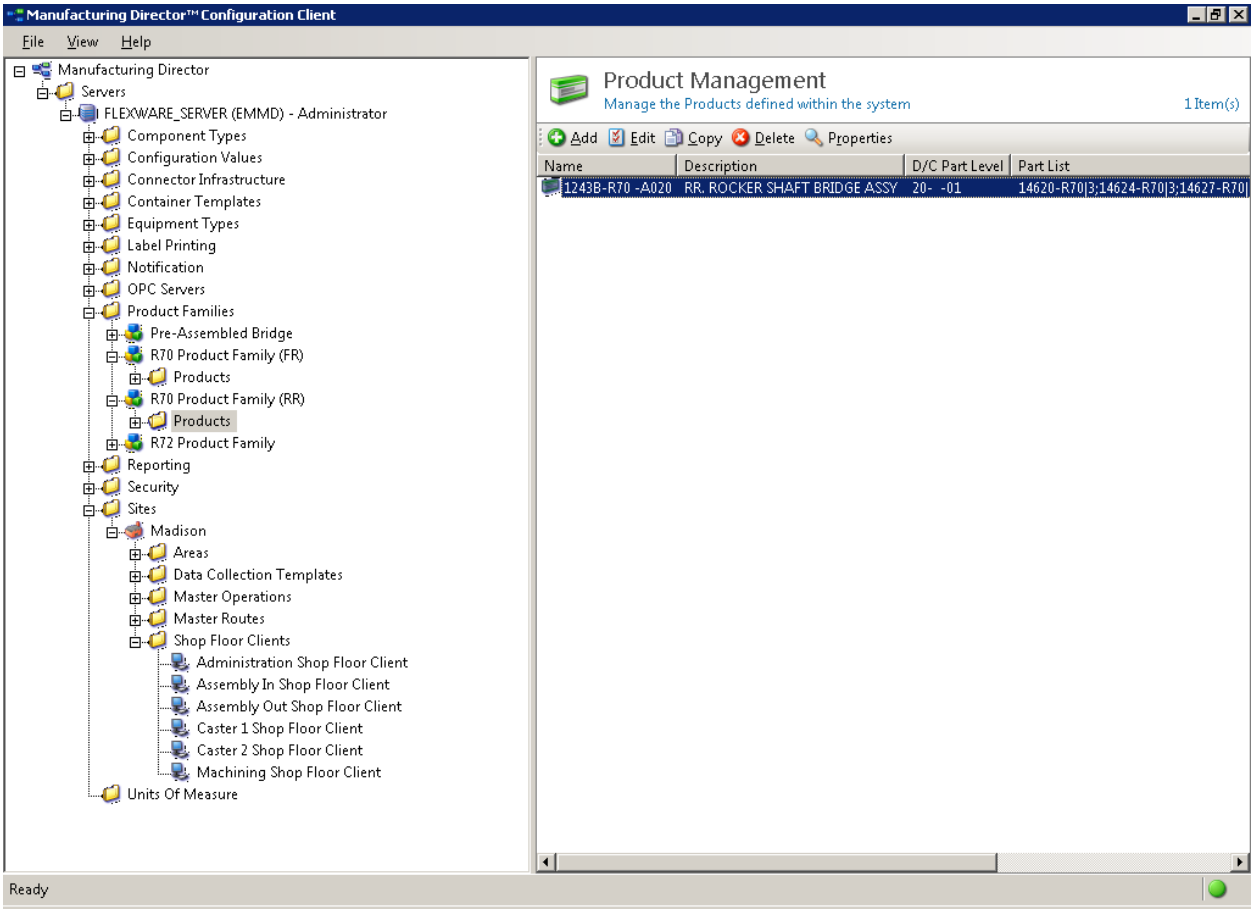
The Configuration Client is used to configure the various aspects of the system. This includes, but is not limited to, defining users and roles, products and product families, physical plant floor components, and tracking components.

3.2.1 Configuring Product Type and Part Level

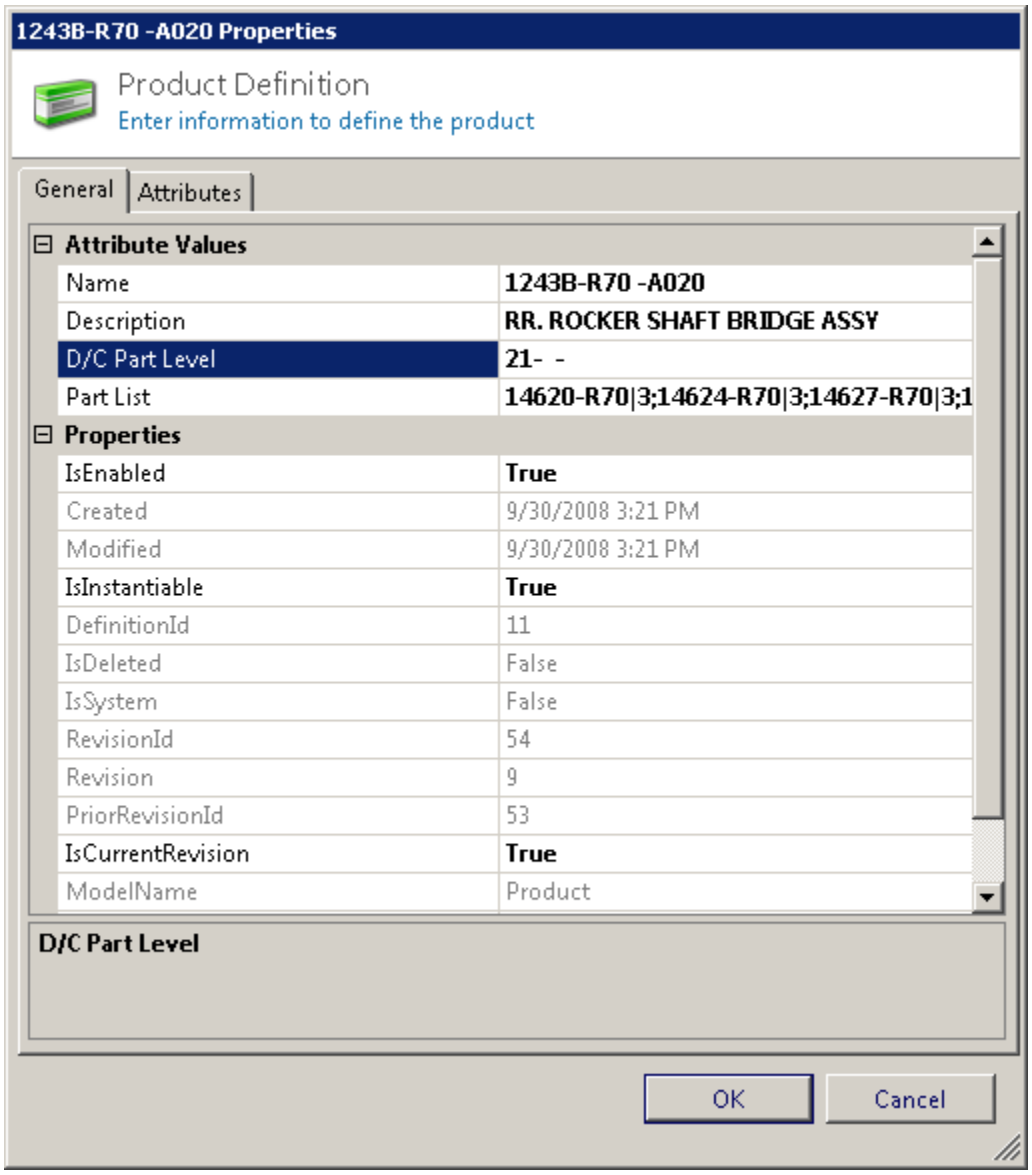
Example: Change D/C Part Level

To change the D/C Part Level of a product (e.g., from 20- -01 to 21- -)

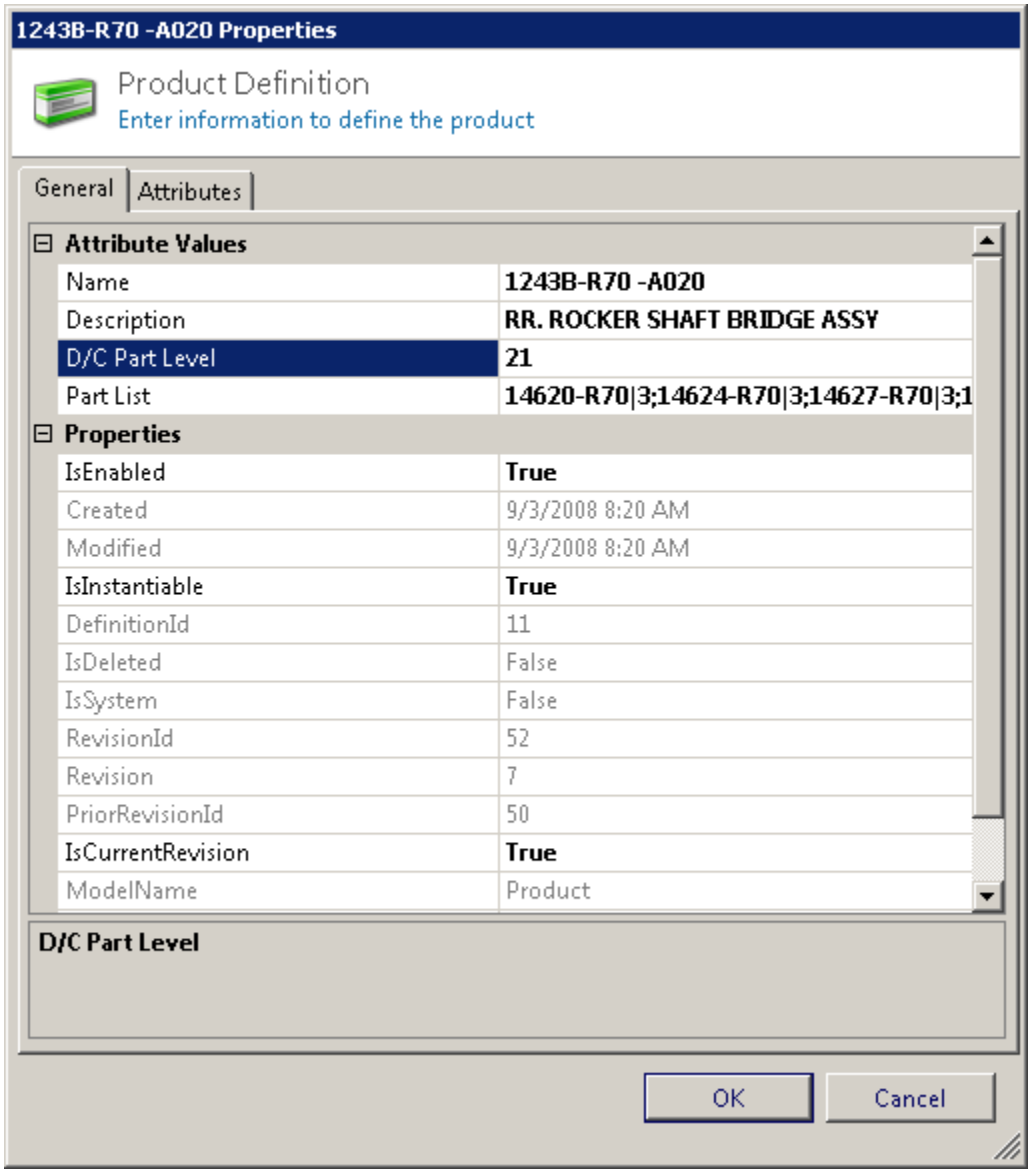
1. Find and select the product you want to update in the **Manufacturing Director™ Configuration Client**.


Manufacturing Director™ Configuration Client
File
View
Help
Manufacturing Director
Servers
FLEXWARE_SERVER (EMMD) - Administrator
Component Types
Configuration Values
Connector Infrastructure
Container Templates
Equipment Types
Label Printing
Notification
OPC Servers
Product Families
Pre-Assembled Bridge
R70 Product Family (FR)
Products
R70 Product Family (RR)
Products
R72 Product Family
Reporting
Security
Sites
Madison
Areas
Data Collection Templates
Master Operations
Master Routes
Shop Floor Clients
Administration Shop Floor Client
Assembly In Shop Floor Client
Assembly Out Shop Floor Client
Caster 1 Shop Floor Client
Caster 2 Shop Floor Client
Machining Shop Floor Client
Units Of Measure
Product Management
Manage the Products defined within the system
1 Item(s)
Add
Edit
Copy
Delete
Properties
Name
Description
D/C Part Level
Part List
1243B-R70 -A020
RR. ROCKER SHAFT BRIDGE ASSY
20- -01
14620-R70|3;14624-R70|3;14627-R70|
Ready

2. Click the **Edit** button to display the product's properties



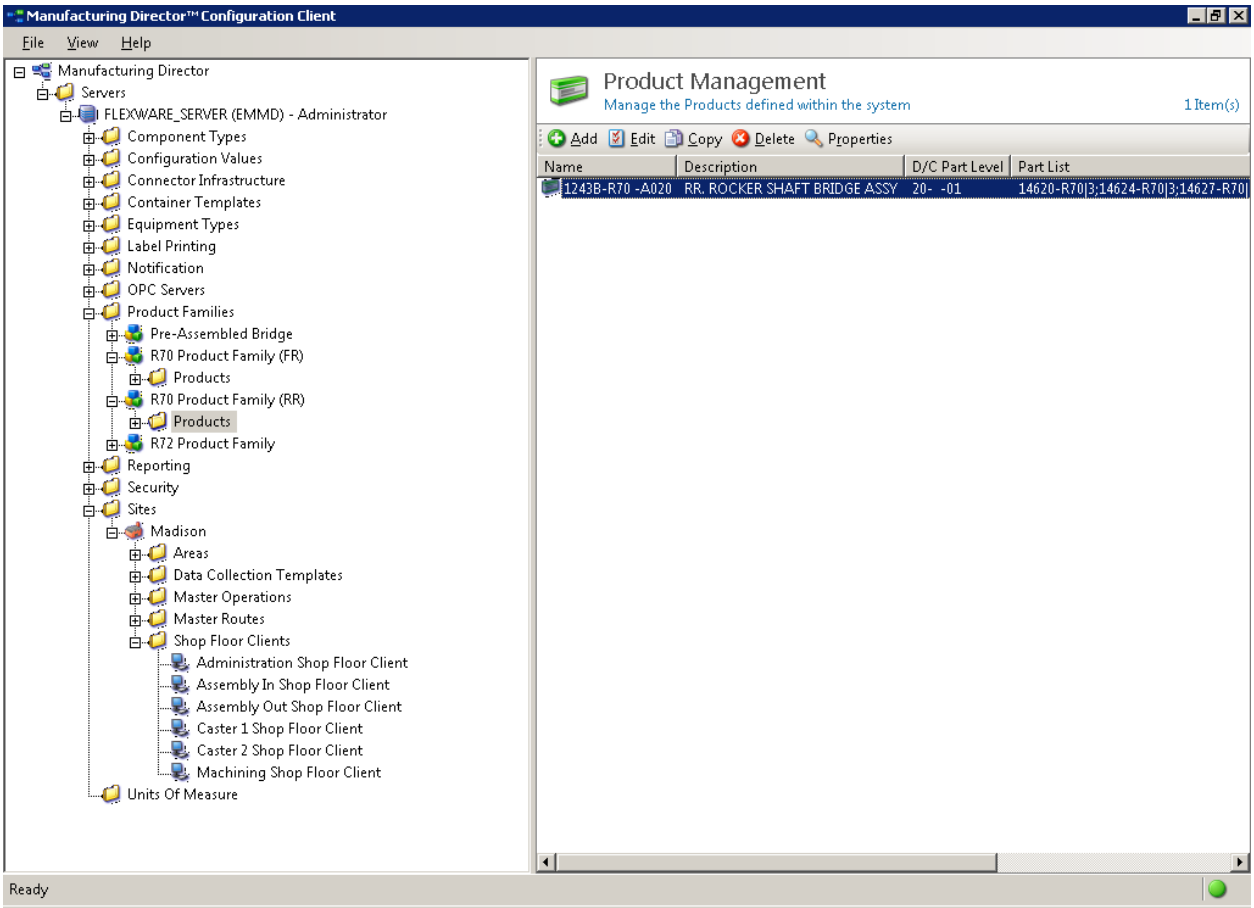
3. Change the **D/C Part Level to 21- -** and click the **OK** button to save the product.



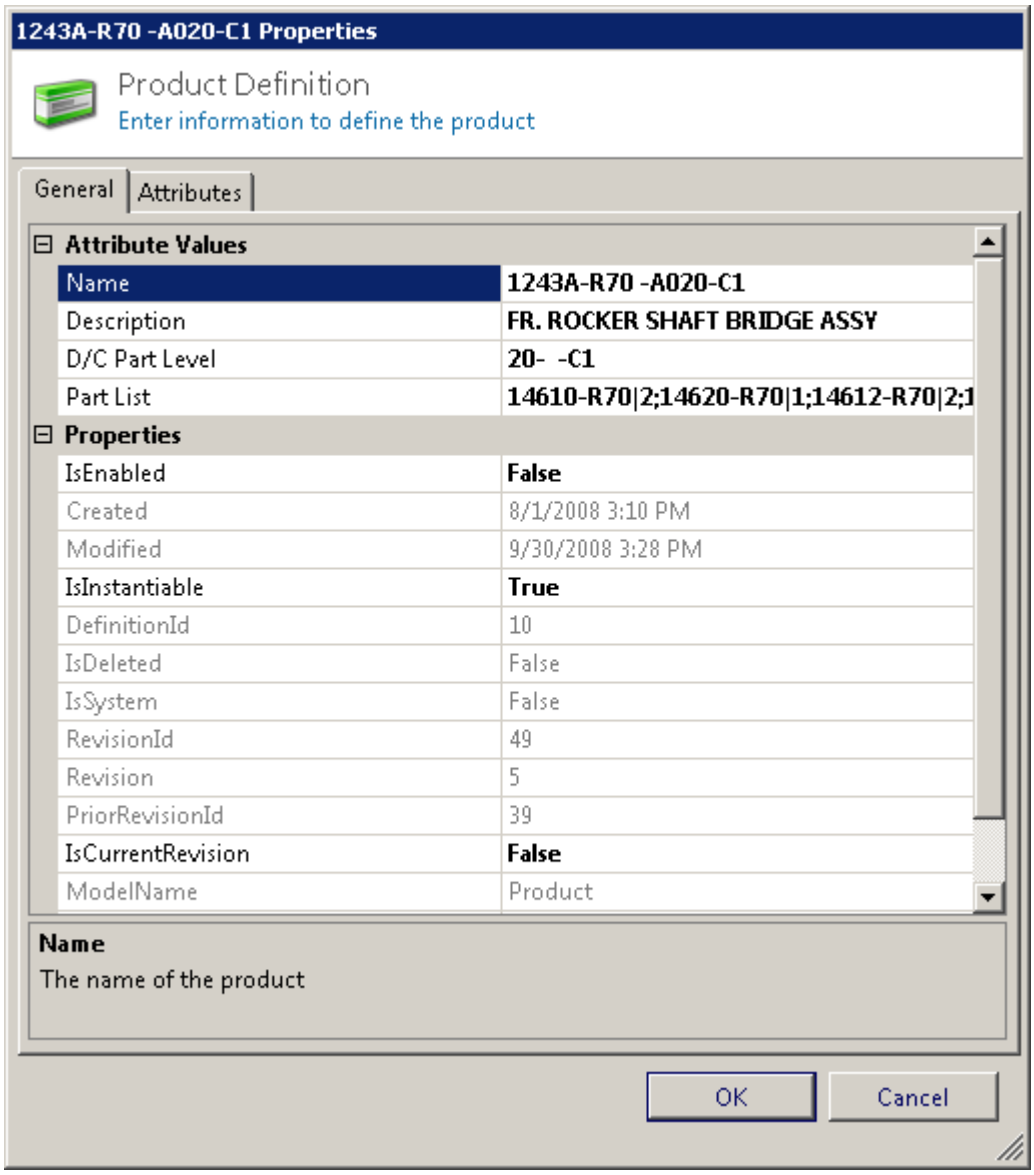
3.2.2 Example: Change Product Name

To change the name of a product (e.g., from **1243A-R70 –A020-C1** to **1243A-R70 – A020-C2**)

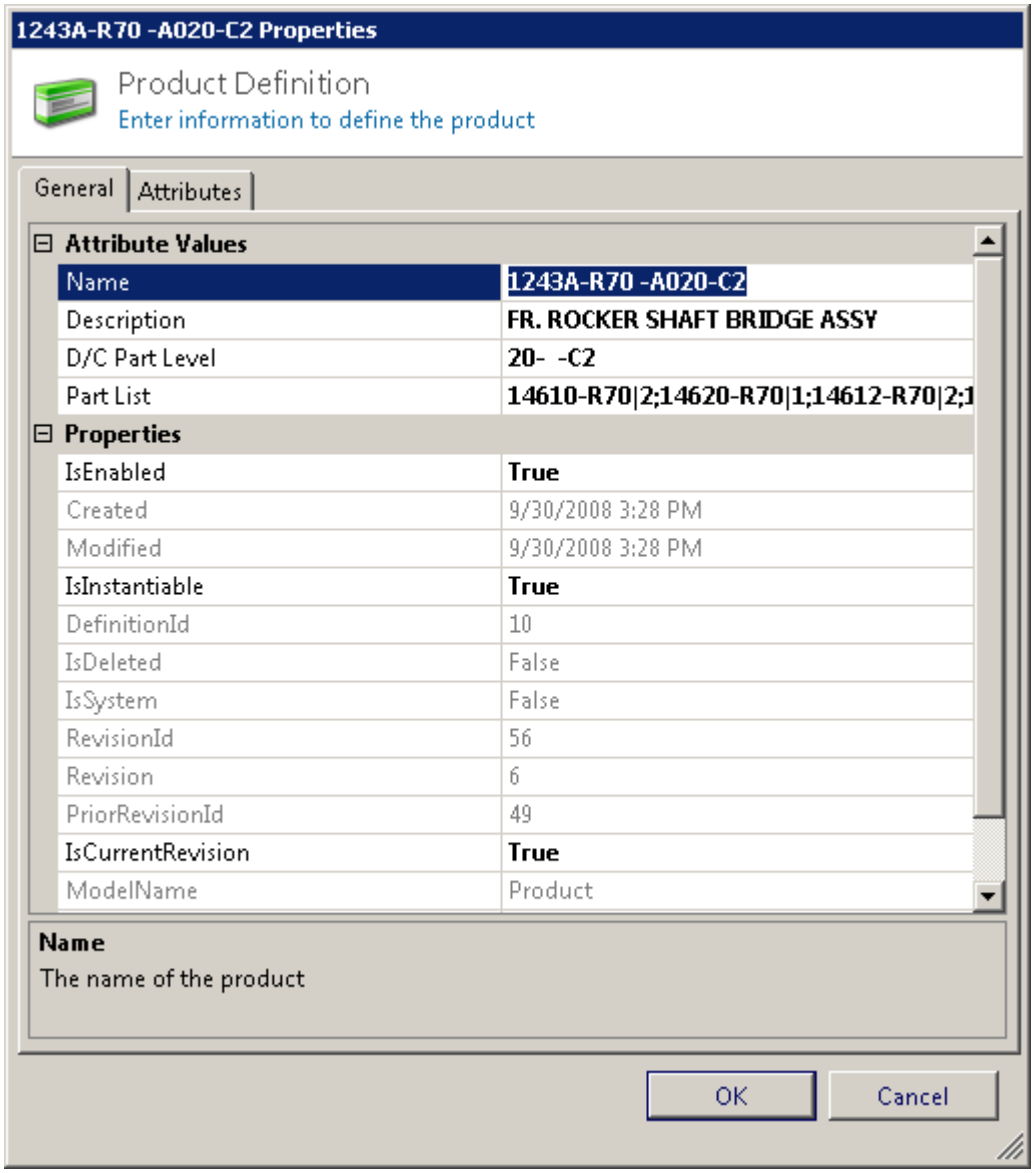
1. Find and select the product you want to update in the **Manufacturing Director™ Configuration Client**.



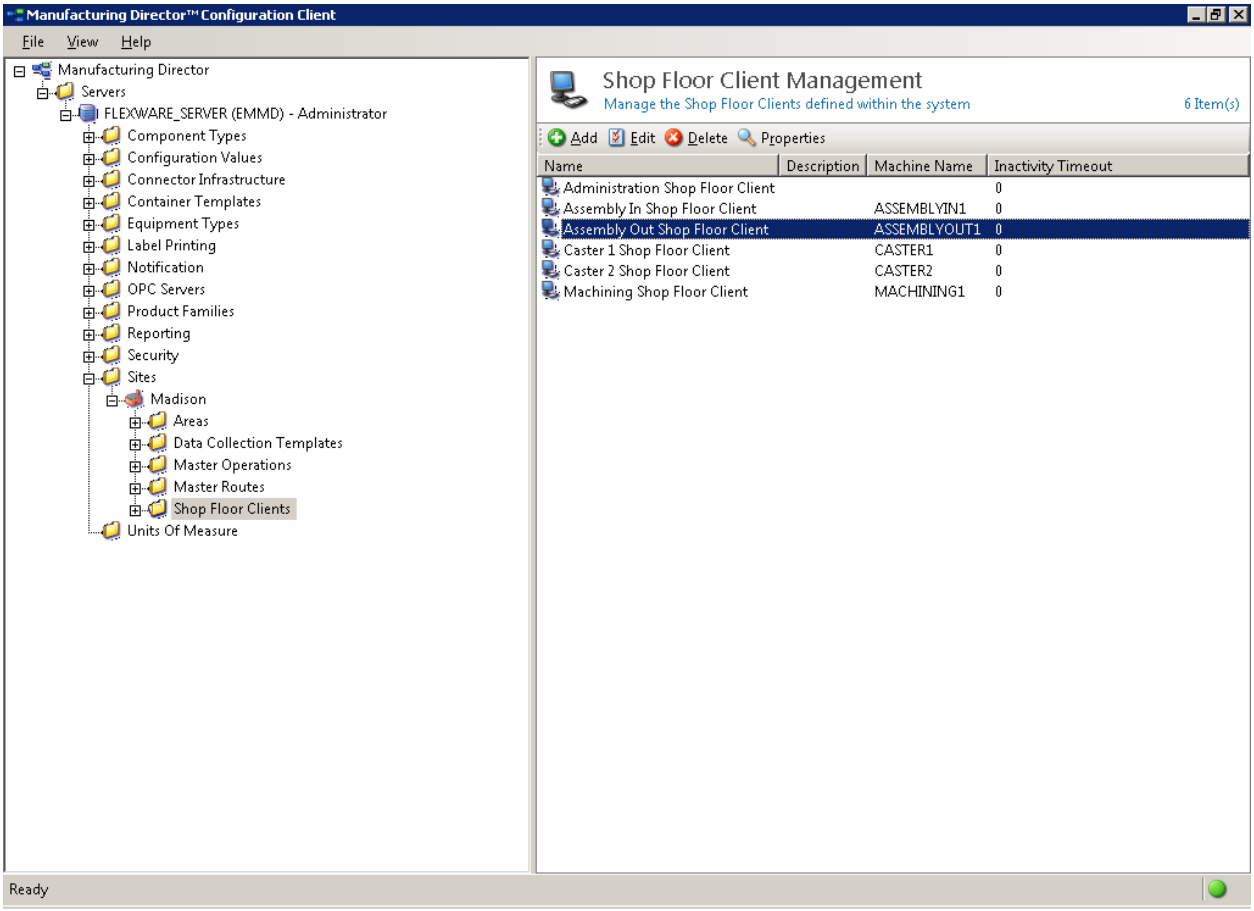
2. Click the **Edit** button to display the product's properties.



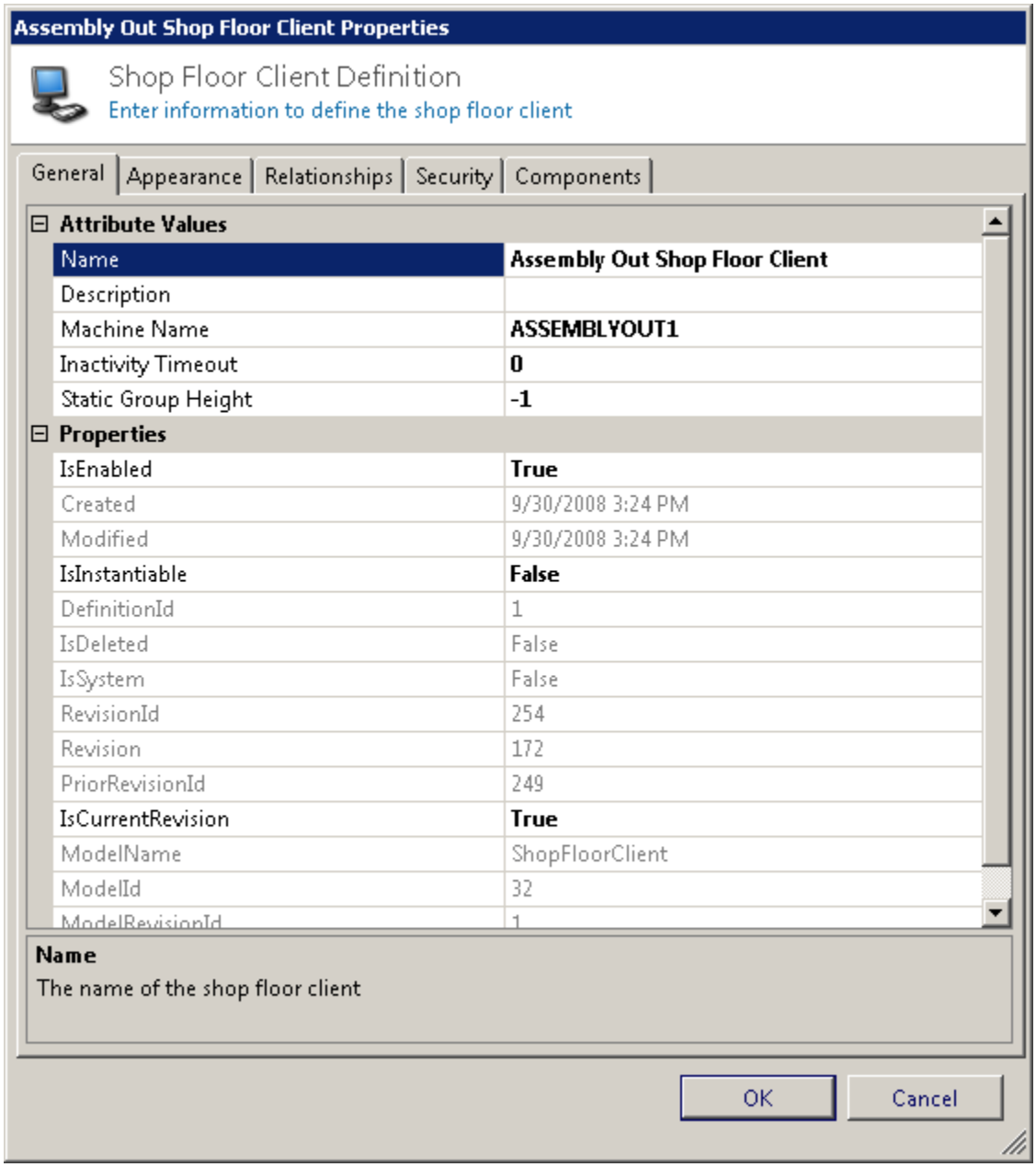
3. Change the Name to 1243A-R70 –A020-C2 and click the **OK** button to save the product.



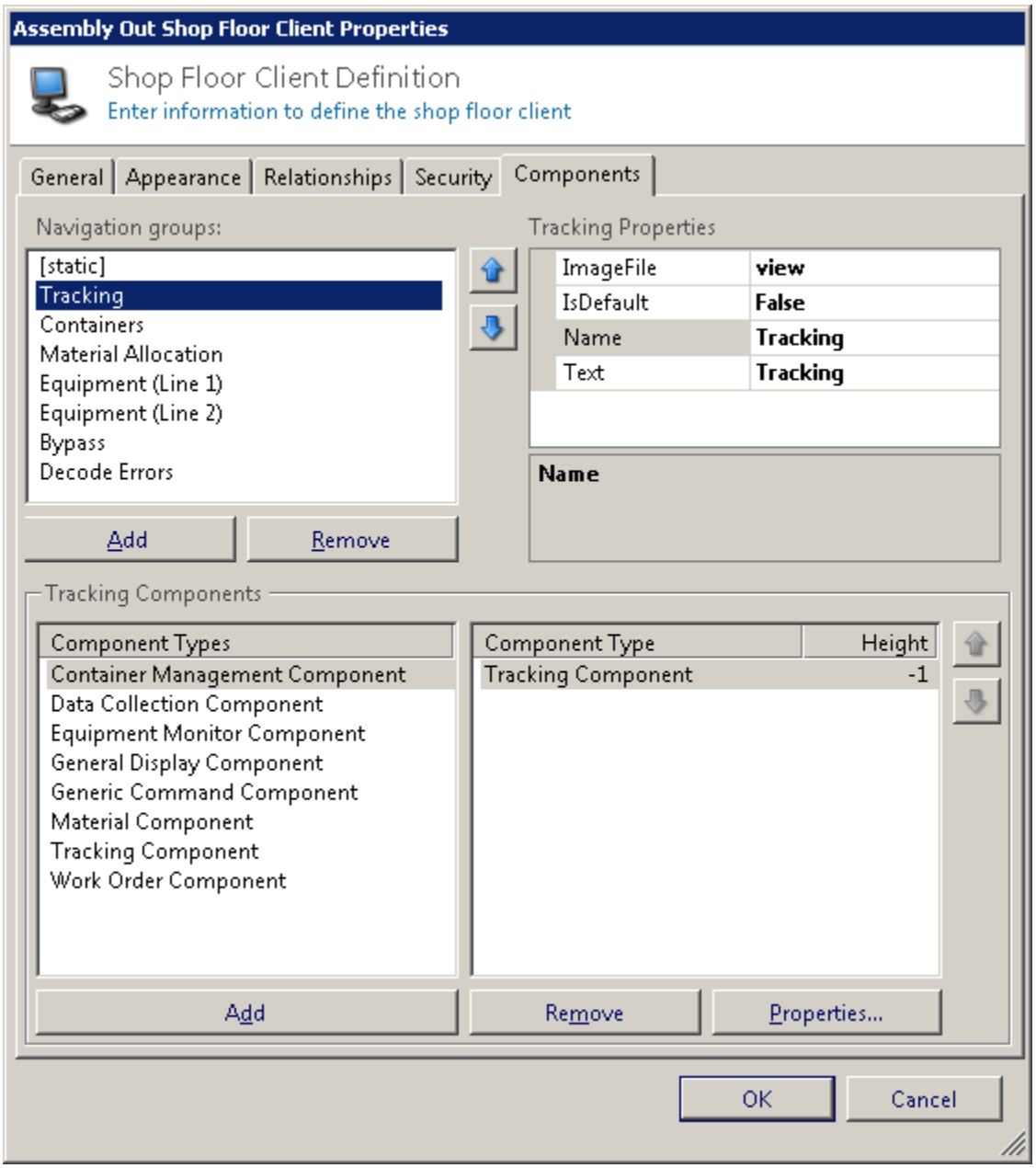
4. Find and select the Assembly Out Shop Floor Client in the **Manufacturing Director™ Configuration Client.**



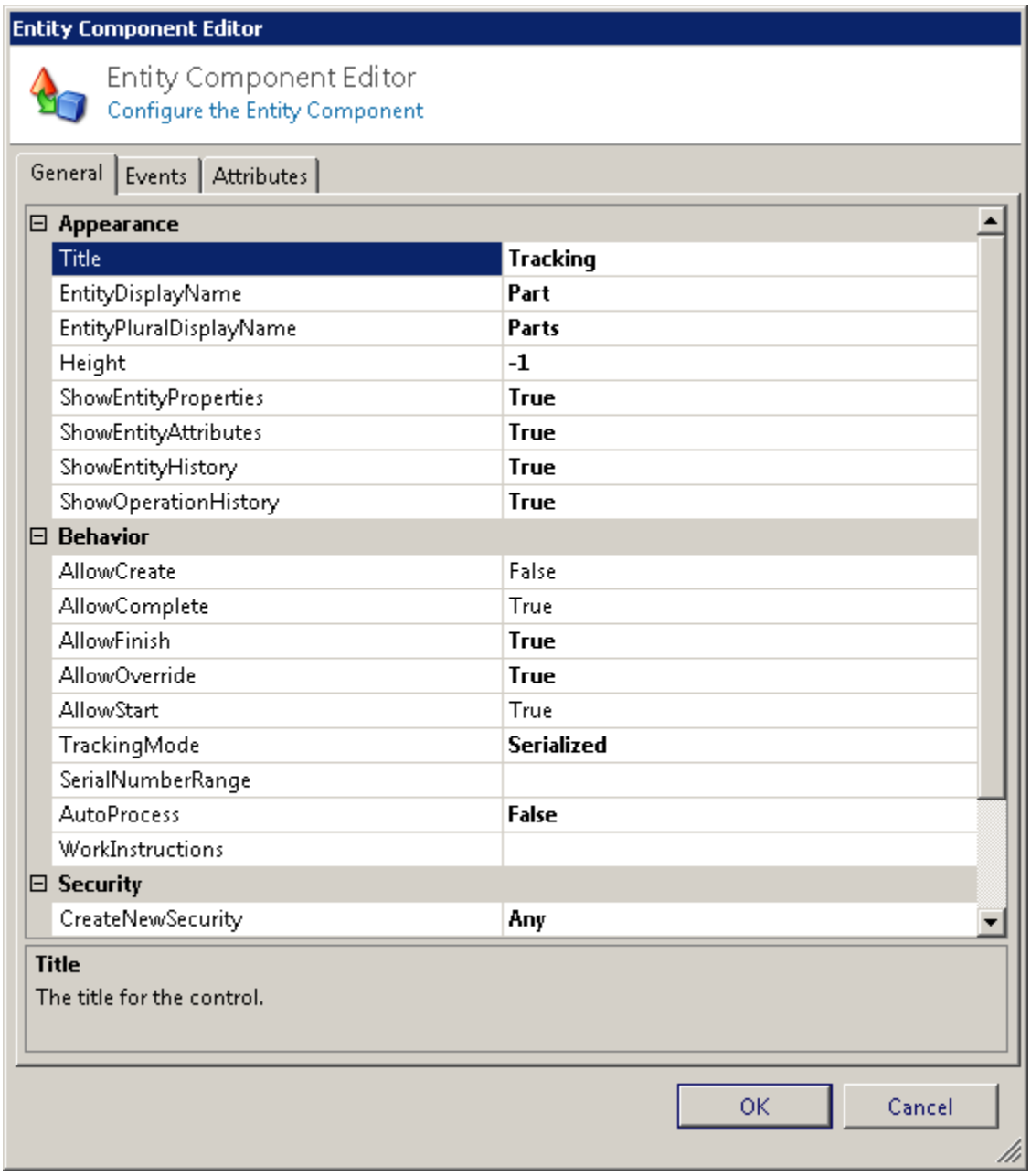
5. Click the **Edit** button to display the shop floor client's properties.



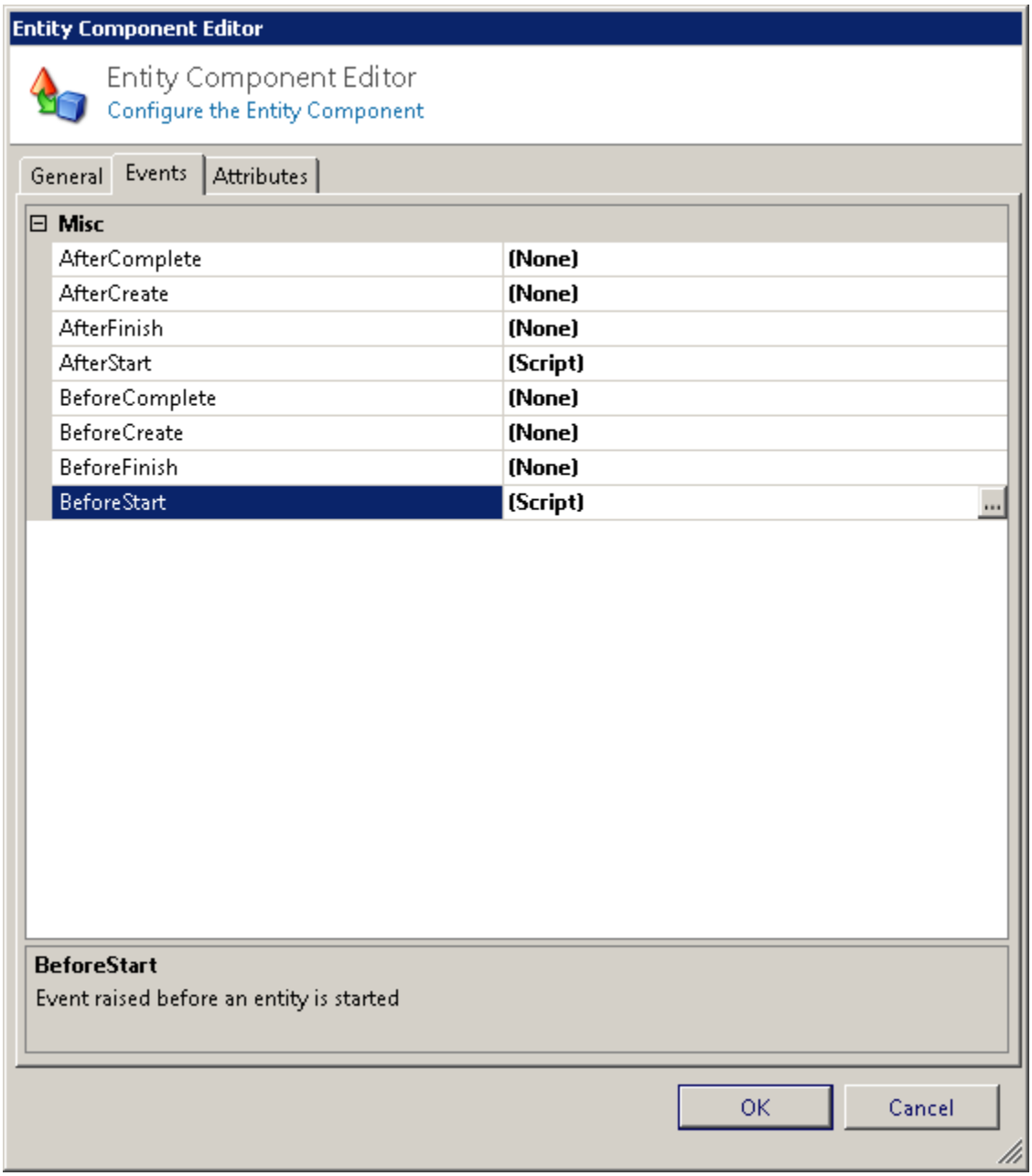
6. Switch to the **Components** tab and select the **Tracking navigation** group in the upper left and select the **Tracking Component** in the lower right.



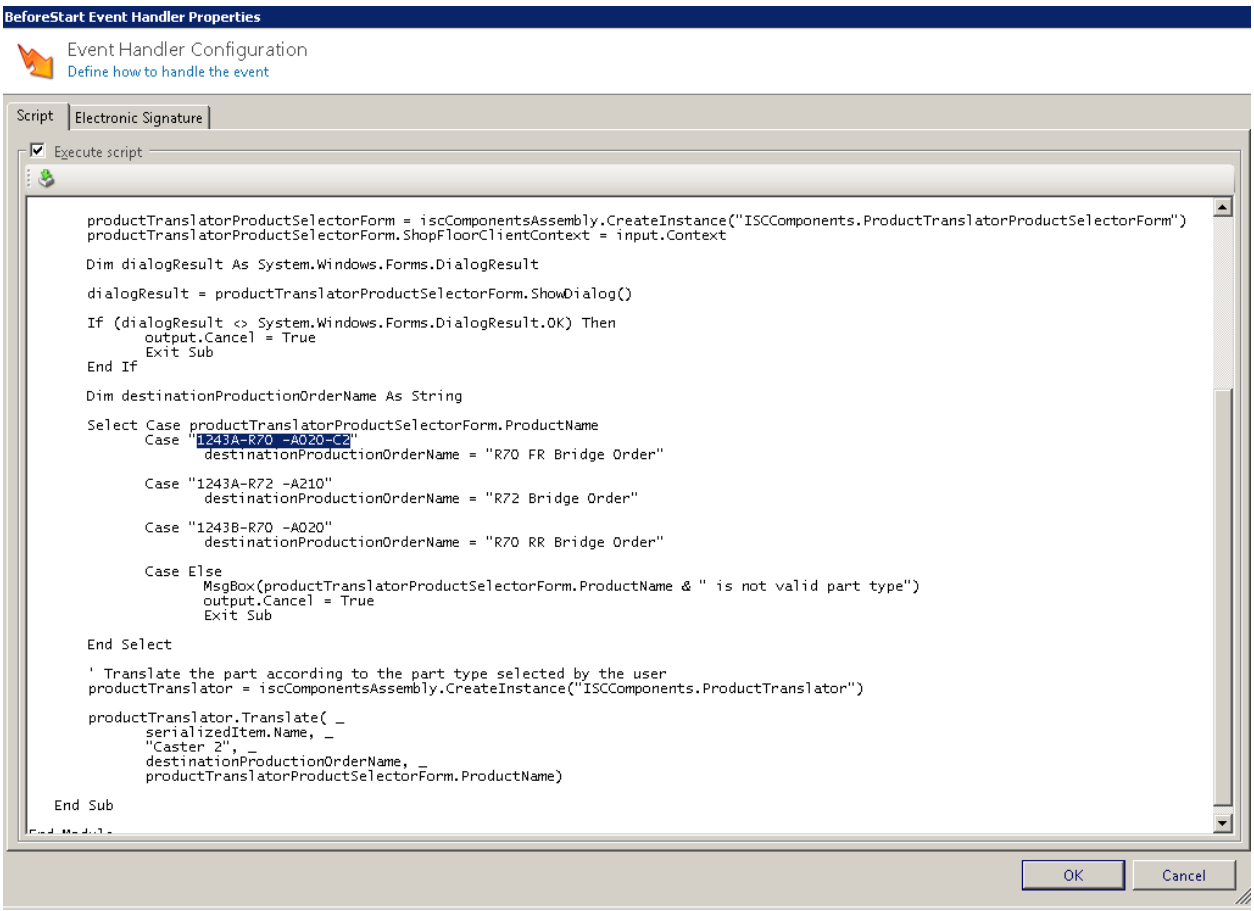
7. Click the **Properties**… button in the lower right.



8. Switch to the **Events** tab.



9. Select the **BeforeStart** event and click the ellipsis **(…)** button to the right to display the **BeforeStart Event Handler Properties** dialog.



10. In the script change **1243A-R70 –A020-C1 to 1243A-R70 –A020-C2**. *This name must match EXACTLY the name of the product as defined in the Configuration Client.*

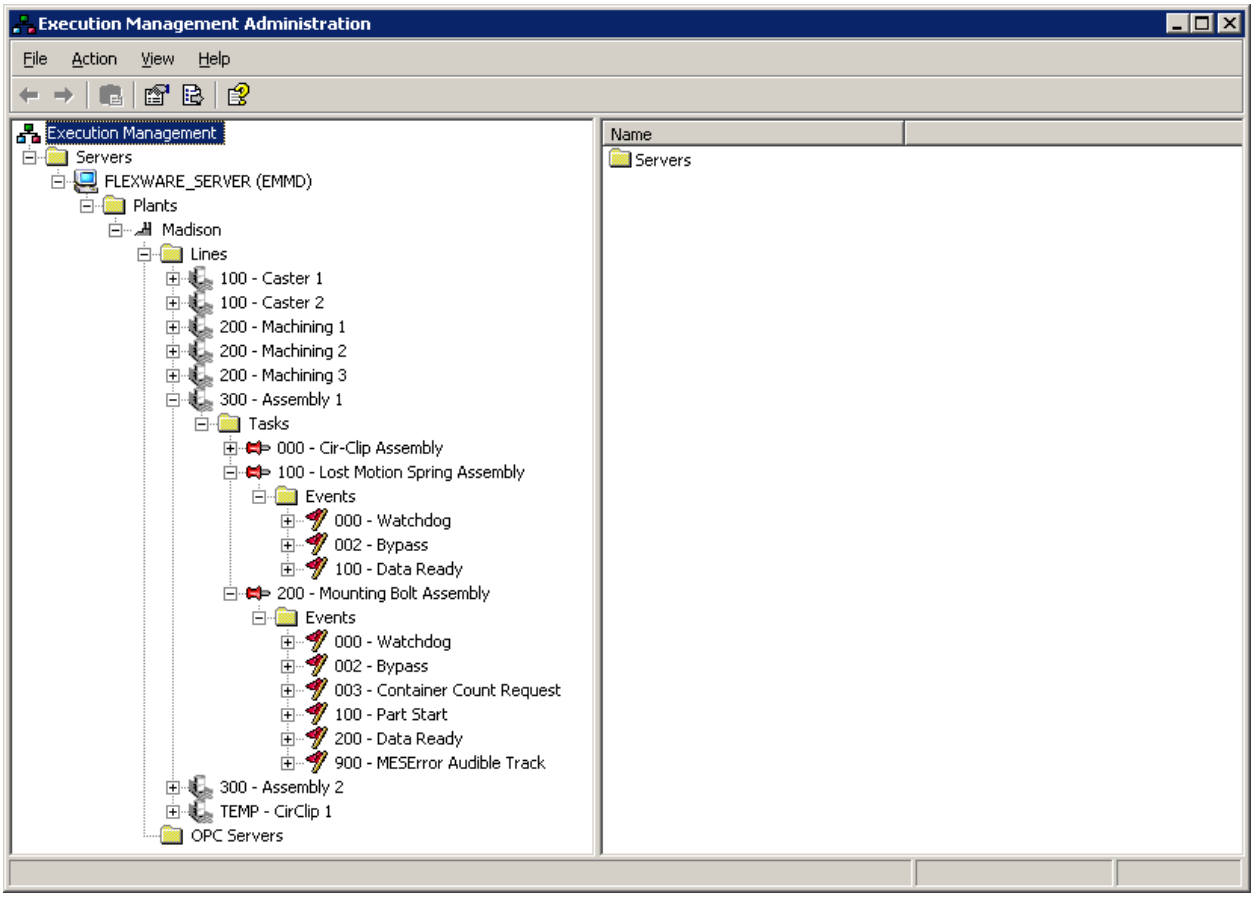
11. Click the **OK** button on the **BeforeStart Event Handler Properties** dialog.

12. Click the **OK** button on the **Entity Component Editor** dialog.

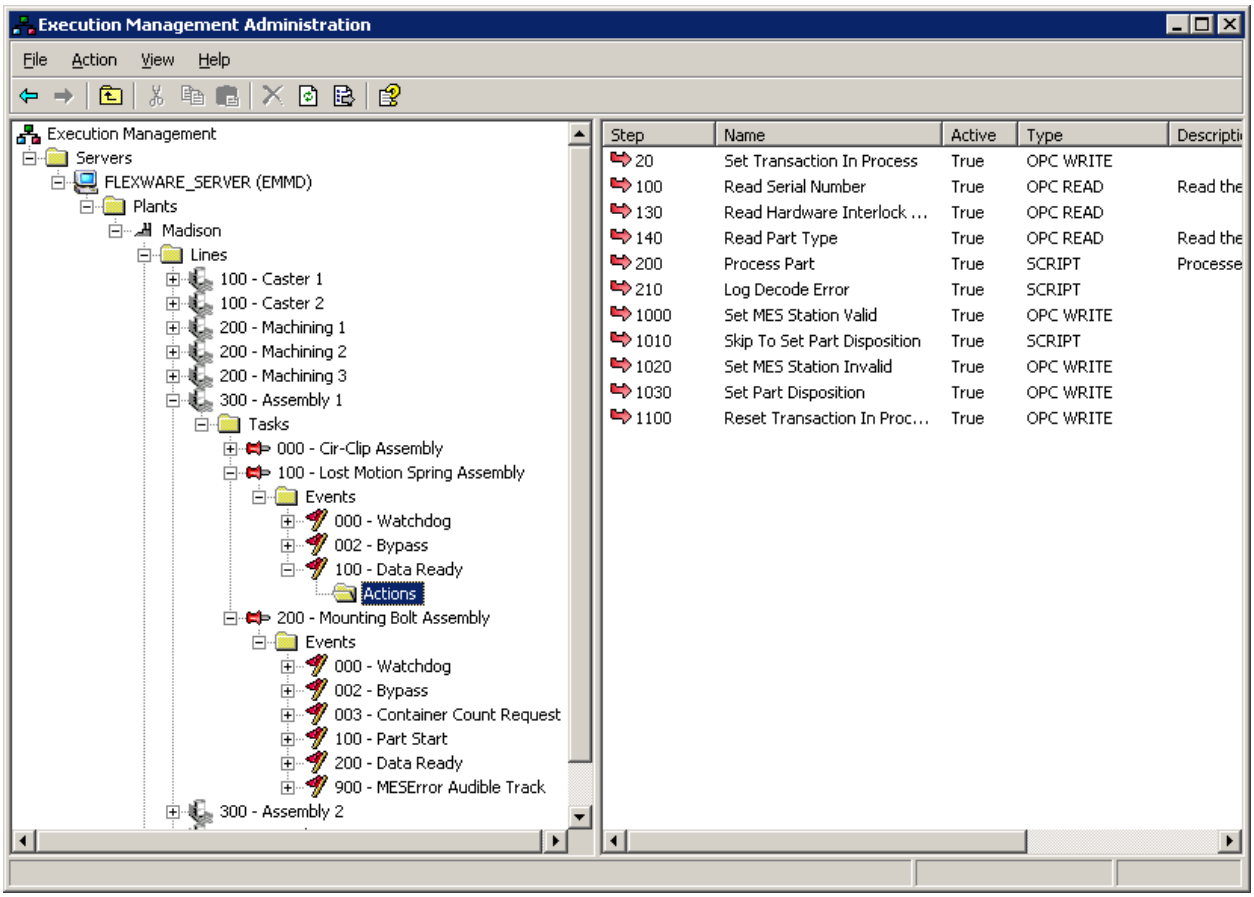
13. Click the **OK** button on the **Assembly Out Shop Floor Client Properties** dialog.

14. Perform the steps 4 – 13 for the Assembly In Shop Floor Client in the **Manufacturing Director™ Configuration Client.**

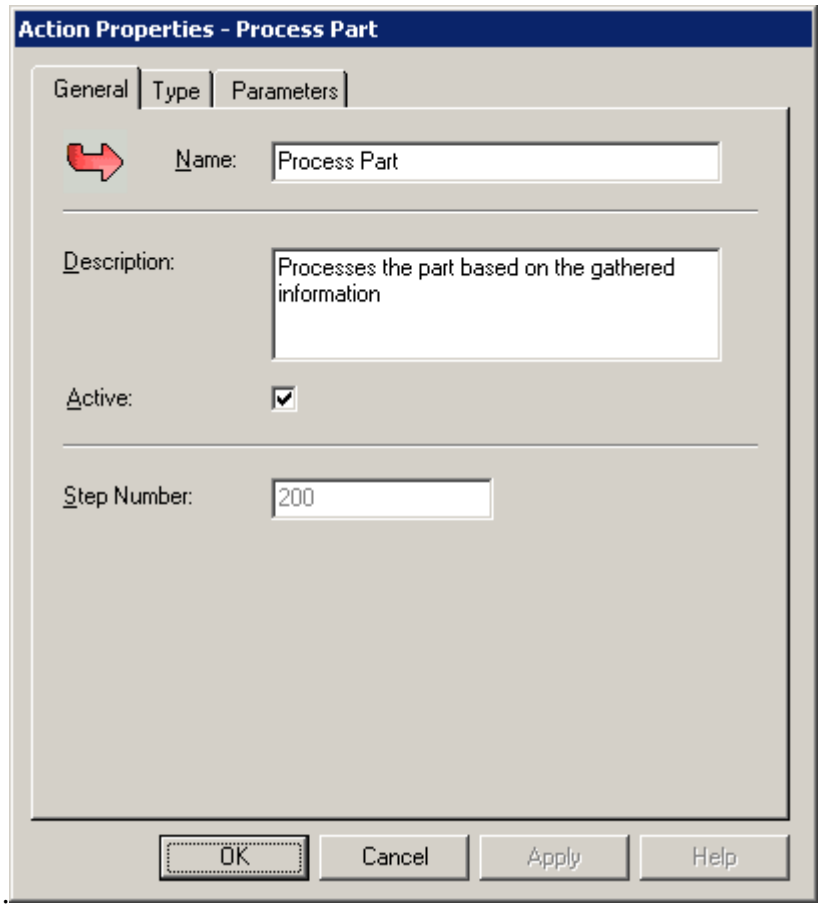
15. Open **Execution Management™ Administration** and expand the tree to see the events for the **100 – Lost Motion Spring Assembly** and **200 – Mounting Bolt Assembly** tasks under the **300 – Assembly 1 line**.



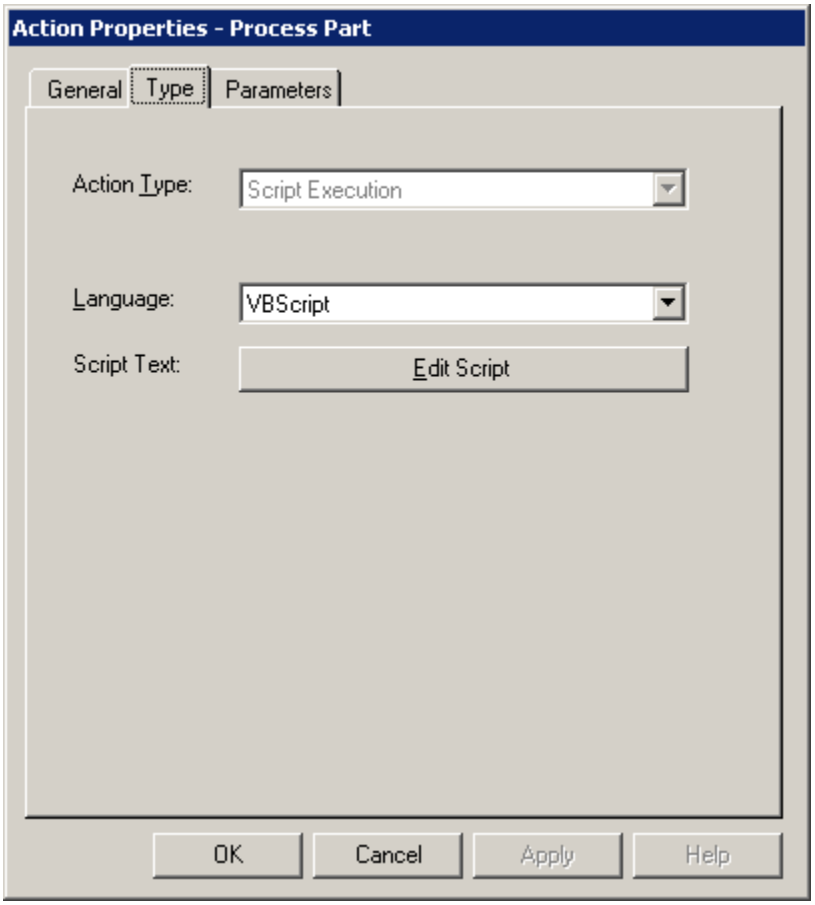
16. Expand the **100 – Data Ready** event under **100 – Lost Motion Spring Assembly** and select the **Actions** folder.



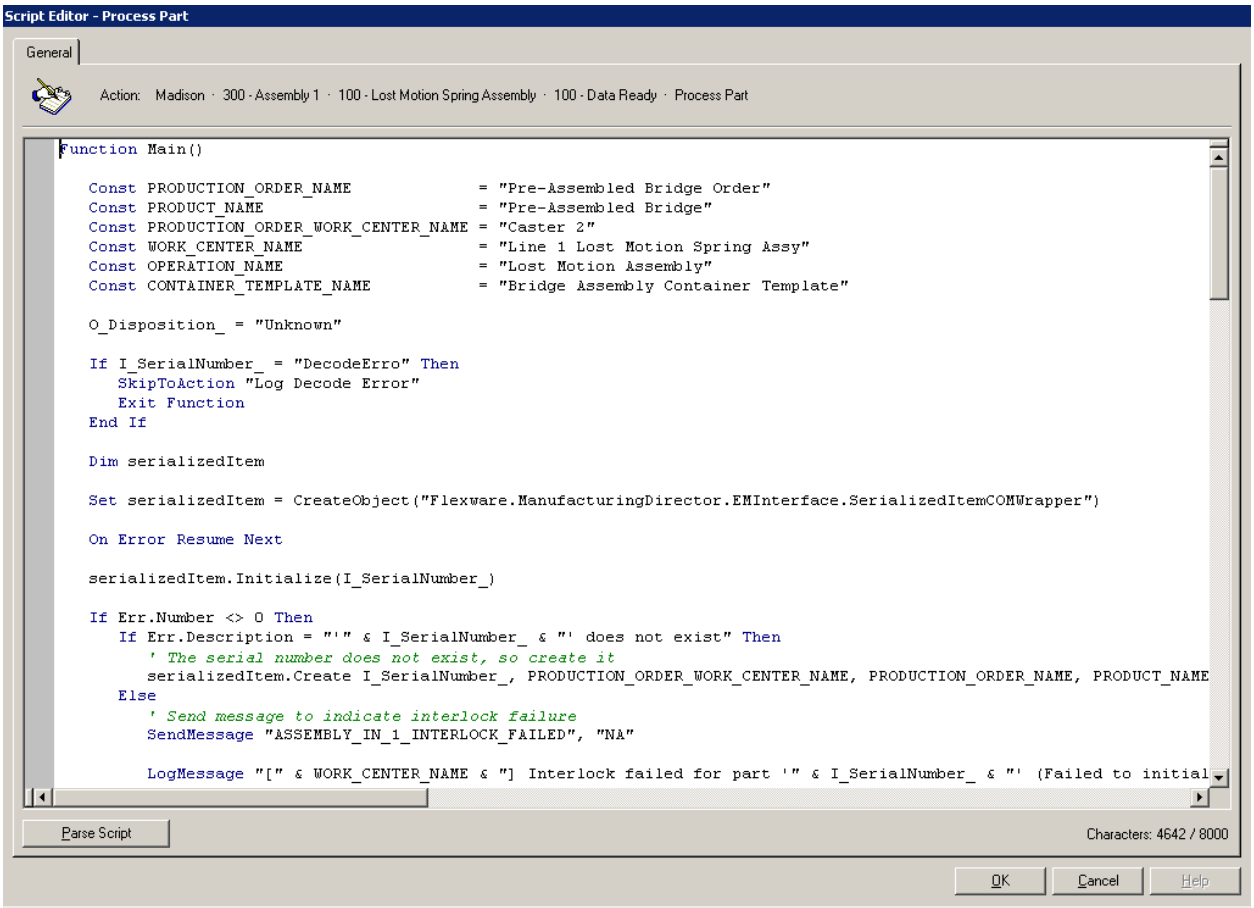
17. Double-click the **Process Part** action on the right side to display the properties for the action.



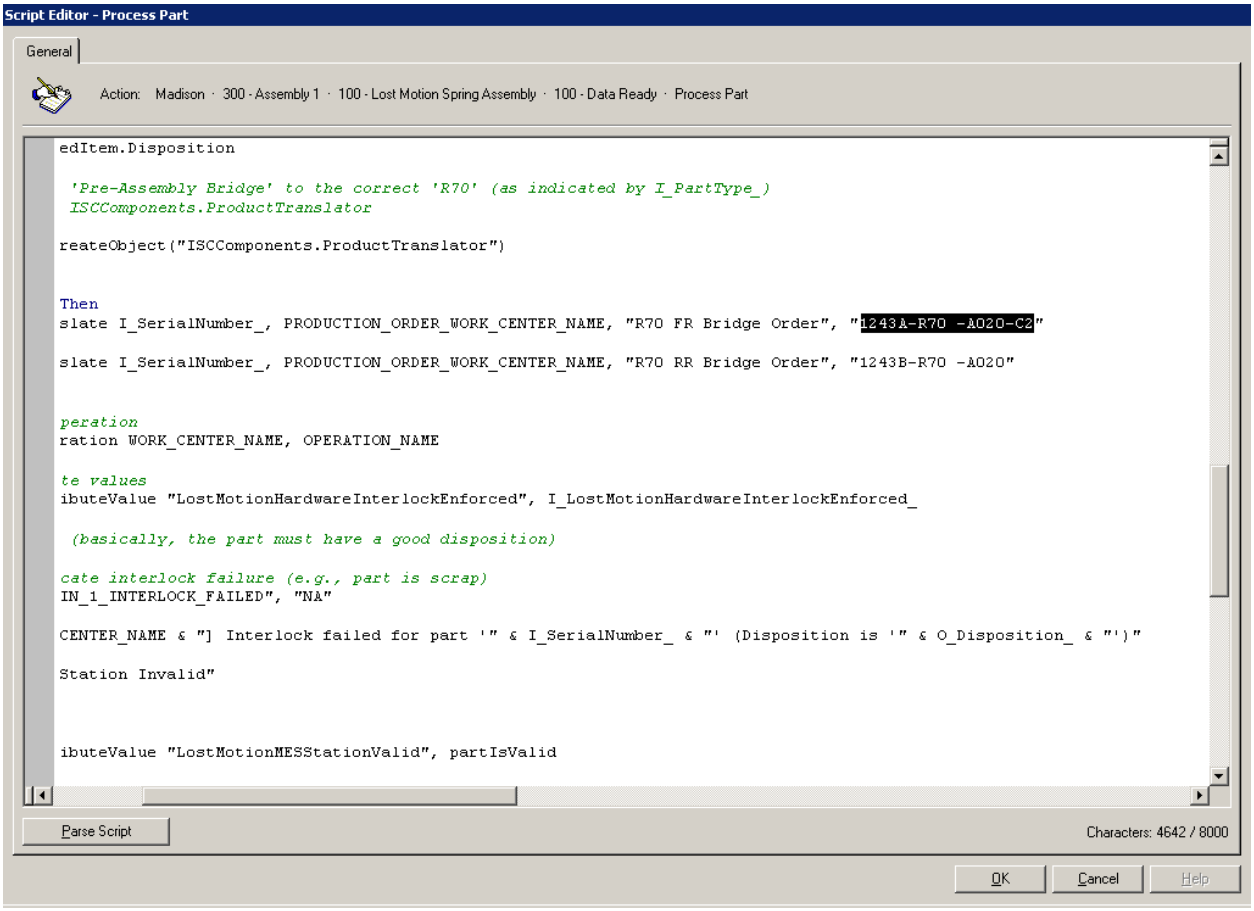
18. Switch to the **Type** tab.



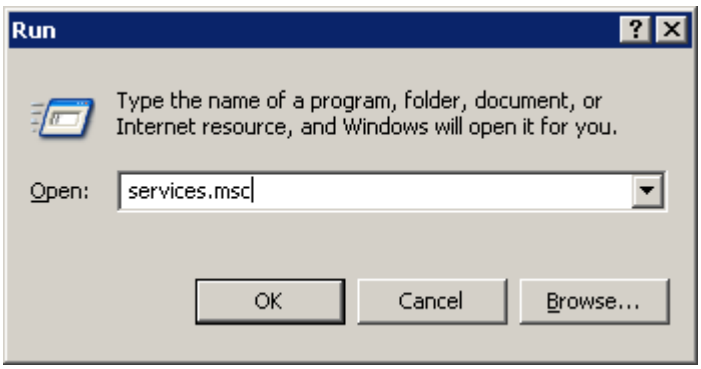
19. Click the **Edit Script** button to display the **Script Editor** for the action.



20. Scroll down to find the product translation part of the script and update the desired product name. *This name must match EXACTLY the name of the product as defined in the Configuration Client.*



21. Click the **OK** button on the **Script Editor** to save the script changes.
22. Click the **OK** button on the **Action Properties** to save the action changes.
23. Repeat steps **15-21** for the **100 – Part Start** event under **200 – Mounting Bolt Assembly.**
24. Click **Start** then **Run...**
25. Type services.msc and click the **OK** button.



26. Find and right-click the **Execution Management Server service.**
27. Click **Restart** to restart the service and load the new configuration.

3.3 Manufacturing Director™ - Shop Floor Client

The Shop Floor Client is the main interface between plant-floor operators and the system. It allows users to perform actions such as starting orders, entering quality data, viewing historical serial number information, and receiving system events.

3.4 Manufacturing Director™ - Dispatch Client

The Dispatching Client is where production schedules are created and managed. Within production schedules, production orders (i.e., manufacturing orders) are created and dispatched to work centers as individual work orders.

3.4.1 Dispatch Client - Implementation Specific Details

Production Schedule: There is one Production Schedule for the VCM implementation; "Current Schedule". This schedule is static and currently does not change. In the future if the production model changes and it is desirable to produce by production schedules this tool would be used to manually and/or automatically manage the schedules from the ERP.

> The Production Schedule "Planned Start Time" and the "Planned End Time" must be set outside of the time of production. Since the schedule is static the time range is opened up to a point far in the future; 1/1/2018.

Production Orders: There is a Production Order defined for each of the part types: Pre-Assembled Bridge Order, R72 Bridge Order, R70 FR Bridge Order, R70 RR Bridge Order. These production orders are static and do not change. In the future if the

production model changes and it is desirable to produce by production orders this tool would be used to manually and/or automatically manage the schedules from the ERP.

> The Production Order "Planned Start Time" and the "Planned End Time" must be set outside of the time of production. Since the orders are static the time range is opened up to a point far in the future; 1/1/2013.
>
> The Production Order must be dispatched to a work center, as a Work Order, for part processing.

Work Orders: Work orders are process at the Shop Floor Client.

See Appendix E for General Use definition of the Dispatch client.

3.5 Execution Management Server

Execution Management (EM) provides an interface layer between traditional business and Manufacturing Execution (MES) systems and manufacturing shop-floor automation systems. EM can be configured to react to changes in shop-floor automation system information from, for example, PLCs, barcode readers, and RFID tag readers, to provide configurable intelligent execution. EM accepts automation input, in the form of OPC events, from one or more OPC servers and responds to those events in such a way as is configured via the Administration Console (EMAdmin) and monitored via the Management Console (EMMgr). These tools are available via "Start>All Programs>Flexware Innovation>Execution Management".

3.6 Manufacturing Director™ Security

Manufacturing Director™ security uses locally defined security accounts and Microsoft network Active Directory user groups. The local and network roles and users are defined below.

Manufacturing Director™ Roles:

The following Manufacturing Director™ roles are required for the current solution.

- Administrator
- Assembly Area Operators
- Assembly Area Supervisors
- Background Processors
- Casting Area Operators
- Casting Area Supervisors
- Machining Area Operators
- Machining Area Supervisors
- Modelers
- Report Viewer
- Schedulers

Local User Accounts:

The following local user accounts can be used instead of the Active Directory user accounts:

Name	User Name	Password
Administrator	Administrator	password
Assembly Operator 1	AO1	ao1
Assembly Supervisor 1	AS1	as1
Casting Operator 1	CO1	co1
Casting Supervisor 1	CS1	cs1
Machining Operator 1	MO1	mo1
Machining Supervisor 1	MS1	ms1
ReportUser	ReportUser	reportuser

3.6.1 Active Directory User Groups and User Roster

In order to add and remove users from the Active Directory Users Groups contact IT and tell them the User Name and what Active Directory Group they are to be added to or removed from.

MES-Operators – Active Directory Group

Users:

- <no users in this group at this time>

MES-Supervisors – Active Directory Group

Users:

- Jeremy Lauderbaugh
- David Ward
- Rob McIntosh
- Arun Viswanathan
- Kevin Parker
- Brandy Douglas
- Brandy Jones

MES-ReportUser – Active Directory Group

Users:

- Tomoyuki Ueno
- Brandi Owens

MES-Administrator – Active Directory Group

Users:

- Russell Cosby

Note: See Appendix: “17.0 Configuring Users” for a general description of configuration user accounts.

3.7 Manufacturing Director Reporting

Manufacturing Director™ reporting utilizes SQL Server Reporting Services as the environment to host reports which are accessible to any computer on the organizations LAN. Users must have an Manufacturing Director™ security account or must have their Windows user account placed in the "MES-ReportUser" domain group managed by IT.

The reports available are:

- Supply Part Reports – Queried by a specified Date Time range
- Tool Change Report – Queried by a specified Date Time range
- Tracking Reports (Container and Genealogy/Serial Number) – Queried by the specific Container or Serial Number

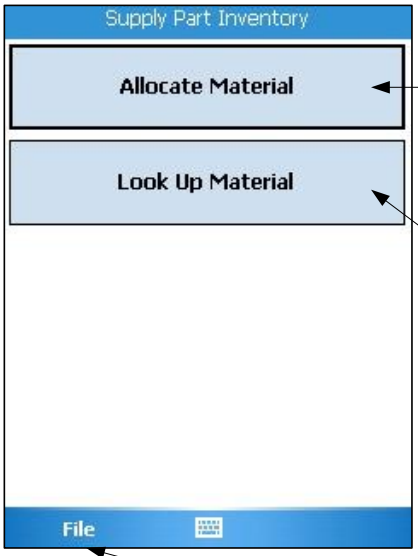
Access the reports by following this link:
http://192.168.1.153/ManufacturingDirector/Reports/Default.aspx

The IPAddress 192.168.1.153 is the Manufacturing Director™ server, "FLEXWARE_SERVER".

3.8 Raw Material Mobile Scanning

3.8.1 Start Application

From the Start menu, click the Supply Part Inventory link to start the application.

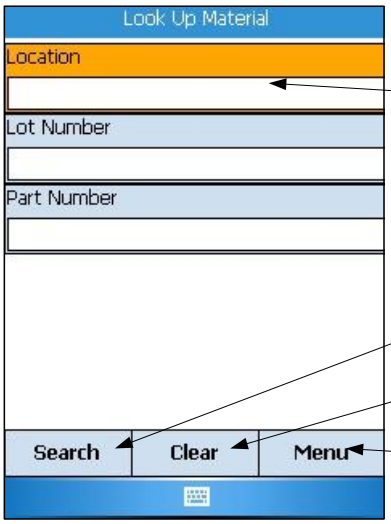


This button begins the process of allocating additional materials to the supply part inventory.

This button begin the process of looking up information on existing supply part inventory records.

The File menu gives an 'exit' option to allow the program to be closed.

3.8.1 Look Up Supply Part Inventory Material Information



Enter information into some, all, or none of the fields. If some or all of the fields are entered, the results will be limited by the information given. If none of the fields are entered, then all results will be returned. The bar code scanner can be used, where possible, to enter data. Note: only 'Active' and "Queued' results are returned.

Press the Search button to find data based on the entered criteria.

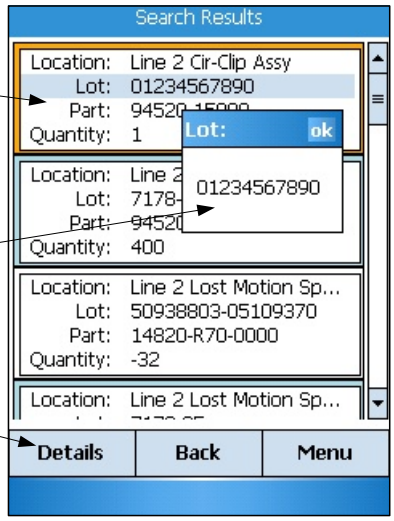
The Clear button will clear all fields of data.

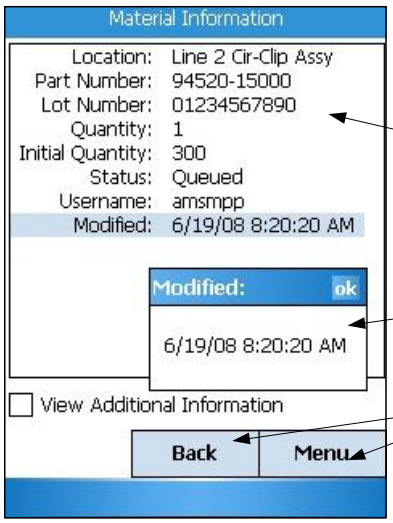
The Menu button will return the program to the main menu.

Results from searching are displayed here. Each box contains an individual record and can be selected to have its details viewed. Note the orange border on the selected record. If there are no results based on the search criteria, the no results will be displayed.

Clicking on an individual field allows its information to be viewed, in case that field is to long to be viewed on a single line.

Click the Details button to view the selected record's information.



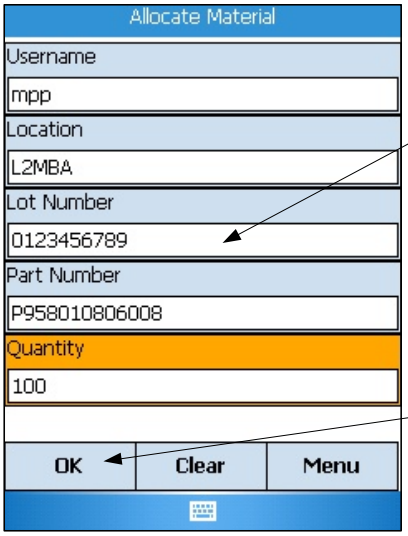


Record details are displayed here.

Clicking on an individual field allows its information to be viewed, in case that field is to long to be viewed on a single line.

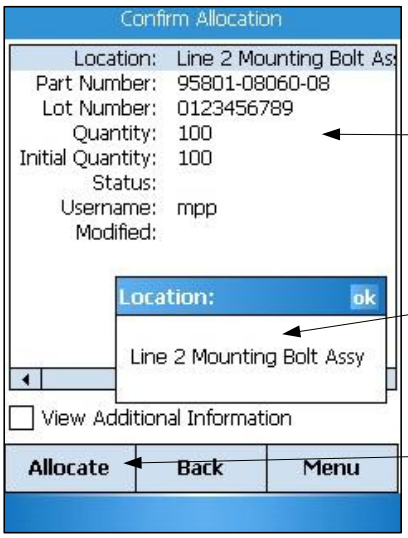
Press the Back button to return to the previous screen and view additional records, or press the Menu button to return to the main menu.

3.8.2 Allocate Supply Part Inventory Materials



Enter appropriate information into all of the fields, using the barcode scanner where possible. Note: at this point, the Part Number data is expected to be in the format that the barcode stores the information. There is another internal step that makes the transformation of the format. It is highly suggested to use the barcode scanner to enter this piece of data.

Press the OK button to continue with allocating these materials into the supply part inventory.

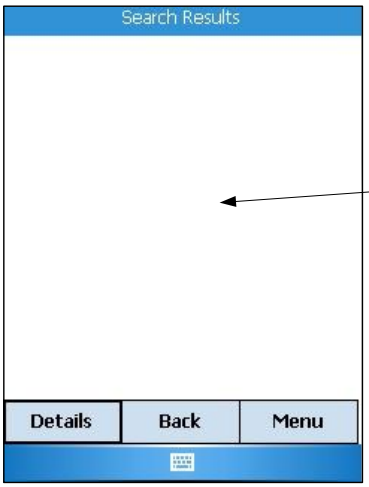


Review the information before making the allocation. Notice the transformed format of the Part Number from the previous format.

Clicking on an individual field allows its information to be viewed, in case that field is to long to be viewed on a single line.

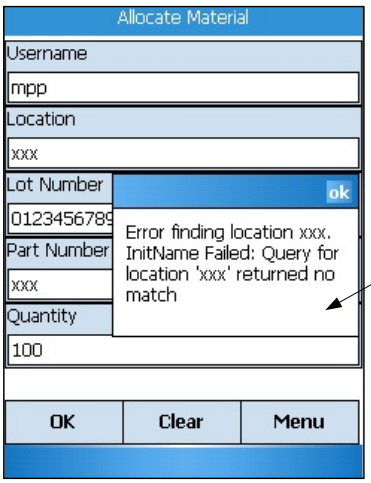
Press the Allocate to make the allocation of materials into the supply part inventory.

3.8.3 Error – No Search Results Displayed



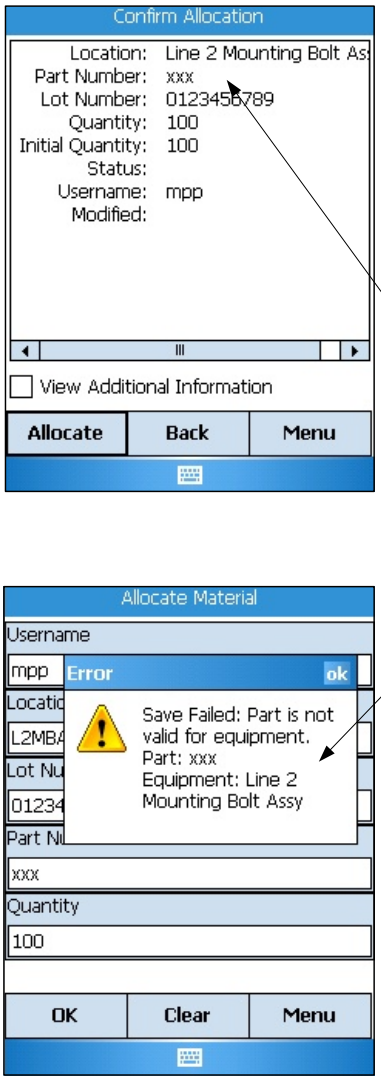
No search results displayed. This is the case that no results were found for the given search criteria. The solution is to change the search criteria to be correct, or it could be the case that there simply are no results to display.

Error Finding Location



Error finding location. The location information entered does not match the location "Short Name" configured in Manufacturing Director™. Enter the correct location information into the field and try again.

3.8.4 Error – Invalid Part Number



Invalid Part Number entered or Part Number not valid for given Location/Equipment. Contact your system administrator if you believe this to be in error.

3.8.5 Barcode Scanner Ceases to Function

If the barcode scanner ceases to function during normal operation of the software, try returning to the Main Menu and begin again from there. The software should reinitialize the barcode reading capabilities.

Error Connecting to Server

An error occurred while attempting to connect to the server. Check network connectivity and configuration information in the application's configuration file.

3.9 Server Installation Details (FLEXWARE_SERVER)

Typically if something critical happens to the Manufacturing Director™ server installation it would be rebuilt from system backups and a step by step installation process would not be required.

3.9.1 Base System

- Microsoft Windows 2003 Server
- Microsoft SQL Server 2005 w/Reporting Services
- IIS Installed and Enabled

3.9.2 Restore System and EMMD Databases

Depending upon the extent in which the system needs recovered the following databases may need to be restored:

- "master"
- "msdb"
- "EMMD"
- "ReportServer"
- Get latest full database backups of these databases from the Manufacturing Director™ Server or from IT file backups
- Restored the database backup files as required
    - EMMD database transaction logs may need to be recovered in addition to the database backup to get back to the closest point in time if needed.
    - Run 'EMMD Schema Logins - Users - Permissions.sql' against EMMD database

3.9.3 Update Log Paths

- Verify that the folder %MD%\Logs exists
- Using notepad, open

  %MD%\Flexware.ExecutionManagement.Agent.Server.exe.config
- Locate 'appSettings/LogDirectory'
- Modify value: enter fully expanded, absolute path to %MD%\Logs
- Save (and close) notepad

- Restart service: Execution Management Server Agent (may not be available yet)
- Using notepad, open

  %MD%\ Flexware.ExecutionManagement.MessageRouter.Server.exe.config
- Locate ‘appSettings/RouterLogDirectory‘
- Modify value: enter fully expanded, absolute path to %MD%\Logs
- Save (and close) notepad
- Restart service: Execution Management Message Router

3.9.4 Restore Reporting Files

Restore implementation specific reporting RDL files. This is done through Reporting Services. Do not attempt this unless familiar with reporting services.

3.9.5 Install EM Agent Service

- Open ‘Command Prompt’ from Start | All Programs | Accessories
- Navigate to %MD% directory

  Verify presence of Flexware.ExecutionManagement.Agent.Server.exe
- Use InstallUtil to install service:

  c:\windows\microsoft.net\framework\v2.0.50727\installutil Flexware.ExecutionManagement.Agent.Server.exe
- Watch for “The Commit phase completed successfully” and ‘The transacted install has completed” messages.

3.9.6 Install MXOPC Server

- Install MX OPC Server from media
- Restore the MXOPC Server configuration file: “MXConfiguratoe_2008_630_01.mdb (or the most recent configuration file)

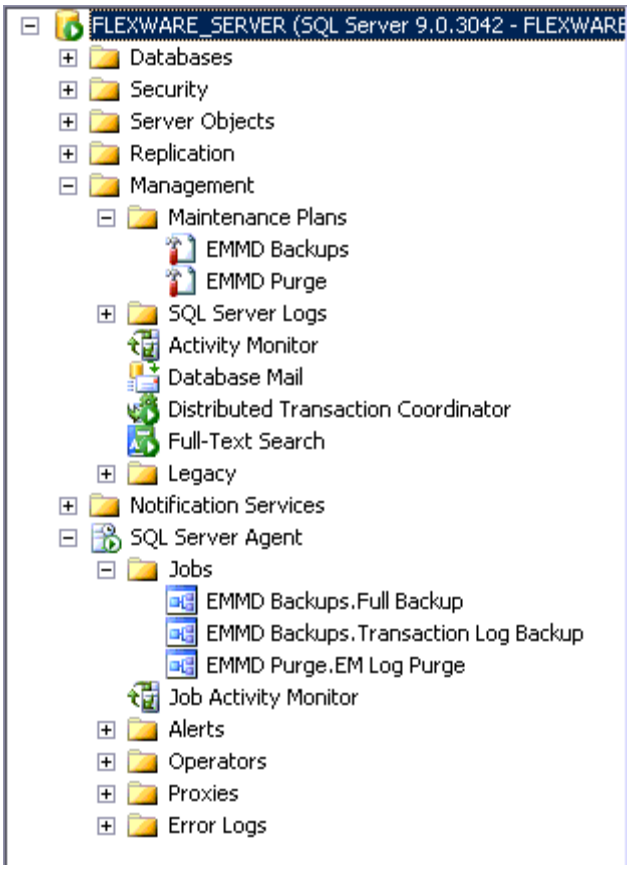
3.9.7 Install InSight OPC Server

- Install MX OPC Server from media
- Restore the MXOPC Server configuration file: “MXConfiguratoe_2008_630_01.mdb (or the most recent configuration file)

3.10 Database Maintenance

3.10.1 Maintenance Plans and Jobs

The Maintenance Plans are configured through Microsoft SQL Server Management Studio.



3.10.1.1 Overview

The MPP solution has two Maintenance Plans:

EMMD Backups – handles full and transaction log backups of the EMMD database. In addition to providing more granular disaster recovery, frequent transaction log backups keep the transaction log file from growing excessively large.

The EMMD Backups Maintenance Plan also backs up the “master”, “msdb”, and “ReportServer” databases.

The configuration of these maintenance plans is stored in the “msdb” database.

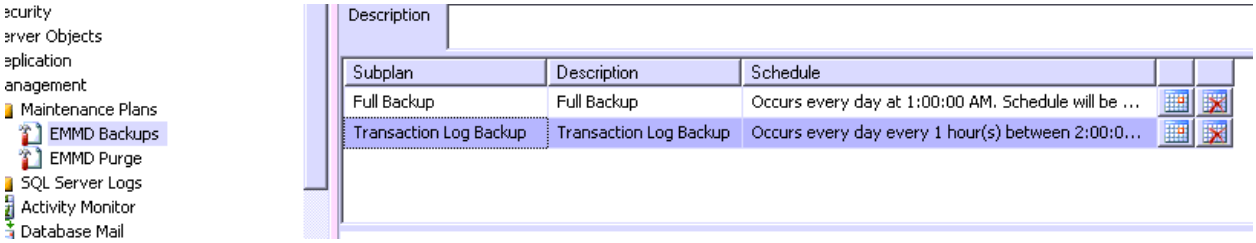
EMMD Purge – handles deletion of production-related log data that is stored in the EMMD database. There are two such log tables: Log_Event, which records all events received from the shop floor, and Log_Action , which records all responses to those events. These tables should be purged on a regular basis to prevent the data file from growing excessively large. The information is useful for system debugging, so it is recommended that at least a month of this data be retained.

3.10.1.2 Scheduling

This section contains conceptual and detailed information. While conceptual information should be accurate, the detailed information (specific times and reoccurrence schedules) could change outside of this document. Reference the Maintenance Plans in SQL Server and their respective jobs in SQL Server Agent for up-to-date information.

New maintenance plan schedules are best created inside the maintenance plan. Existing maintenance plan schedules can be edited inside the maintenance plan, or by modifying the SQL Agent job itself.

EMMD Backups – This maintenance plan has two sub-plans: a Full Backup sub-plan and a Transaction Log Backup sub-plan.



The full backup (EMMD Backups.Full Backup) is scheduled to run daily, at 10:00pm. This yields one self-contained BAK file, per day, which can be used to restore to up to the last 10:00pm. There is nothing particular about 10:00pm. Any time when production is at a minimum is sufficient.

The transaction log backup (EMMD Backups .Transaction Log Backup) is scheduled to run hourly, for every hour except 10:00pm. This yields one TRN file, per hour (23 per day) which can be used to restore up to any given hour. There is nothing particular about 1 hour. Business requirements must determine the interval. The frequency of transaction backup directly affects:

- File size: more often means small log file
- Duration: more often means quicker backup
- Impact: more often means less work each backup and reduced effect on performance
- Recovery: more often means more granular recovery and less data loss
- Ease of recovery: more often means more files to restore during recovery

Regardless of the frequency, be sure to skip the interval when the full backup is executing.

Please see SQL Server documentation or MSDN Library for more information on backup strategies.

EMMD Purge – This maintenance plan is scheduled to run at 9:30pm. There is nothing particular about 9:30pm, other than it is just before the full backup. Since the purge is deleting the oldest data (which has already been backed-up many times) it can occur before the full backup, thereby reducing the backup size. As with any purge, it should be executed during a down or off-peak time in order to reduce impact on the live system. Breaks, shift changes, and late nights/early mornings are good candidates.

This plan executes a stored procedure in the EMMD database. The number of days of data to retain is passed as an argument to the stored procedure. The default is 28 days. This number may be increased if disk space allows. Reducing this number is not recommended.

3.10.1.3 Monitoring

The SQL Server Agent Job Activity Monitor (in SQL Server Management Studio) should be used regularly to check job execution history. Any failures are out of norm and should be investigated immediately.

3.10.1.4 File Backup

The EMMD Backup maintenance plan only generates files on the local hard drive. Other IT processes or software should be used to copy or archive the backup files to another location.

Backup files are output to an EMMD folder in the standard SQL Server backup location:

C:\Program Files\Microsoft SQL Server\MSSQL.1\MSSQL\Backup

The actual database files should be excluded from any direct access by backup software. These files are located in the standard SQL Server data location:

C:\Program Files\Microsoft SQL Server\MSSQL.1\MSSQL\Data

3.10.1.5 File Cleanup

The EMMD Backup maintenance plan (Full Backup sub-plan) handles file cleanup. Whenever a full backup is created, any BAK or TRN files older than 4 weeks are deleted.

The cleanup of transaction log files is handled with the full backup for usability purposes. A TRN file without the prior backup file is useless.

There is a possibility of data loss by allowing SQL Server to handle file clean up. Should the external process fail, SQL Server will continue to delete files that have expired. Ideally, the external process that archives the files should somehow indicate its success or failure. If desired, the file cleanup steps can be removed from the maintenance plan.

4.0 System Details and User Accounts

4.1 Application Server

Application Server	
Computer Name:	FLEXWARE_SERVER
Domain:	MPPNET.com
IP Address:	192.168.1.153
Subnet Mask:	255.255.255.0
Default Gateway:	192.168.1.1
Preferred DNS Server:	153.121.200.154
Alternate DNS Server:	153.121.200.21
User Accounts	
Username:	flexware
Password:	flexdogs
Group:	(Local) Administrators
Username:	FlexAdmin
Password:	flexdogs
Group:	(Local) Administrators
Username:	administrator
Password:	password
Group:	(Local) Administrators
Software	
Operating System:	Windows 2003 Server R2 Service Pack 2
Database:	SQL Server 2005 Service Pack 2
OPC Servers:	TOP Server v4.280.435 Mitsubishi Suite User Configurable Driver
Manufacturing Director:	Build 3.2.183.0
Support	
Support Folder:	C:\Flexware\Support

4.2 Macola Server (ERP)

Macola Server	
Computer Name:	DELLSQLSERVER
IP Address:	153.121.200.163
SQL Server Login:	Contact IT
Database	
Database Name:	Base2

4.3 Cognex Insight Cameras

Device:	Assembly Out #1 (Bolt Assembly)
Name:	Assembly1_Out
Model Number:	InSight 5100
Firmware Version:	3.40.00 (318)
MAC Address:	00-d0-24-01-ce-c7

IP Address:	192.168.1.129
Serial Number:	Z72236369
Monitor Version:	2.01
Device:	Assembly Out #2 (Bolt Assembly)
Name:	Assembly2_Out
Model Number:	InSight 5110
Firmware Version:	4.01.00 (226)
MAC Address:	00-d0-24-02-1a-c5
IP Address:	192.168.1.147
Serial Number:	Z81524961
Monitor Version:	2.03
Device:	Casting #1
Name:	Casting1Cognex
Model Number:	InSight 5110
Firmware Version:	3.30.01 (346)
MAC Address:	00-d0-24-01-85-7d
IP Address:	192.168.1.121
Serial Number:	Z62877921
Monitor Version:	2.00
Device:	Casting #3
Name:	Casting2Cognex
Model Number:	InSight 5110
Firmware Version:	3.40.02 (464)
MAC Address:	00-d0-24-01-f1-b8
IP Address:	192.168.1.139
Serial Number:	Z74366477
Monitor Version:	2.01

4.4 Shop Floor Clients

MD accounts aren't documented anywhere, so you'd have to look in the MD configuration client

Device:	Casting 1 Client PC
Computer Name:	NEO-4D5E48
Domain:	WORKGROUP
IP Address:	192.168.1.109
Local Username:	MESUser
Local Password:	password
Startup Group Contain	Shop Floor Client RDP Shortcut
RDP Terminal Session	MESUser
RDP Terminal Session to FLEXWARE_SERVER UserName:	MD_CAST_U_1
RDP Terminal Session to FLEXWARE_SERVER Password:	password
User Profile Configuration	MD_CAST_U_1 user profile on FLEXWARE_SERVER contains the SFC shortcut on the desktop and in the startup group:
Shortcut to SFC –"Target":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director\Flexware.ManufacturingDirector.ShopFloor.Client.exe" /S:FLEXWARE_SERVER /D:EMMD /m:CASTER1 /w:f
Shortcut to SFC – "Start In":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director"

Device:	Casting 2 Client PC
Computer Name:	NEO-4D5E66
Domain:	WORKGROUP
IP Address:	192.168.1.134
Local Username:	MESUser
Local Password:	password
Startup Group Contain	Shop Floor Client RDP Shortcut
RDP Terminal Session	MESUser
RDP Terminal Session to FLEXWARE_SERVER UserName:	MD_CAST_U_2
RDP Terminal Session to FLEXWARE_SERVER Password:	password
User Profile Configuration	MD_CAST_U_2 user profile on FLEXWARE_SERVER contains the SFC shortcut on the desktop and in the startup group:
Shortcut to SFC –"Target":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director\Flexware.ManufacturingDirector.ShopFloor.Client.exe" /S:FLEXWARE_SERVER /D:EMMD /m:CASTER2 /w:f
Shortcut to SFC – "Start In":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director"
Device:	Machining Client PC
Computer Name:	NEO-4D5E57
Domain:	WORKGROUP
IP Address:	192.168.1.110
Local Username:	MESUser
Local Password:	password
Startup Group Contain	Shop Floor Client RDP Shortcut
RDP Terminal Session	MESUser
RDP Terminal Session to FLEXWARE_SERVER UserName:	MD_MACHINING_U_1
RDP Terminal Session to FLEXWARE_SERVER Password:	password
User Profile Configuration	MD_MACHINING_U_1 user profile on FLEXWARE_SERVER contains the SFC shortcut on the desktop and in the startup group:
Shortcut to SFC –"Target":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director\Flexware.ManufacturingDirector.ShopFloor.Client.exe" /S:FLEXWARE_SERVER /D:EMMD /m:MACHINING1 /w:f
Shortcut to SFC – "Start In":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director"
Device:	Assembly In Client PC
Computer Name:	NEO-4D5E7E
Domain:	WORKGROUP
IP Address:	192.168.1.107
Local Username:	MESUser
Local Password:	password
Startup Group Contain	Shop Floor Client RDP Shortcut
RDP Terminal Session	MESUser
RDP Terminal Session to FLEXWARE_SERVER UserName:	MD_AI_U_1
RDP Terminal Session to FLEXWARE_SERVER Password:	password

User Profile Configuration	MD_AI_U_1 user profile on FLEXWARE_SERVER contains the SFC shortcut on the desktop and in the startup group:
Shortcut to SFC –"Target":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director\Flexware.ManufacturingDirector.ShopFloor.Client.exe" /S:FLEXWARE_SERVER /D:EMMD /m:ASSEMBLYIN1 /w:f
Shortcut to SFC – "Start In":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director"
Device:	Assembly Out Client PC
Computer Name:	NEO-4D5FC1
Domain:	WORKGROUP
IP Address:	192.168.1.108
Local Username:	MESUser
Local Password:	password
Startup Group Contain	Shop Floor Client RDP Shortcut
RDP Terminal Session	MESUser
RDP Terminal Session to FLEXWARE_SERVER UserName:	MD_AO_U_1
RDP Terminal Session to FLEXWARE_SERVER Password:	password
User Profile Configuration	MD_AO_U_1 user profile on FLEXWARE_SERVER contains the SFC shortcut on the desktop and in the startup group:
Shortcut to SFC –"Target":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director\Flexware.ManufacturingDirector.ShopFloor.Client.exe" /S:FLEXWARE_SERVER /D:EMMD /m:ASSEMBLYOUT1 /w:f
Shortcut to SFC – "Start In":	"C:\Program Files (x86)\Flexware Innovation\Manufacturing Director"

4.5 Vorne Displays

Device:	Casting 1 Vorne
Model Number:	XL800
IP Address:	192.168.1.101
Device:	Casting 2 Vorne
Model Number:	XL800
IP Address:	192.168.1.142
Device:	Machining Vorne 1
Model Number:	XL800
IP Address:	192.168.1.102
Device:	Machining Vorne 2
Model Number:	XL800
IP Address:	192.168.1.103
Device:	Machining Vorne 3
Model Number:	XL800
IP Address:	192.168.1.141
Device:	Assembly Out 1 Vorne
Model Number:	XL800
IP Address:	192.168.1.104
Device:	Assembly Out 2 Vorne
Model Number:	XL800
IP Address:	192.168.1.140
User Accounts	

Username:	Supervisor
Password:	porthos
Username:	Administrator
Password:	aragorn

4.6 Assembly Raw Material Scanner

Device:	Hand Held Scanner #1
Name:	Hand Held Scanner #1
Model Number:	Motorola MC-9090-G
IP Address:	192.168.1.118
Device:	Hand Held Scanner #2
Name:	Hand Held Scanner #2
Model Number:	Motorola MC-9090-G
IP Address:	192.168.1.119
Device:	Hand Held Scanner #3
Name:	Hand Held Scanner #3
Model Number:	Motorola MC-9090-G
IP Address:	192.168.1.120

4.7 Zebra Label Printer

Device:	Assembly Out 1 Zebra Printer
Model Number:	Zebra 2844
IP Address:	192.168.1.144
Device:	Assembly Out 2 Zebra Printer
Model Number:	Zebra 2844
IP Address:	192.168.1.145

5.0 Casting

5.1 Drawings

Refer to Drawing Documentation.

5.2 Architecture

Refer to **Appendix B – System Architecture.**

5.3 Device Communication Settings

5.3.1 Insight Camera

The Insight camera is configured for Ethernet to communication the Insight OPC Server. Refer to **Appendix A - IP Addresses.**

6.0 Machining

6.1 Drawings

Refer to Drawing Documentation.

6.2 Architecture

Refer to **Appendix B – System Architecture**.

6.3 Device Communication Settings

6.3.1 Machining PLC (FX3U)

PLC communication is configured for Ethernet communication. Refer to **Appendix A - IP Addresses.**

Serial Port Settings	Settings
Communication Type	Serial
Baud Rate	19.2
Data Bits	8
Stop Bits	1
Parity	None

6.3.2 DM 7500 Hand Held Scanner

The DM100 fixed scanner is configured to send Serial Number data to the Machining FX PLC thru the serial port on the GOT 1020.

Parameter	Settings
Communication Type	Serial
Baud Rate	19.2
Data Bits	8
Stop Bits	1
Parity	None

7.0 Line 1 Assembly

7.1 Drawings

Refer to Drawing Documentation.

7.2 Architecture

Refer to **Appendix B – System Architecture**.

7.3 Process Data Flow

7.3.1 Assembly Cir-Clip

Refer to **Appendix D – Process Data Flow Diagram (Assembly In).**

7.3.2 Assembly (Lost Motion)

Refer to **Appendix D – Process Data Flow Diagram (Assembly In).**

7.3.3 Assembly Out (Bolt Assembly)

Refer to **Appendix C – Process Data Flow Diagram (Assembly Out).**

7.4 Device Communication Settings

7.4.1 Assembly Cir-Clip

7.4.1.1 Cir-Clip PLC (A Series)

PLC communication is configured for Ethernet communication. Refer to **Appendix A - IP Addresses.**

7.4.1.2 DM 100 Fixed Scanner

The DM100 fixed scanner is configured to send Serial Number data to the Cir-Clip PLC (FX3U) thru the serial port.

Serial Port Settings	Settings
Communication Type	Serial
Baud Rate	9600
Data Bits	8
Stop Bits	1
Parity	None

7.4.1.3 GOT 940

The Cir-Clip GOT 940 is only configured to monitor data from the Cir-Clip PLC.

7.4.1.4 Cir-Clip PLC (FX3U)

PLC communication is configured for Ethernet communication. Refer to **Appendix A - IP Addresses.**

Serial Port Settings	Settings
Communication Type	Serial
Baud Rate	9600
Data Bits	8
Stop Bits	1
Parity	None

7.4.2 Assembly In (Lost Motion)

7.4.2.1 Lost Motion PLC (Q Series)

PLC communication is configured for Ethernet communication. Refer to **Appendix A - IP Addresses.**

7.4.2.2 DM 100 Fixed Scanner

The DM100 fixed scanner is configured to send Serial Number data to the PLC thru the GOT1000 serial port.

Serial Port Settings	Settings
Communication Type	Serial
Baud Rate	9600
Data Bits	8
Stop Bits	1
Parity	None

7.4.2.3 DM 7500 Hand Held Scanner

Parameter	Settings
Communication	USB

7.4.2.4 GOT 1000

Serial Port Settings	Settings
Communication Type	Serial
Baud Rate	9600
Data Bits	8
Stop Bits	1

Parity	None

7.4.3 Assembly Out (Bolt Assembly)

7.4.3.1 Bolt Assembly PLC (Q Series)

PLC communication is configured for Ethernet communication. Refer to **Appendix A - IP Addresses.**

7.4.3.2 InSight Camera

Parameter	Settings
Communication Type	Serial
Baud Rate	9600
Data Bits	8
Stop Bits	1
Parity	None

7.4.3.3 DM 7500 Hand Held Scanner

Parameter	Settings
Communication	USB

8.0 Line 2 Assembly

8.1 Drawings

Refer to Drawing Documentation.

8.2 Architecture

Refer to Refer to **Appendix B – System Architecture**.

8.3 Process Data Flow

8.3.1 Assembly Cir-Clip

Refer to **Appendix D – Process Data Flow Diagram (Assembly In).**

8.3.2 Assembly Out (Bolt Assembly)

Refer to **Appendix C – Process Data Flow Diagram (Assembly Out).**

8.4 Device Communication Settings

8.4.1 Line 2 Main PLC (Q Series)

Refer to **Appendix A - IP Addresses.**

8.4.2 Assembly In (Cir-Clip)

8.4.2.1 Cir-Clip PLC (Q Series)

PLC communication is configured for MelSec Network thru Ethernet communication on the Line 2 Main PLC.

Parameter	Settings
MELSEC Network	3
Station Number	2

8.4.2.2 DM 100 Fixed Scanner

The DM100 fixed scanner is configured to send Serial Number data to the PLC thru the GOT1000 serial port.

Serial Port Settings	Settings
Communication Type	Serial

Baud Rate	9600
Data Bits	8
Stop Bits	1
Parity	None

8.4.2.3 GOT 1000

Serial Port Settings	Settings
Communication Type	Serial
Baud Rate	9600
Data Bits	8
Stop Bits	1
Parity	None

8.4.3 Assembly Out (Bolt Assembly)

8.4.3.1 Bolt Assembly PLC (Q Series)

PLC communication is configured for MelSec Network thru Ethernet communication on the Line 2 Main PLC.

Parameter	Settings
MELSEC Network	3
Station Number	8

8.4.3.2 DM 7500 Hand Held Scanner

Parameter	Settings
Communication	USB

8.4.3.3 Insight Camera

Parameter	Settings
Communication Type	Serial
Baud Rate	9600
Data Bits	8
Stop Bits	1
Parity	None

9.0 Device Configuration

9.1 Cognex In-Sight Sensor

The following job files are loaded on startup into the Cognex In-Sight Sensors using the In-Sight Explorer 3.3.2 Software.

File Name	Location
Casting1Cognex.job	Casting Line 1
Casting2Cognex.job	Casting Line 2
Assembly1_Out.job	Assembly Out - Line 1
Assembly2_Out.job	Assembly Out - Line 2

Source Code is to be backed up and maintained by Madison Precision.

9.2 DataMan 100 Fixed Scanner

The following configuration files are loaded into the Cognex DataMan ID Readers using the DataMan 3.1.0 Setup Tool.

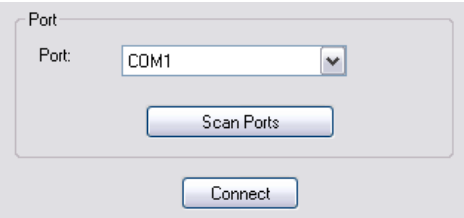
File Name	Location
Assembly1_DataMan.cfg	Assembly In - Line 1
Assembly2_DataMan.cfg	Assembly In – Line 2

Vision Sensor Source Code is to be backed up and maintained by Madison Precision.

9.2.1 Load Configuration File

To configure the DataMan 100 fixed scanners:

- Connect the DM100 to the PC via the USB Communications cable
- Open **Start | All Programs | Cognex | DataMan Setup Tool v3.1.0 | Setup Tool**
- Click **Scan Ports** to find the correct port information
- Click **Connect**



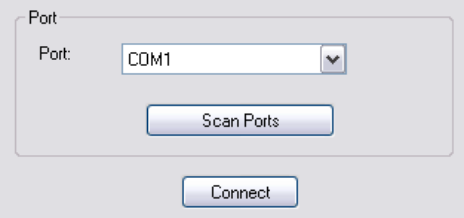
- The “Status” icon should now display “Connected” and menu items will display in the left pane.

- Select **File | Open Configuration**
- Select **System | Save settings**
- Select **Connect To Reader** in left window
- Click on **Disconnect**
- Disconnect USB Communications cable from DM100

9.2.2 Modify Device Settings

To configure the DataMan 100 fixed scanners:

- Connect the DM100 to the PC via the USB Communications cable
- Open **Start | All Programs | Cognex | DataMan Setup Tool v3.1.0 | Setup Tool**
- Click **Scan Ports** to find the correct port information
- Click **Connect**



- The "Status" icon should now display "Connected" and menu items will display in the left pane.
- To modify device setting, click on menu selections on left pane and change desired parameters.
- Select **System | Save settings**
- Select **Connect To Reader** in left window
- Click on **Disconnect**
- Disconnect USB Communications cable from DM100

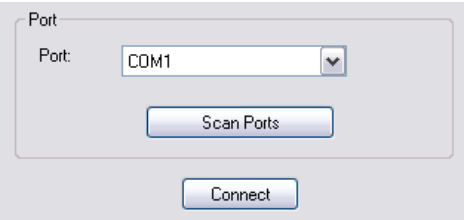
9.3 DataMan 7500 – Handheld Scanner (Machining)

To Reset the DM7500 Handheld scanner to default settings, scan the "Reset" configuration code followed by the "Configure" code located by the hand scanners at all three lines at Machining station.

9.3.1 Load Configuration File

To configure the DataMan 7500 Handheld scanners:

- Connect the DM7500 to the PC via the RS232 Communications cable
- Open **Start | All Programs | Cognex | DataMan Setup Tool v3.1.0 | Setup Tool**
- Click **Scan Ports** to find the correct port information
- Click **Connect**

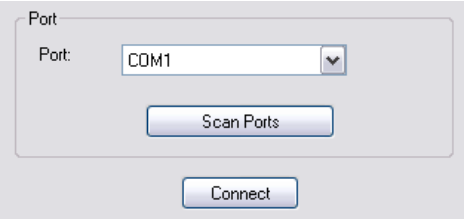


- The "Status" icon should now display "Connected" and menu items will display in the left pane.
- Select **File | Open Configuration**
- Select **System | Save settings**
- Select **Connect To Reader** in left window
- Click on **Disconnect**
- Disconnect USB Communications cable from DM7500

9.3.2 Modify Device Settings

To configure the DataMan 7500 Handheld scanners:

- Connect the DM7500 to the PC via the RS232 Communications cable
- Open **Start | All Programs | Cognex | DataMan Setup Tool v3.1.0 | Setup Tool**
- Click **Scan Ports** to find the correct port information
- Click **Connect**



- The "Status" icon should now display "Connected" and menu items will display in the left pane.
- To modify device setting, click on menu selections on left pane and change desired parameters.

- Select **System | Save settings**
- Select **Connect To Reader** in left window
- Click on **Disconnect**
- Disconnect USB Communications cable from DM7500

9.4 GOT 1000/1020 HMI Display

To configure the serial port setting on the GOT HMI displays:

9.4.1 Upload Application

- Open **Start | All Programs | Melsoft Application | GT Designer2**
- Select **Open**
- Browse to GOT project application
- Click **Open**
- Connect to GOT display using the USB communication cable
- Select **Communication | To/From GOT**
- Select **Project Upload -> Computer**
- Select **Get Lastest**
- Select **Upload**
- Select **Project | Save As** to save current backup of GOT application

9.4.2 Download Application Changes

- Make Application changes
- Connect to GOT display using the USB communication cable
- Select **Communication | To/From GOT**
- Select **Project Download -> GOT**
- Select modified items in project tree
- Select **Download**
- Select **Close**
- Select **Project | Save As** to save current backup of GOT application

9.5 GOT 940 HMI Display

To configure the serial port setting on the GOT HMI displays:

9.5.1 Upload Application

- Open **Start | All Programs | Melsoft Application | GT Designer2**
- Select **Open**
- Browse to GOT project application
- Click **Open**
- Connect to GOT display using the Serial communication cable
- Select **Communication | To/From GOT**
- Select **Project Upload -> Computer**
- Select **Get Lastest**
- Select **Upload**
- Select **Project | Save As** to save current backup of GOT application

9.5.2 Download Application Changes

- Make Application changes
- Connect to GOT display using the Serial communication cable
- Select **Communication | To/From GOT**
- Select **Project Download -> GOT**
- Select modified items in project tree
- Select **Download**
- Select **Close**
- 1111Select **Project | Save As** to save current backup of GOT application

9.6 FX3U PLC

The FX Configurator-EN application is used to view and make changes to the Ethernet module communication settings.

9.6.1 Load Ethernet Configuration File

- Open Start | All Programs | MelSoft Application | FX Configurator-EN
- Select Open | File

- Browse to configuration file location
- Click Open
- Click Transfer Setup
- Select connecting interface settings
- Click OK
- Click Write

9.6.2 Modify Device Ethernet Settings

- Open Start | All Programs | MelSoft Application | FX Configurator-EN
- Click Transfer Setup
- Select connecting interface settings
- Click Read
- Modify desired device settings: Operational Settings (PLC IP Address) or Open Settings (Configure TCP connections)
- Click Write
- Select File | Save

9.7 Vorne XL800 Display Board

The Vorne XL800 display board must be properly configured to conform to pre-defined standards and formats. This is done by exporting configuration files from existing displays and importing the file into a display that is un-configured.

Note: The XL800 must be using a firmware version of 3.0 or greater in order to be able to export a configuration file. Instructions on upgrade the device firmware is outlined in section 4.8.1.3.

9.7.1 Importing Configuration Files

- Open a standard web browser
- Enter the IP address into the web browser address bar **http://192.168.1.xxx/**
- Click **Login**
- Select **Administrator** from drop down list
- Enter password
- Click **Log In**

- Click **Communication | More Options**
- Click **Import Configuration**
- Locate and import .xml configuration file
- Click **Production Monitor** to view current data

9.7.2 List of Existing Configuration Files

The following configuration files are pre-loaded into the designated Vorne XL display boards.

File Name	Location
"XL_Configuration_Casting.xml"	Casting Lines 1 and 2
"XL_Configuration_Assembly.xml"	Assembly Out Lines 1 and 2
"XL_Configuration_Machining.xml"	Machining Lines 1, 2 and 3

9.7.3 Exporting Configuration Files

- Open a standard web browser
- Enter the IP address into the web browser address bar **http://192.168.1.xxx/**
- Click **Login**
- Select **Administrator** from drop down list
- Enter password
- Click **Log In**
- Click **Communication | More Options**
- Click **Export Configuration to File**
- Choose name and location for .xml file

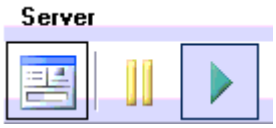
9.7.4 Update Firmware to XL Release 3.0

Firmware will only need to be done updated if the firmware of the device is older than 3.0. Current device firmware is displayed when the device is rebooted.

- Load the XL Tools CD in the CD-ROM drive
- Click **Firmware Update**
- Run **Update XL Firmware to Release 3.bat**
- Enter IP Address
- Click **Enter**
- Click **Enter** to start the update

9.8 In-Sight OPC Server

The Cognex In-Sight OPC Server 4.1.0 is used for EM communication with the In-Sight Sensors located at Casting Lines 1 and 2. Changes can be made to the configuration of the server by opening the "Configuration Editor" by clicking the icon in the Server tab.



The currently configured sensors will be displayed in the Configuration Editor and their respective OPC tags. Although both sets of OPC tags have the same tag name, they are only configured to one Cognex Sensor and its IP address. Below is the list of configured tags for each sensor and what they will look like during normal conditions.

Configured Tags

Casting1Cognex Configured ...	Access	Data Type	Value	Quality
DataRdy	ReadWrite	Float	0	Good
DataRdyReset	ReadWrite	Float	0	Good
MESStnInvalid	ReadWrite	Float	0	Good
MESStnValid	ReadWrite	Float	0	Good
PartGrade	ReadWrite	String	C	Good
PartSN	ReadWrite	String	8807E1123T	Good
PartStatus01	ReadWrite	Float	1	Good
SoftIntrLkByPass	ReadWrite	Float	0	Good
TransInProc	ReadWrite	Float	0	Good
WatchDog	ReadWrite	Float	1	Good

9.9 Mitsubishi OPC Server

The Mitsubishi OPC Server is used for EM communications with PLCs located in Machining and Assembly. Each PLC location corresponds to a unique device name in the OPC server with its own OPC tags. All changes to the Mitsubishi OPC Server is made by using the MXConfigurator software with the database of "MXConfiguratoe_2008_630_01.mdb".

9.9.1 Running the Server

The Mitsubishi OPC Server is configured to go into mode on startup and to use the "MXConfigurator_2008_630_01.mdb" database file. The play, stop, and light bulb icons located in the menu bar provide indication for the current server state. The green arrow will be depressed when the server is in run mode. The red square will be depressed when the server is stopped. No communication between the Server and PLCs will take place when the server is stopped. The light bulb is depressed to make the current database active for the server. Finally, the glasses icon is used to monitor OPC tags. Select the desired device to monitor and click the glasses icon. The OPC tags will be displayed below.

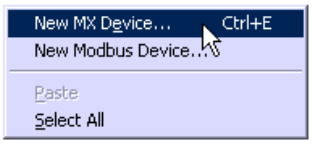
9.9.2 Devices

Below is a list of the devices that are configured in the Mitsubishi OPC Server.

Device Name	Description	Enable	PC Side IF	Comment	Back...
CClip		No	Ethernet board	AssemblyOut1	No
L1_AssemblyIn		Yes	Ethernet board	AssemblyIn1	No
L1_AssemblyOut		Yes	Ethernet board	AssemblyOut1	No
L1_CClip		Yes	Ethernet board	CirClip1	No
L1_Machining		Yes	Ethernet board	Machining Line 3	No
L2_AssemblyIn		Yes	Ethernet board	AssemblyIn2	No
L2_AssemblyOut		Yes	Ethernet board	BoltAssy2	No
L2_Machining		Yes	Ethernet board	Machining Line 2	No
L3_Machining		Yes	Ethernet board	Machining Line 1	No

9.9.2.1 Adding a New Device

New devices can be created in the MXConfigurator tool by right-clicking in the window and selecting “New MX Device”.



Clicking this option will open the Communication Setting Wizard. The Wizard is used to configure the communication parameters for both the PC and PLC side.

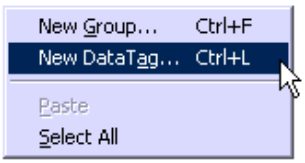
9.9.3 Tags

Once a device is created, OPC tags need to be added to send/receive data to the PLC. Shown below are the tags that have been configured for the L1_Machining device. Many attributes need to be selected while adding new tags such as PLC address, Access Rights, Data Type, and Polling Method. The tags shown below demonstrate typical tag properties for all PLC devices.

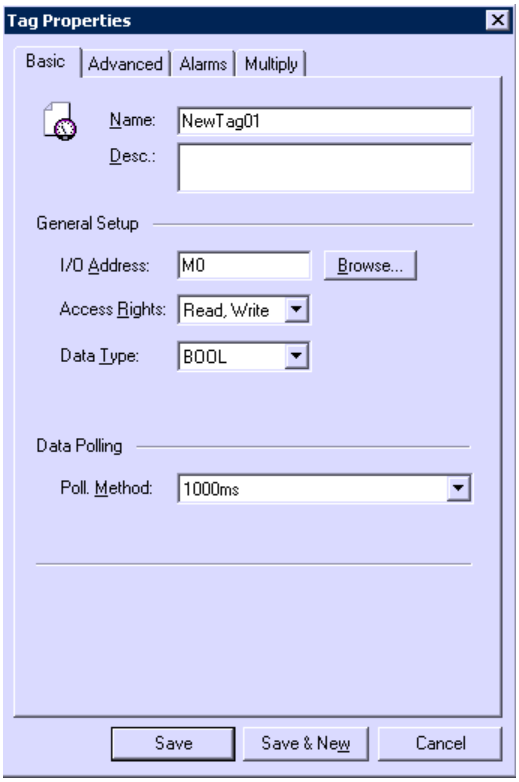
Name	Enable	Simul...	Address	Access Rights	Alarms	Data Type	Polling Met...
Dynamic Tags							
DataRdy	Yes	No	M950	Read, Write	No	BOOL	100ms
HardwareInterlockEnforced	Yes	No	M954	Read, Write	No	BOOL	100ms
MESStnInValid	Yes	No	M958	Read, Write	No	BOOL	100ms
MESStnValid	Yes	No	M955	Read, Write	No	BOOL	100ms
PartDisposition	Yes	No	D872	Read, Write	No	STRING (10)	100ms
PartSN	Yes	No	D800	Read, Write	No	STRING (10)	100ms
SoftIntrLkByPass	Yes	No	M953	Read, Write	No	BOOL	100ms
TransInProc	Yes	No	M951	Read, Write	No	BOOL	100ms
WatchDog	Yes	No	M957	Read, Write	No	BOOL	100ms

9.9.3.1 Adding a New Tag

New tags can be created in the MXConfigurator tool by right-clicking in the window and selecting “New DataTag”.



Clicking this option will open the Tag Properties Window. The window is used to configure the parameters associated with the tag such as I/O Address and Data Type.



9.10 Zebra Label Printer

The zebra label template is located in the Manufacturing Director™ program folder. Changes to the label are to be made with a ZPL editor (i.e., Loftware).

Refer to the Zebra Manuals for more information regarding printer setup and configuration of network settings.

10.0 Tools And Software

The following tools and software are available and needed to make modifications or to monitor devices on the line:

Device	Tools and Software
DM100	DataMan Setup Tool v3.1.0
DM7500	DataMan Setup Tool v3.1.0
Insight Camera	In-Sight Explorer 3.3.2
Vorne XL800	Web Browser (Microsoft Internet Explorer 6, Microsoft Internet Explorer 7, or Mozilla Firefox 2. Although Internet Explorer 6 is supported, it is not recommended, as performance is substantially improved with Internet Explorer 7.)
GOT 940	GT Designer2 v2.77F
GOT 1000	GT Designer2 v2.77F
GOT 1020	GT Designer2 v2.77F
A Series PLC	GX Developer v8.55H
Q Series PLC	GX Developer v8.55H
FX3U PLC	GX Developer v8.55H FX Configurator-EN v1.00 (Ethernet IP Configuration)

11.0 Support

Flexware Innovation, Inc. offers high-quality technical support. If you have a question, please contact us at:

Flexware Innovation, Inc.

9128 Technology Lane

Fishers, IN 46038

Office: 317-813-5400

Fax: 317-813-2121

http://www.flexwareinnovation.com

Services and prices are subject to Flexware's then current prices, terms and conditions and are subject to change without notice.

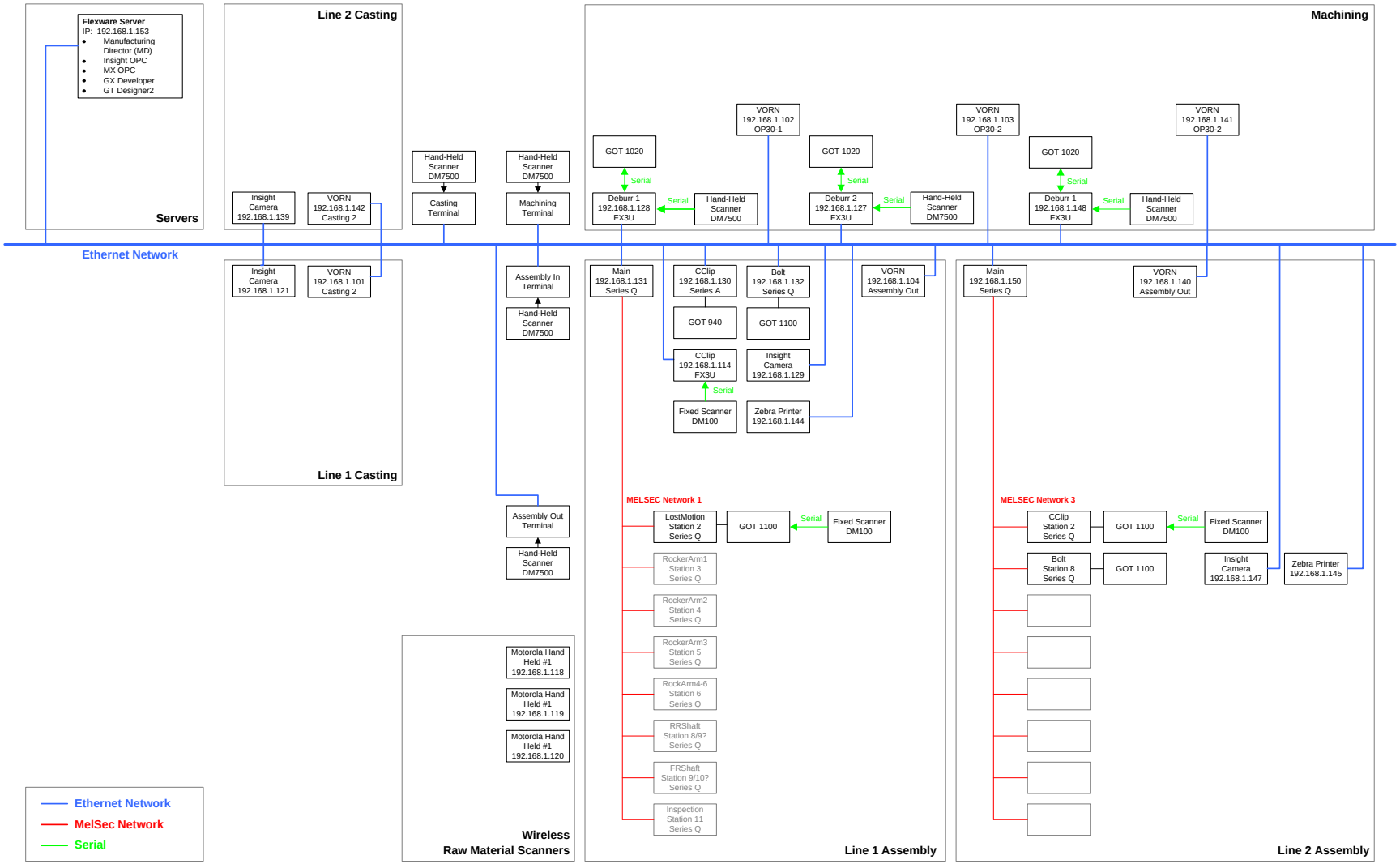
12.0 Appendix A – IP Addresses

Area	Name	Line #	MAC Address	IP Address
Lost Motion	LEP			153.121.200.192
Lost Motion	LEP Printer			153.121.200.193
Casting	Casting Vorne XL	1		192.168.1.101
Machining	Maching Vorne XL #1	1		192.168.1.102
Machining	Maching Vorne XL #2	1		192.168.1.103
Assembly	Assy Vorne XL	1		192.168.1.104
Assembly	Cisco Access Point			192.168.1.105
Assembly	Assy Comtrol	1		192.168.1.106
Assembly	Assy OIT 1	1		192.168.1.107
Assembly	Assy OIT 2	1		192.168.1.108
Casting	Casting OIT 1	1		192.168.1.109
Machining	Machining OIT 1	1		192.168.1.110
Casting	Casting GB #1	1	00:30:DE:01:43:33	192.168.1.111
Machining	Machining GB 1	1	00:30:DE:01:43:80	192.168.1.112
Machining	Machining GB 2	1	00:30:DE:01:41:67	192.168.1.113
Assembly	Cir Clip FX	1		192.168.1.114
Assembly	Assy GB #2	1	00:30:DE:01:43:25	192.168.1.115
Assembly	Assy LEP	1		192.168.1.116
Assembly	Assy LEP Printer	1		192.168.1.117
Assembly	Hand Held #1	1		192.168.1.118
Assembly	Hand Held #2	1		192.168.1.119
Assembly	Hand Held #3	1		192.168.1.120
Casting	Cognex Camera	1		192.168.1.121
Assembly	C-Clip OIT Lantronix	1		192.168.1.122
Assembly	Washer OIT Lantronix	1		192.168.1.123
Casting	Casting OIT Lantronix	1		192.168.1.124
Casting	Data Collection Lantronix	1		192.168.1.125
Machining	Machining OIT Lantronix	1		192.168.1.126
Machining	Machining FX PLC	2		192.168.1.127
Machining	Machining FX PLC	1		192.168.1.128
Assembly	Bolt Install Cognex	1		192.168.1.129
Assembly	Assy C-Clip PLC	1		192.168.1.130
Assembly	Assy Main PLC	1		192.168.1.131
Assembly	Assy Bolt PLC	1		192.168.1.132
Casting	Repair OIT Lantronix	1		192.168.1.133

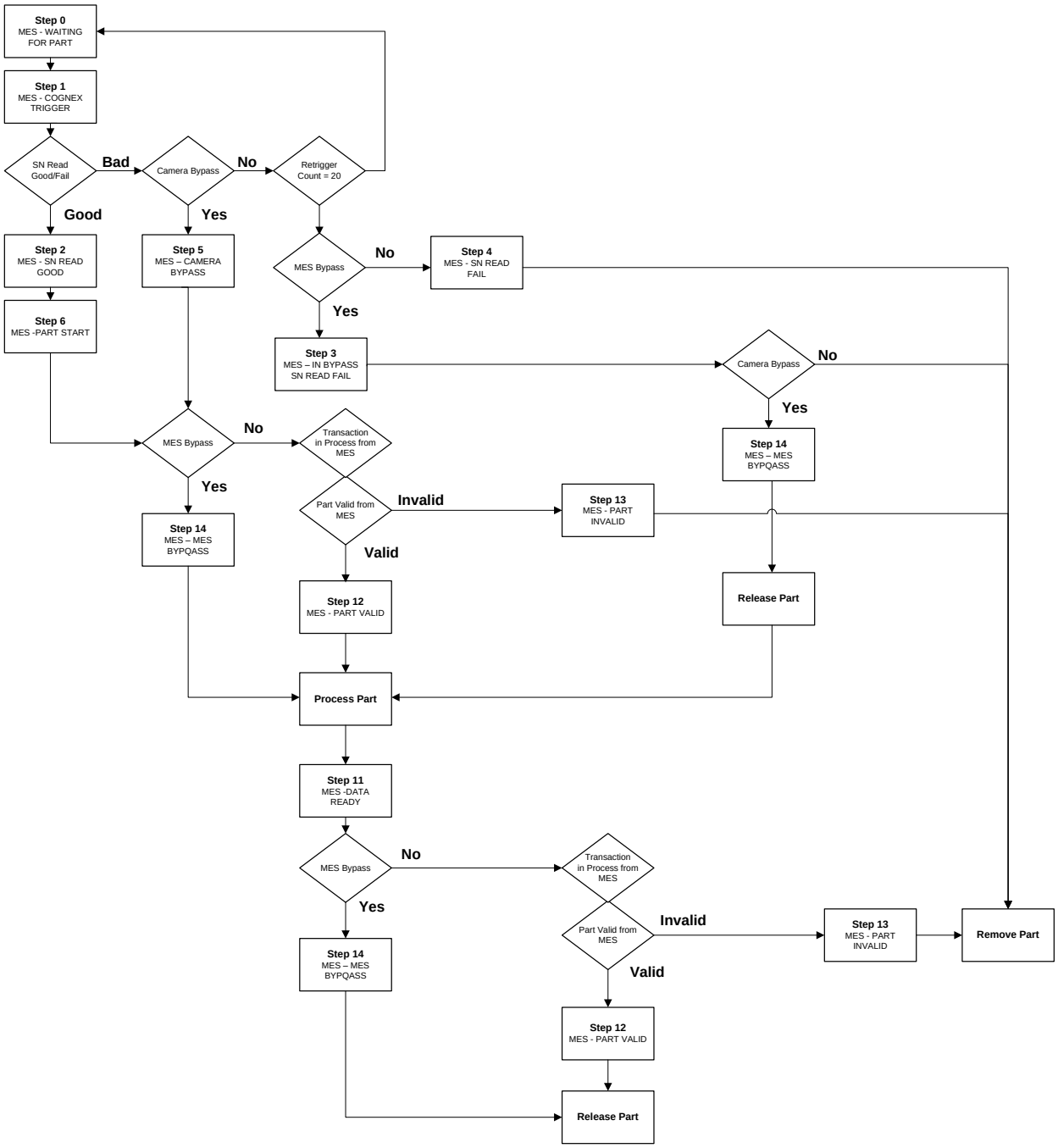
Casting	Repair OIT	1		192.168.1.134
R70	LEP		00:30:DE:01:C2:8A	192.168.1.135
R70	LEP Printer			192.168.1.136
Thermo Case	LEP		00:30:DE:01:37:CE	192.168.1.137
Thermo Case	LEP Printer			192.168.1.138
Casting	Cognex Camera	2		192.168.1.139
Assembly	Bolt Assembly (out) Vorne XL	2		192.168.1.140
Machinning	OP30 Vorne XL	2		192.168.1.141
Casting	Casting 2 Vorne XL	2		192.168.1.142
Spare	Spare			192.168.1.143
Assembly	Assembly Out Zebra Printer	1		192.168.1.144
Assembly	Assembly Out Zebra Printer	2		192.168.1.145
Spare	Spare			192.168.1.146
Assembly	Assembly Out Cognex Camera	2		192.168.1.147
Machining	Maching FX #3	3		192.168.1.148
Spare	Spare			192.168.1.149
Assembly	Assembly Line Main PLC	2		192.168.1.150
Spare	Spare			192.168.1.151
Spare	Spare			192.168.1.152
Server	Flexware Server			192.168.1.153
Spare	Spare			192.168.1.154
Spare	Spare			192.168.1.155

Highlighted IP Addresses could not be found with a network ping. These items were offline or no longer exist in the system.

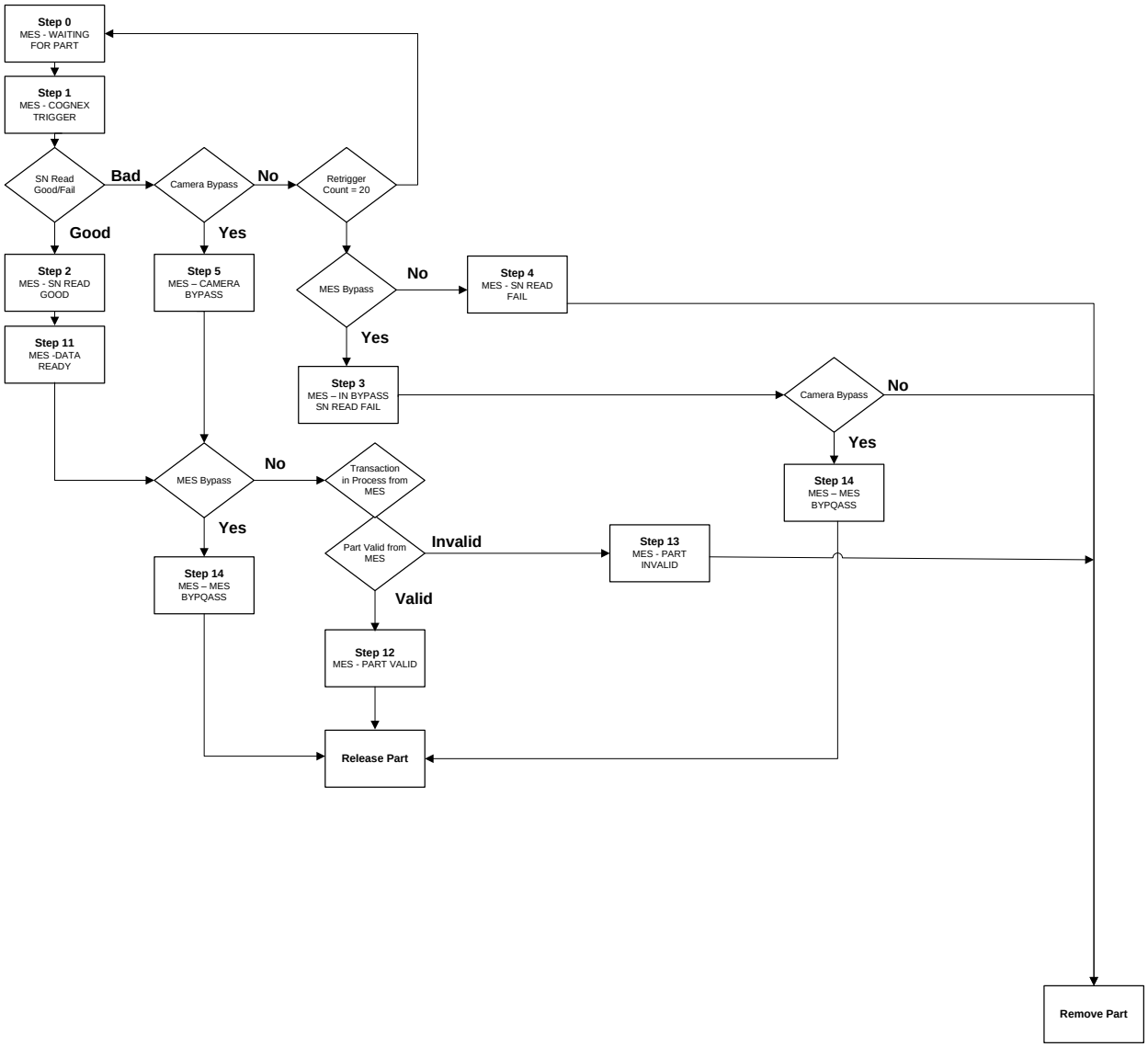
13.0 Appendix B – System Architecture



14.0 Appendix C – Process Data Flow Diagram (Assembly Out)



15.0 Appendix D – Process Data Flow Diagram (Assembly In)



16.0 Appendix E –Creating a New Production Order for the New Product

Manufacturing Director contains the Dispatching Client where **production schedules** are created and managed. Within production schedules, **production orders** (i.e., **manufacturing orders**) are created and **dispatched** to **work centers** as individual **work orders**.

16.1 Using the Dispatching Client

In the **Dispatching Client** the schedulable level (e.g., Sites) will display in the server explorer tree. In this system only one schedule is used and it is always open. Click on the "Production Schedules" folder to display the **Production Schedule Management** screen. From here the production schedule can be viewed.

In the server explorer tree, click on the production schedule to view the **Production Order Management** screen and the **Work Center Grid**.

Click **Add** to create a new production order, or select a production order and click edit to modify. This will display the Production Order Definition window. It is not necessary to use the Comment or Attributes tab. On the General tab fill in:

- Name
- Planned Start Time
- Planned End Time
- State (set to Scheduled)
- Quantity Required

On the **Product Specification** tab select the appropriate product specification.

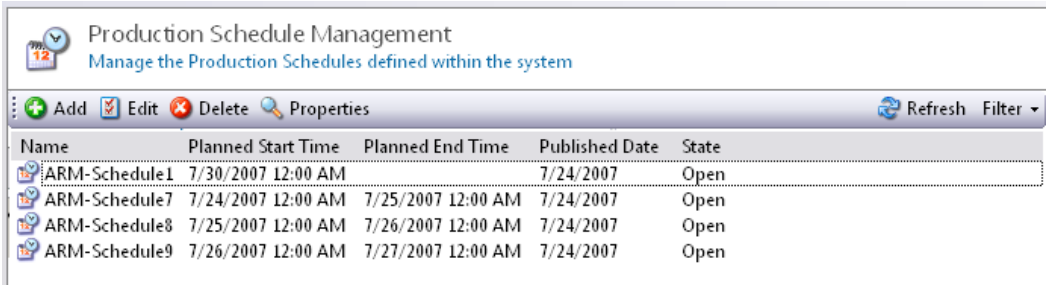
On the **Route** tab select the only route available and click **Select**.

The following sections describe in more details each of the screen mentioned above:

- Production Schedule Management
- Production Order Management
- Production Order Definition (General)
- Production Order Definition (Product Specification)
- Production Order Definition (Route)

16.2 Production Schedule Management

There is only one production schedule in the system. It will always be open.



16.2.1 How to Access:

From the Dispatching Client:

1. Select a **Production Schedules** folder in the explorer tree. The Production Schedule Management screen displays on the right.

16.2.2 Overview:

The Production Schedule Management window allows for the creating, viewing, modifying, deleting of production schedules.

16.2.3 Details:

The production schedules can be filtered based upon state.

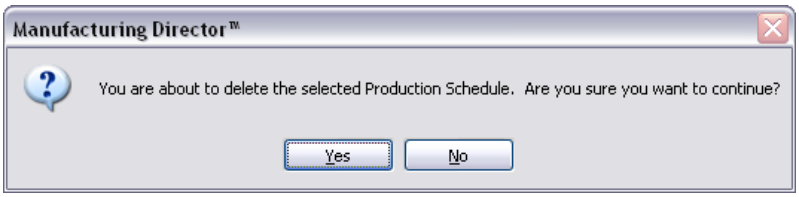
The functionality for the Add, Edit, Delete, and Properties buttons described in the Fields & Buttons section can also be accessed with the right click option.

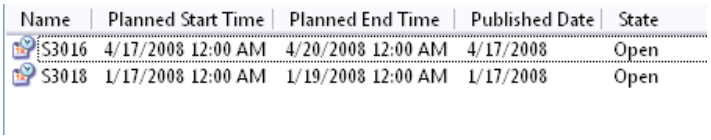
16.2.4 Prerequisites:

See the Dispatching Client Overview section.

16.2.5 Fields & Buttons:

Name & Type	Description & Use
Add Type: Button	Description: Add a production schedule. Use: Click **Add**. The Production Schedule Definition window will display.

Name & Type	Description & Use
Edit Type: Button	Description: Displays the Production Schedule Definition window pre-populated with the selected production schedule. Use: 1. Select a production schedule from the list. 2. Click **Edit**.
Delete Type: Button	Description: Delete a production schedule. **NOTE:** Once it is deleted there is no getting it back (unless you go to a database backup). Use: 1. Select a production schedule from the list. 2. Click **Delete**. Confirmation is required via a pop-up.  3. Click **Yes** to delete the entity or click **No** to cancel.
Properties Type: Button	Description: Displays the Production Schedule Definition window pre-populated with the selected production schedule in read only mode. Use: 1. Select a production schedule from the list. 2. Click **Properties**.
Refresh Type: Button	Description: Refreshes the list of production schedules. Use: Click **Refresh**.

Name & Type	Description & Use
Filter Type: Drop-Down	Description: Defines the schedules which will be displayed. If a state is checked with the Filter drop-down the schedules with that state will be displayed. Use: Click **Filter**. Click a state to toggle it to either to be checked or not checked.
Selection List Type: List	 Description: The list of all the production schedules that are configured. The columns which display are defined in the Modeling Client. Use: Select an order and click **Edit**, **Copy**, **Delete,** or **Properties**.

16.2.6 Do Next:

After production schedules are created, click on a production schedule in the tree view to display the Production Order Management screen. Use the Production Order Management screen to view/manage the production orders associated with the production schedule.

16.2.7 See Also:

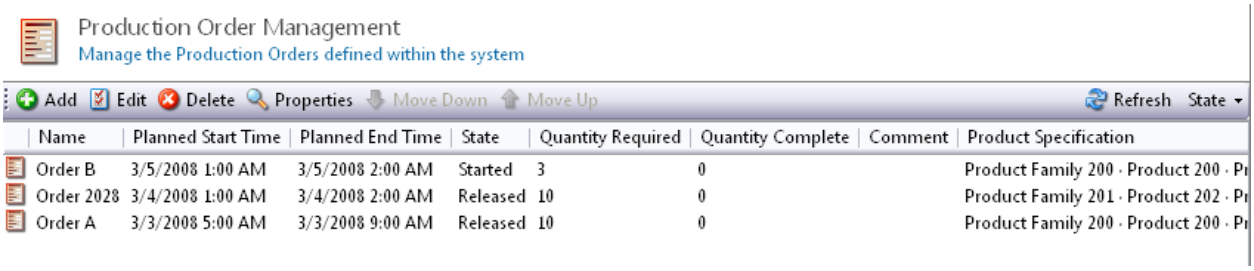
Parent Window:
Dispatching Client

Other Tabs:
None

Child Windows:
Production Schedule Definition (General)

Other:
Dispatching Client Overview

16.3 Production Order Management



16.3.1 How to Access:

From Dispatching Client:

- Select a specific production schedule in the server explorer tree. The Production Order Management screen displays on the right.

16.3.2 Overview:

The Production Order Management screen allows for creating, viewing, modifying, and deleting production orders.

16.3.3 Details:

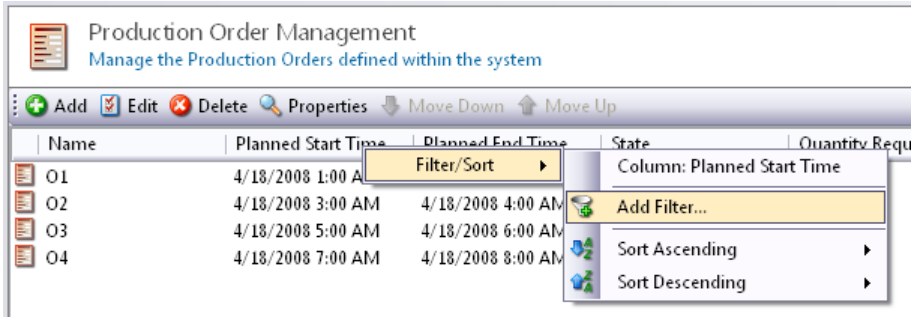
The right side of the Dispatching Client window, when Production Order Management is displayed, is divided into three panes, horizontally.

- The top pane allows management of production orders (this is the section described in this section.
- The middle pane shows a grid of the work centers configured for the selected order. (See also, Work Center Grid)
- The bottom pane shows the details of a work order when selected in the grid.

The orders can be filtered based upon state with the State drop-down list.

Production orders can be re-ordered by using the **Move Up** and **Move Down** buttons.

Production orders can be sorted using the **Filter/Sort** right click option when a production order is selected or when a header field is selected.

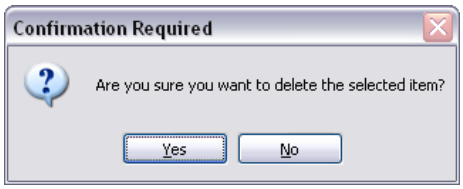


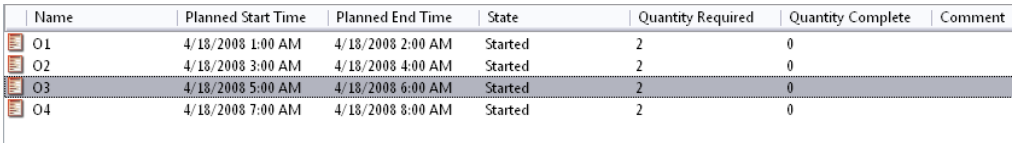
16.3.4 Prerequisites:

Create schedules.

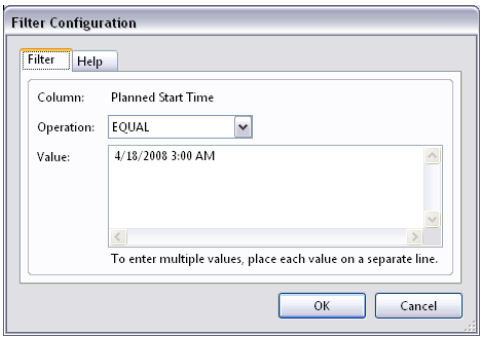
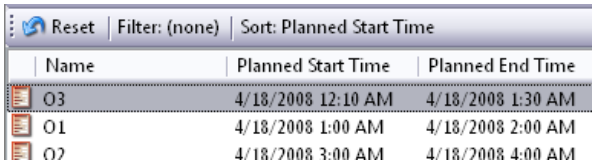
16.3.5 Fields & Buttons:

Name & Type	Description & Use
Add Type: Button	Description: Add a production order. Use: Click **Add**. The Production Order Definition window will display.
Edit Type: Button	Description: Displays the Production Order Definition window pre-populated with the selected production order. Use: 1. Select a production order from the list. 2. Click **Edit**.

Name & Type	Description & Use
Delete Type: Button	Description: Delete a production order. **NOTE:** Once it is deleted there is no getting it back (unless you go to a database backup). Use: 1. Select a production order from the list. 2. Click **Delete**. The **Confirmation Required** pop-up will display.  3. Click **Yes** to delete the entity or click **No** to cancel to deletion.
Properties Type: Button	Description: Displays the Production Order Definition window pre-populated with the selected production order in read only mode. Use: 1. Select a production order from the list. 2. Click **Properties**.
 Move Up Type: Button	Description: Moves the selected order up one position. Use: 1. Select an order in the list. 2. Select the Move up button. The order changes position with the order above it. The top order cannot be moved up.

Name & Type	Description & Use
 Move Down Type: Button	Description: Moves the selected order down one position. Use: 1. Select an order in the list. 2. Select the **Move Down** button. The order changes position with the order below it. The bottom order cannot be moved down.
Refresh Type: Button	Description: Refreshes the list of production orders. Use: Click **Refresh**.
State Type: Drop-Down	Description: Defines the orders which will be displayed. If a state is checked with the State drop-down the schedules with that state will be displayed. Use: 1. Click **State**. 2. Click a state to toggle it to either to be checked or not checked.
Selection List Type: List	 Name | Planned Start Time | Planned End Time | State | Quantity Required | Quantity Complete | Comment O1 | 4/18/2008 1:00 AM | 4/18/2008 2:00 AM | Started | 2 | 0 O2 | 4/18/2008 3:00 AM | 4/18/2008 4:00 AM | Started | 2 | 0 O3 | 4/18/2008 5:00 AM | 4/18/2008 6:00 AM | Started | 2 | 0 O4 | 4/18/2008 7:00 AM | 4/18/2008 8:00 AM | Started | 2 | 0 Description: The list of all the production orders that are configured. The columns which display are defined in the Modeling Client. Use: • Select an order and click **Edit**, **Copy**, **Delete,** or **Properties**. • Or right click and select one of the following options: • **Edit** – displays the Production Order Definition window • **Delete** – removes the production order

Name & Type	Description & Use

	• Filter/Sort
	  ○ **Column: xxx** - where xxx is the column. The orders will be sorted by the selected column.
	  ○ **Add Filter…** - displays the Filter Configuration window. Refer to the Help tab in this window for details on using the filter features.
	 
	
	  ○ **Sort Ascending**
	  ▪ **This column** – sort all orders sorting on this column only
	  ▪ **Include this column** – sort all orders retaining the sort order of other columns and then sorting on this column
	  ▪ **NOTE**: When sorting is defined an addition menu bar will display. Use the **Reset** button on this menu to clear or defined sorting. The default sorting will be displayed.
	 
	
	  ○ **Sort Descending**
	  ▪ **This column** – sort all orders sorting on this column only
	  ▪ **Include this column** – sort all orders retaining the sort order of other columns and then sorting on this column
	• **Change State** – changes the state of the production order to and from Scheduled and Released
	• **Change Hold** – puts the production order on hold. To remove the hold, go to the Production Order Definition window.
	• **Dispatch** – displays the Production Order Dispatch window

Name & Type	Description & Use
	• **WIP Status** – displays the WIP Status window • **Properties** – displays the Production Order Definition window in read only mode

16.3.6 Do Next:

Define and modify the production order definitions on the Production Order Definition window.

16.3.7 See Also:

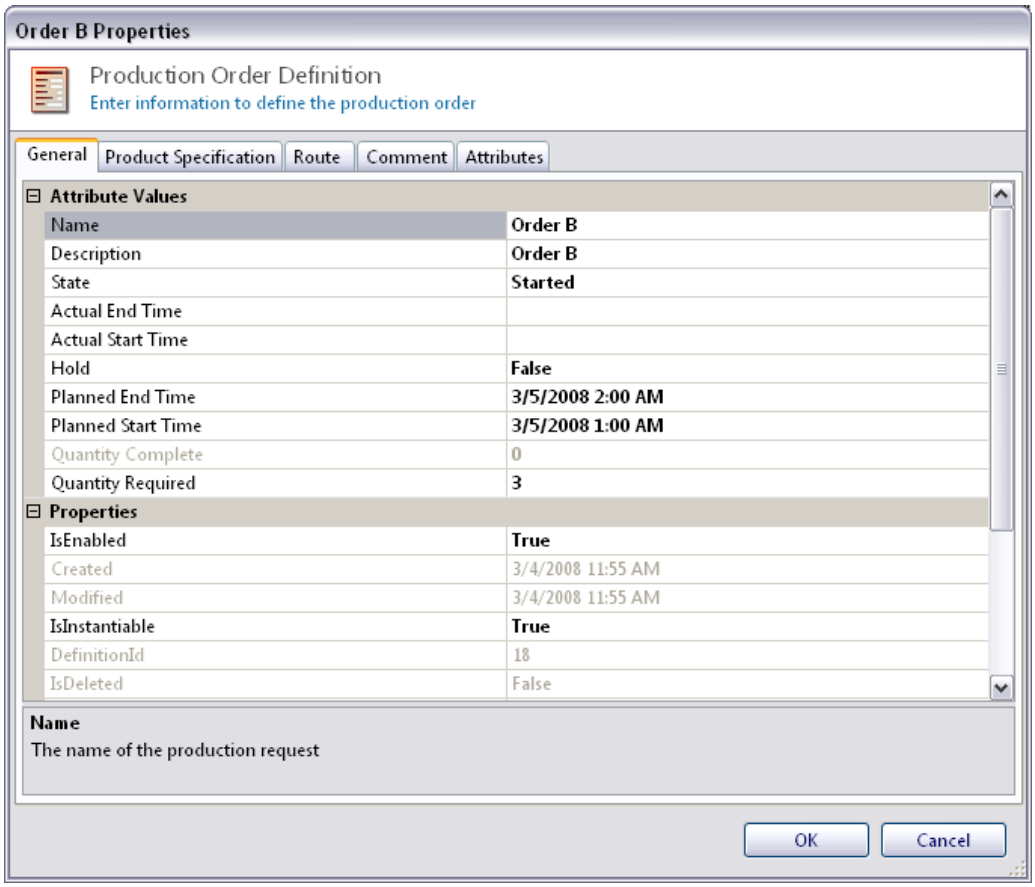
Parent Window:
Dispatching Client

Other Tabs:
None

Child Windows:
Production Order Definition (Attributes)
Production Order Definition (Comment)
Production Order Definition (General)
Production Order Definition (Product Specification)
Production Order Definition (Route)
Production Schedule Definition (General)
Dispatch
Work Center Grid

Other:
Dispatching Client Overview

16.4 Production Order Definition (General)



16.4.1 How to Access:

From Dispatching Client:

1. Select a specific production schedule in the explorer view. The Production Order Management screen displays on the right.
2. From the Production Order Management screen, use one of the following methods:
   - Click **Add**.
   - Select an existing production order and click **Edit** or **Properties** (button or right clicking).
     - **Edit** will allow modifications to be made.
     - **Properties** will display the window as read only.
   - Double click on an existing production order and it will display as read only.
3. From the Production Order Definition window, select the **General** tab (it is displayed upon startup).

16.4.2 Overview:

The Production Order Definition window defines the details of a production order.

16.4.3 Details:

The General tab configures the basic information regarding the order.

16.4.4 Prerequisites:

None

16.4.5 Fields & Buttons:

Name & Type	Description & Use
Name Type: Text Field	Description: The name of the production order. Use: Click in the field and type.
Description Type: Text Field	Description: The description of the production order. Use: Click in the field and type.
Planned Start Time Type: Date and Time Field	Description: The date and time when processing of the production order is planned to start. Use: • Type a properly formatted date (and optionally the time) • Click the drop-down, and use the calendar. Time, if desired, must be typed manually.

Name & Type	Description & Use
Planned End Time Type: Date and Time Field	Description: The date and time when processing of the production order is planned to end. Use: • Type a properly formatted date (and optionally the time) • Click the drop-down, and use the calendar. Time, if desired, must be typed manually.
State Type: Drop-Down Field	Description: Sets the state of the production order. The available states include: • Scheduled – not visible to the Shop Floor Client • Released – available to Shop Floor Client for processing • Queued – ready to process; may be partially processed • Started – currently being processed • Suspended – processing prohibited • Complete – the required quantity has been satisfied • Closed – no further action allowed Use: • Type the first letter of a state. • Or, click the drop-down list and select the desired state.
Actual Start Time Type: Date and Time Field	Description: The date and time when processing of the production order actually started. Use: • Type a properly formatted date (and optionally the time) • Click the drop-down, and use the calendar. Time, if desired, must be typed manually.

Name & Type	Description & Use
Actual End Time Type: Date and Time Field	Description: The date and time when processing of the production order actually ended. Use: • Type a properly formatted date (and optionally the time) • Click the drop-down, and use the calendar. Time, if desired, must be typed manually.
Quantity Required Type: Number Field	Description: The quantity that is required to be created in order to satisfy the production order. Use: Click in the field and type.
Quantity Complete Type: Number Field	Description: The quantity that is complete. Use: Click in the field and type.
Hold Type: True / False	Description: If True, the production order is on hold. If False, the production order is not on hold. Use: • Type “T” or “F”. • Or, click the drop-down list and select either True or False.
Is Enabled Type: True/False	Description: Determines if the production order is enabled. Use: • Type “T” or “F”. • Click the drop-down list and select either True or False.

Name & Type	Description & Use
Is Instantiable Type: True/False	Description: Determines if the production order is instantiable, that is, if there can be instances of the production order. Use: • Type "T" or "F". • Click the drop-down list and select either True or False.
OK Type: Button	Description: Closes the window and saves any configuration changes. Use: Click **OK**. The window will close and changes will be saved.
Cancel Type: Button	Description: Closes the window and does not save any configuration changes. Use: Click **Cancel**. The window will close and changes will not be saved.

16.4.6 Do Next:

Define the configuration on each tab.

Then dispatch the order.

16.4.7 See Also:

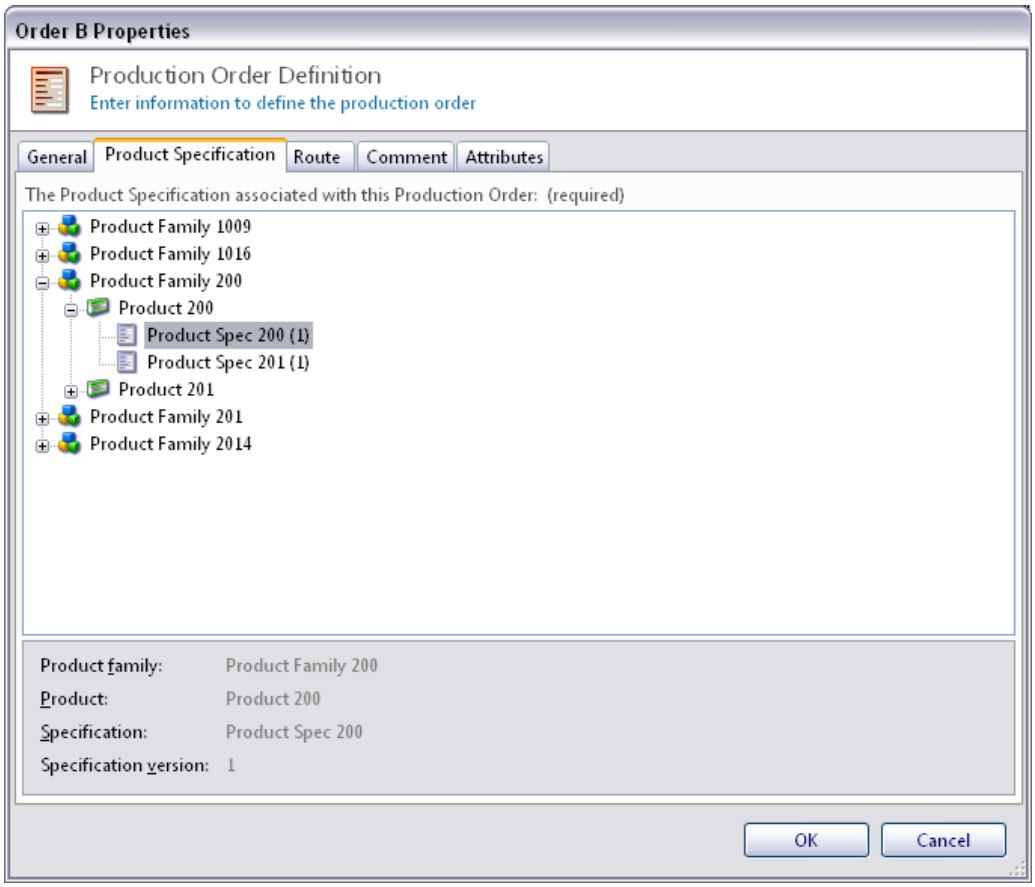
Parent Window:
Production Order Management

Other Tabs:
Production Order Definition (Attributes)
Production Order Definition (Comment)
Production Order Definition (Product Specification)
Production Order Definition (Route)

Child Windows:
None

Other:
Dispatching Client Overview

16.5 Production Order Definition (Product Specification)



16.5.1 How to Access:

From Dispatching Client:

1. Select a specific production schedule in the explorer view. The Production Order Management screen displays on the right.
2. From the Production Order Management screen, use one of the following methods:
   - Click **Add**.
   - Select an existing production order and click **Edit** or **Properties** (button or right clicking).
     - **Edit** will allow modifications to be made.
     - **Properties** will display the window as read only.
   - Double click on an existing production order and it will display as read only.
3. From the Production Order Definition window, select the **Product Specification** tab.

16.5.2 Overview:

The Production Order Definition window defines the details of a production order.

16.5.3 Details:

The Product Specification tab defines what is being produced for the production order.

Multiple versions may be available for each product specification, so use care when selecting a specification.

16.5.4 Prerequisites:

Define product specification in the Configuration Client on the Product Specification Management screen.

16.5.5 Fields & Buttons:

Name & Type	Description & Use
Production Specification Type: List	Description: Contains the tree view of all available product specifications. The selection from this list determines "what" will be created by the production order. The selected product specification is listed at the bottom of the window. Use: Expand the tree and select the desired product specification.
OK Type: Button	Description: Closes the window and saves any configuration changes. Use: Click **OK**. The window will close and changes will be saved.
Cancel Type: Button	Description: Closes the window and does not save any configuration changes. Use: Click **Cancel**. The window will close and changes will not be saved.

16.5.6 Do Next:

Define the configuration on each tab.

Then dispatch the order.

16.5.7 See Also:

Parent Window:
Production Order Management

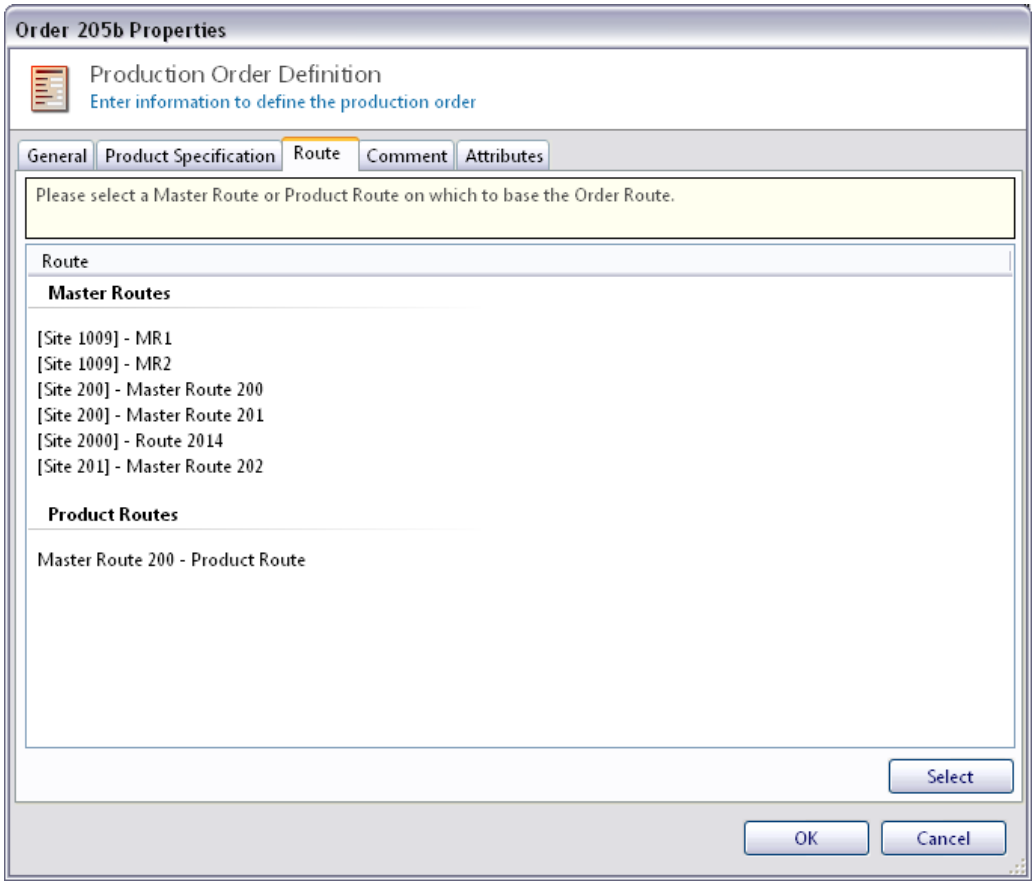
Other Tabs:
Production Order Definition (Attributes)
Production Order Definition (Comment)
Production Order Definition (General)
Production Order Definition (Route)

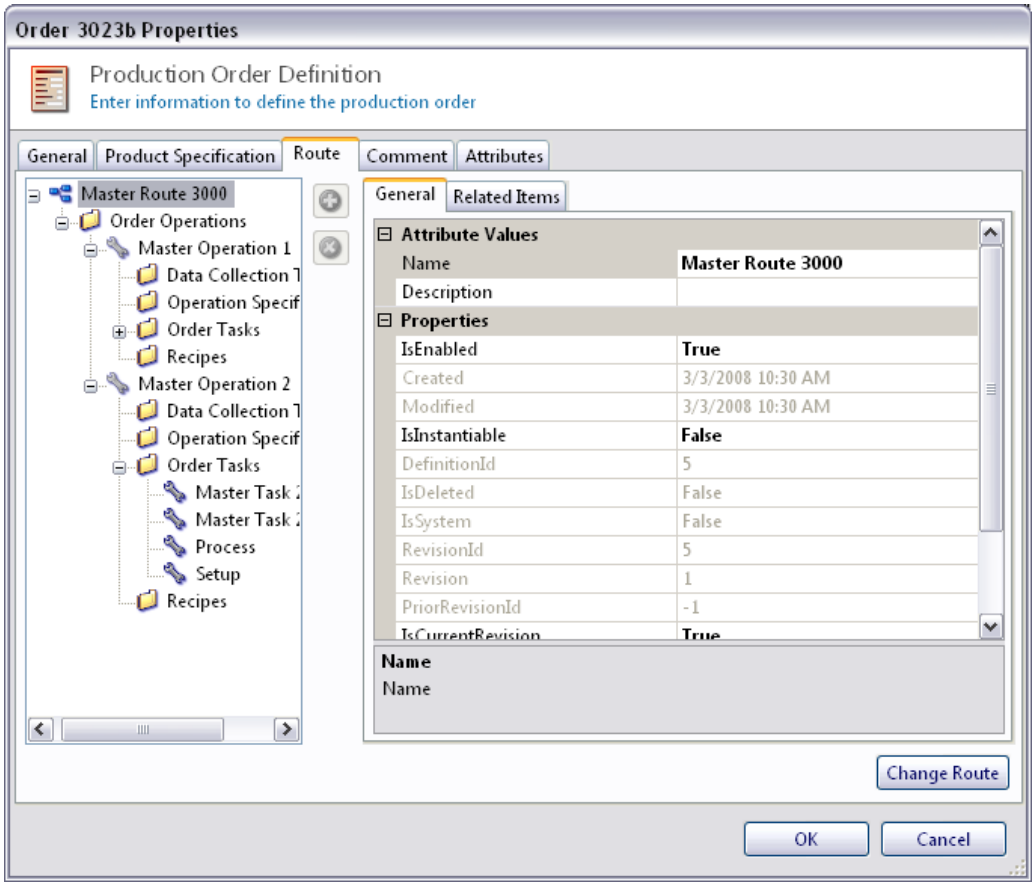
Child Windows:
None

Other:
Dispatching Client Overview

16.6 Production Order Definition (Route)

There will be just one master route to select on this screen. Select the master route and click **Select**.





> **NOTE:** The details of how to use the Route tab when the selected route is displayed is not explained in this section but in the Master Route Definition section.
>
> See the subsection "Operation & Routes Overview" in the "Configuration Client Overview" section.

16.6.1 How to Access:

From Dispatching Client:

1. Select a specific production schedule in the explorer view. The Production Order Management screen displays on the right.
2. From the Production Order Management screen, use one of the following methods:
   - Click **Add**.
   - Select an existing production order and click **Edit** or **Properties** (button or right clicking).
     - **Edit** will allow modifications to be made.
     - **Properties** will display the window as read only.

- Double click on an existing production order and it will display as read only.

3. From the Production Order Definition window, select the **Route** tab.

16.6.2 Overview:

The Production Order Definition window defines the details of a production order.

16.6.3 Details:

The Route tab defines where and how the production order is going to be executed.

When first creating the production order the available master routes and product routes will be listed. Select a route and click the **Select** button. The selected route will display in the tab. At this point the route can be modified. If any fields are modified, changes made at the master and product levels will not override the changes at this level (the order level). But if changes are not made to a field, changes made at the master and product level will show up here, until the state of this production order is no longer set to Scheduled.

See the subsection "Operation & Routes Overview" in the "Configuration Client Overview" section.

16.6.4 Prerequisites:

Define master routes and/or product routes in the Configuration Client.

16.6.5 Fields & Buttons:

Name & Type	Description & Use
Route Type: List	 Description: List of all available routes for this production order. Use: Select the route and click **Select**.

Name & Type	Description & Use
Select Type: Button	Description: Saves the selected route. **NOTE:** The details of how to use the Route tab when the selected route is displayed is not explained in this section but in the Master Route Definition section. Use: Select the route and click **Select**.
Change Route Type: Button	Description: Displays the list of all available routes for this production order. Use: Click **Change Route**.
OK Type: Button	Description: Closes the window and saves any configuration changes. Use: Click **OK**. The window will close and changes will be saved.
Cancel Type: Button	Description: Closes the window and does not save any configuration changes. Use: Click **Cancel**. The window will close and changes will not be saved.

16.6.6 Do Next:

Define the configuration on each tab.

Then dispatch the order.

16.6.7 See Also:

Parent Window:
Production Order Management

Other Tabs:

Production Order Definition (Attributes)
Production Order Definition (Comment)
Production Order Definition (General)
Production Order Definition (Product Specification)

Child Windows:
Master Route Definition (General)
Master Route Definition (Related Items)

Other:
Dispatching Client Overview
Master Route Definition

16.7 Creating a New Work Order for the New Production Order

The work center grid displays the **work orders** in the work centers to which they have been dispatched. Dispatching is done in two different ways. Either, by dragging and dropping a production order to the work center grid; or, by right clicking on a production order and selecting Dispatch. The work orders are displayed in the time frame they are planned.

This system has two work cells so the work order will need to be dispatched to each work center.

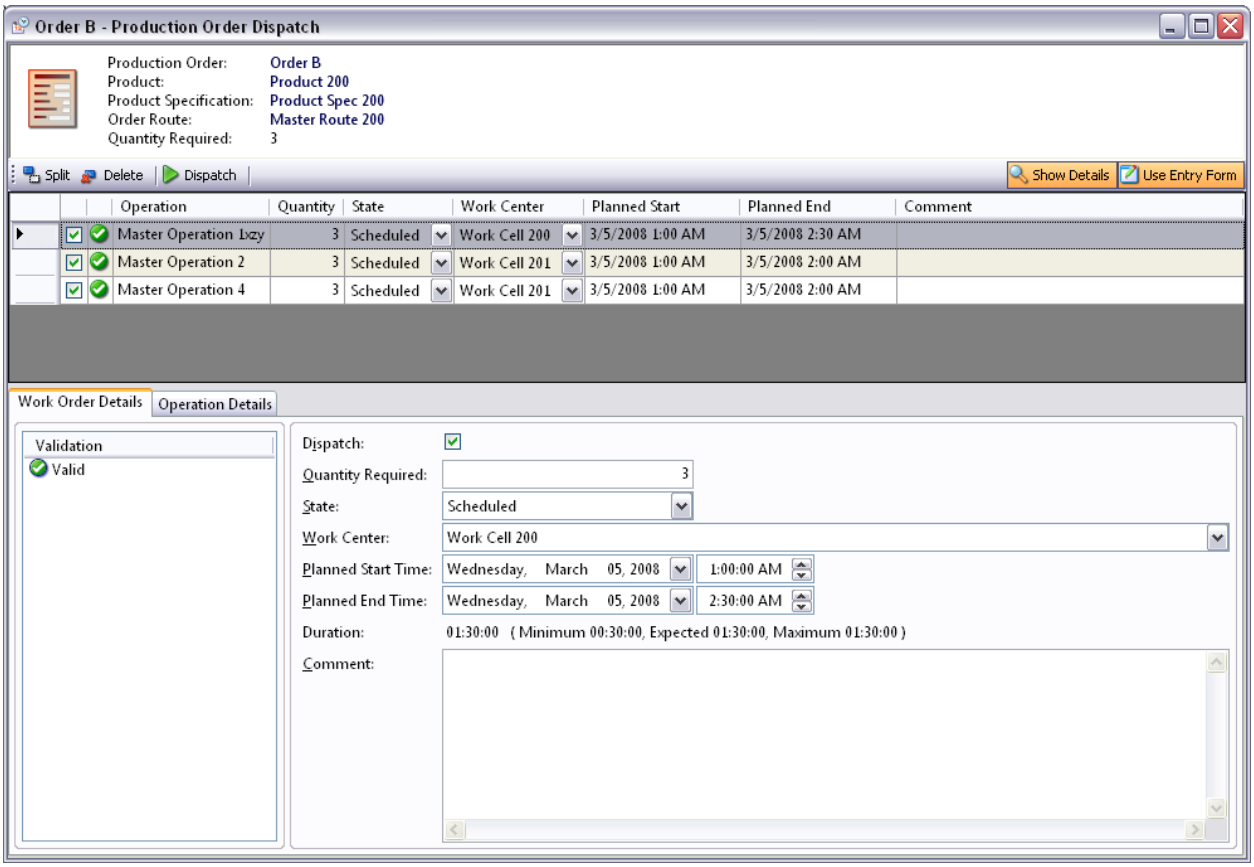
The **Work Center Dispatch** form displays when the production order is dropped to the work center grid or when a production order is right clicked and Dispatch is selected. On this form, all operations required by the production order are displayed. Only those operations supported by the work center will be enabled. Of the supported operations, one or more may be configured for dispatch.

Once dispatched, the attributes of a work order can be modified by right clicking on the work order in the work center grid and selecting Edit. The Work Order window will display.

The following sections describe in more details each of the screen mentioned above:

- Dispatch
- Dispatching Client
- Work Center Grid
- Work Order (Attribute)
- Work Order (General)
- Work Order (Tasks)

16.8 Dispatch



16.8.1 How to Access:

This window can be accessed in two ways and each way displays basically the same window:

- **Production Order Dispatch**: On the Production Order Management screen, right click a production order and select **Dispatch**.
- **Work Center Dispatch**: On the Production Order Management screen, drag a production order to the grid and drop it in the appropriate work cell at the time the order is to start.

On the Work Center Dispatch only the operations for the work center into which the order was dropped will be active. And the Work Center column will not appear.

16.8.2 Overview:

The Dispatch window is where production orders are dispatched to work centers as work orders.

16.8.3 Details:

See the Dispatching Client Overview.

16.8.4 Prerequisites:

See the Dispatching Client Overview.

16.8.5 Fields & Buttons:

Name & Type	Description & Use
Production Order Type: Link	Description: The production order from which this work order was created. Use: Click the production order and the Production Order Definition will display in read only mode.
Product Type: Link	Description: The product from which this work order was created. Use: Click the product and the Product Definition will display in read only mode.
Product Specification Type: Link	Description: The product specification from which this work order was created. Use: Click the product specification and the Product Specification Definition will display in read only mode.
Order Route Type: Link	Description: The route from which this work order was created. Use: Click the route and the Route Definition will display in read only mode.
Quantity Required Type: Text Only	Description: The required quantity which was defined in the production order. Use: This is informational only.

Name & Type	Description & Use
Work Center Type: Text Only	Description: The work center in which the work order was dropped into on the grid. Work Center: Work Cell - Area 3000\Line 3000b\Work Cell 3000b1 **This option is only listed on the Work Center Dispatch window, not on the Production Order Dispatch window.** Use: This is informational only.
  Split Type: Button	Description: Splits the current row. Use this to create additional operations that can be dispatched separately, to different work centers, at different times. The selected operation's quantity will be split in half. Then the quantities can be adjusted by hand as necessary. Shortcut Key: Ctrl +S Use: Select the operation to be split, and then click the **Split** button.
  Delete Type: Button	Description: Deletes the current row. Shortcut Key: Ctrl +L Use: Select the operation to be deleted, and then click the **Delete** button.
  Dispatch Type: Button	Description: Dispatches all the rows that have a checkmark in the dispatch check box (at the front of each row in the list of operations). Shortcut Key: Ctrl +D Use: 1. Click **Dispatch**. The Dispatch Confirmation dialog displays. 2. Review the information in the Dispatch Confirmation dialog and either click **OK** or **Cancel**.

Name & Type	Description & Use
Show Details **Show Details** Type: Button	Description: Controls visibility of the Work Order Details and Operation Details tabs at the bottom of the window. The button toggles the tabs visible or invisible. Shortcut Key: Ctrl +H Use: Click **Show Details**.
Use Entry Form **Use Entry Form** Type: Button	Description: Determines where work order details are entered, either in-line or in the details section. The button toggles where the details are entered. By clicking the **Use Entry Form** button the fields in the grid will be active or the fields on the Work Order Details tab will be active. The **Show Details** button must be "on" for the **Use Entry Form** button to be active. Shortcut Key: Ctrl +E Use: Click use Entry Form.
Dispatch check box Type: Check Box	Description: Indicates if the operation will be dispatched. Work orders with a check mark will be dispatched. Work orders without a checkmark will not be dispatched. The checkmark can be added either in the list or on the Work Order Details tab. See "Use Entry Form" button. Use: Click on in the check box.
Operation Type: Text Only	Description: The operation that will be used to create the work order. Use: This is informational only.

Name & Type	Description & Use
Quantity Type: Text Field	Description: The quantity that will be processed by this operation. As a default, the quantity is set to the entire required quantity for each operation. Use: Enter a value.
State Type: Drop-Down	Description: Defines the state of the work order after dispatched. Use: Click on the down arrow and select a state.
Work Center Type: Drop-Down	Description: The work center where this operation will be preformed. **The Work Center field does not display on the Work Center Dispatch window, only on the Production Order Dispatch window.** Use: Click on the down arrow and select a work center.
Planned Start Type: Date Field	Description: The planned start date and time for the operation. Use: In the Date field: 1. Click in the field and the drop-down date picker will display. 2. On the calendar select the day. 3. At the bottom of the calendar type the time value, or use the up/down arrows to modify the time value.

Name & Type	Description & Use
Planned End Type: Date Field	Description: Set the planned end date and time for the operation. Use: In the Date field: 1. Click in the field and the drop-down date picker will display. 2. On the calendar select the day. 3. At the bottom of the calendar type the time value, or use the up/down arrows to modify the time value.
Comment Type: Text	Description: Comments can be entered that will be displayed on the Work Order, Attributes tab. Use: Type in the comment field.
Work Order Details tab Type: Area	Description: Displays information about the selected operation. The **Validation** column displays current errors and warnings or displays "Valid" if there are no errors or warnings. If the **Use Entry Form** option is selected the fields to the right will displayed. The fields can be modified and the changes will display in the list above. This fields work the same as in the list. See the above help for details on each specific field. If the **Use Entry Form** option is off the fields will not appear, only the Validation column is displayed. Use: Click on the **Work Order Details** tab. (The Show Details button must be on.)
Operation Details tab Type: Area	Description: Displays information about the selected operation. Use: Click on the **Operation Details** tab.

16.8.6 Do Next:

Use the Dispatching Client grid or Shop Floor Client to view and manage the work orders.

16.8.7 See Also:

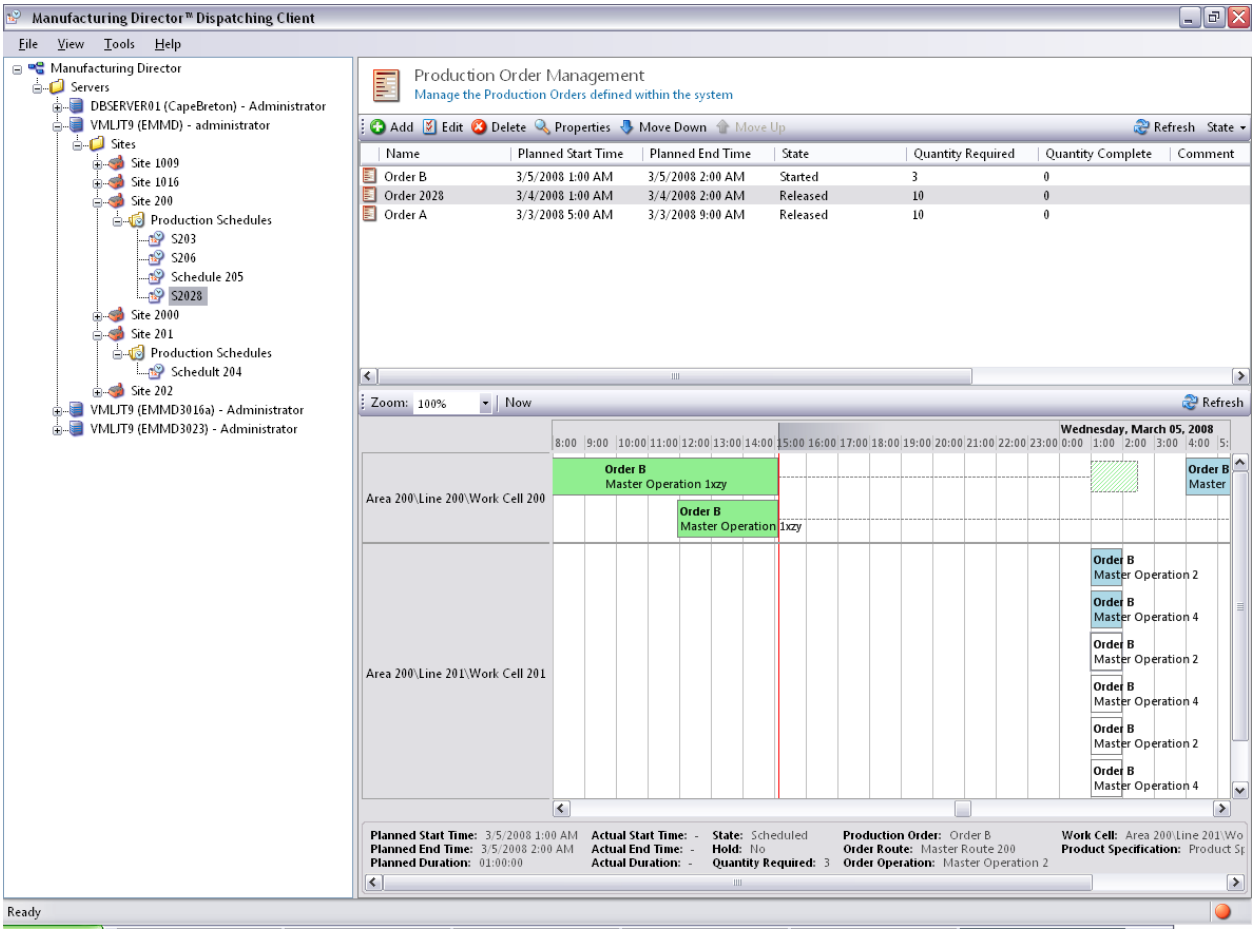
Parent Window:
Production Order Management

Other Tabs:
None

Child Windows:
Dispatching Client
Production Order Definition (Attributes)
Production Order Definition (Comment)
Production Order Definition (General)
Production Order Definition (Product Specification)
Production Order Definition (Route)
Product Definition
Product Specification Definition (General)
Product Specification Definition (Parameter Details)
Order Route Definition

Other:
Dispatching Client Overview

16.9 Dispatching Client



16.9.1 How to Access:

To start the Dispatching Client:

- From the Start Menu, select:

  All Programs → Flexware Innovation → Manufacturing Director → Dispatching Client

- Or, click on the file **Flexware.ManufacturingDirector.Dispatching.Client.exe**. In a default installation, it will be in the following directory:

  C:\Program Files\Flexware Innovation\Manufacturing Director

16.9.2 Overview:

The Dispatching Client is where production schedules are created and managed. Within production schedules, production orders (i.e., manufacturing orders) are created and dispatched to work centers as individual work orders.

16.9.3 Details:

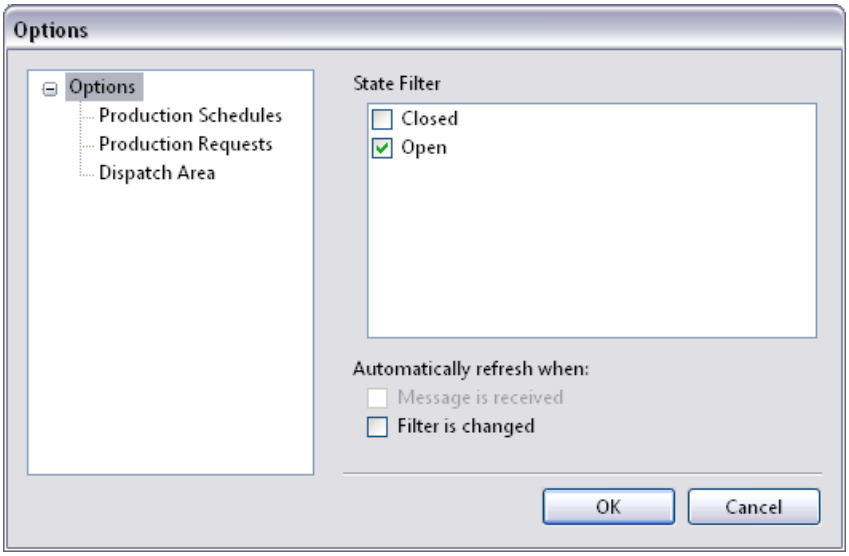
See the Dispatching Client Overview section.

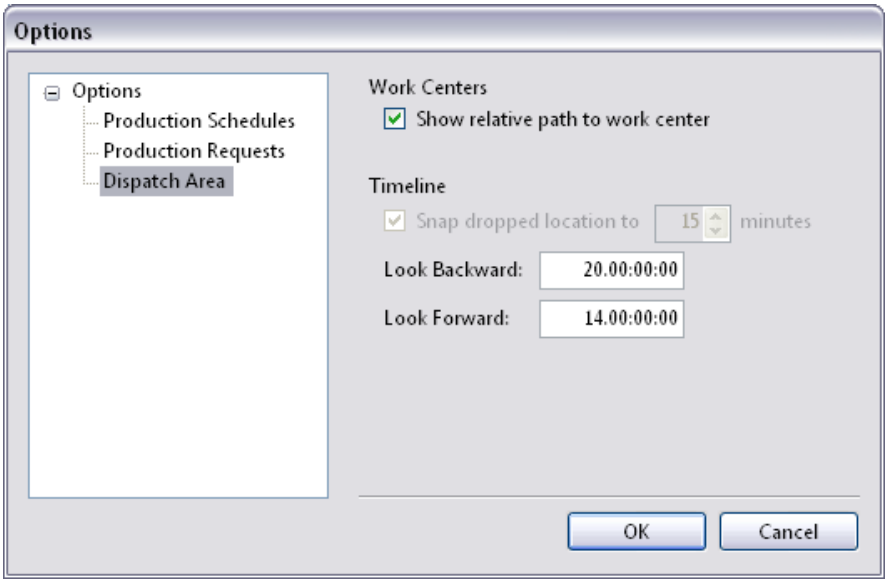
16.9.4 Prerequisites:

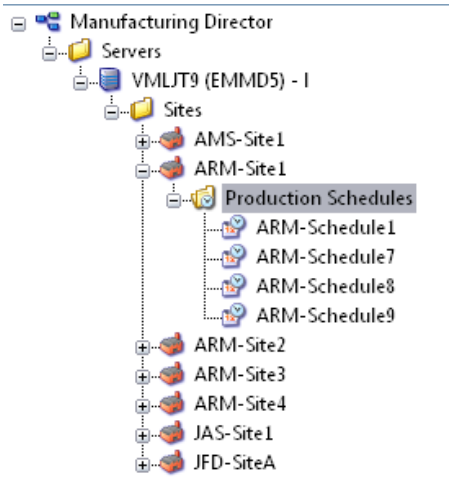
See the Dispatching Client Overview section.

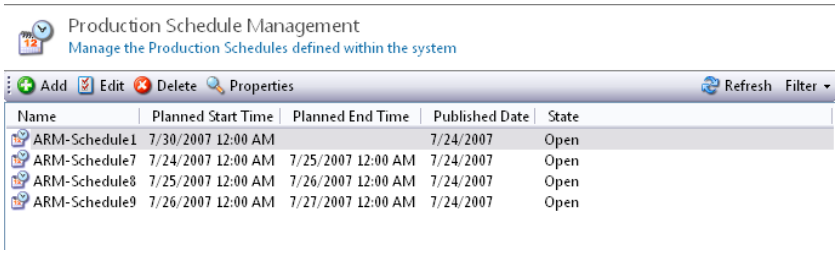
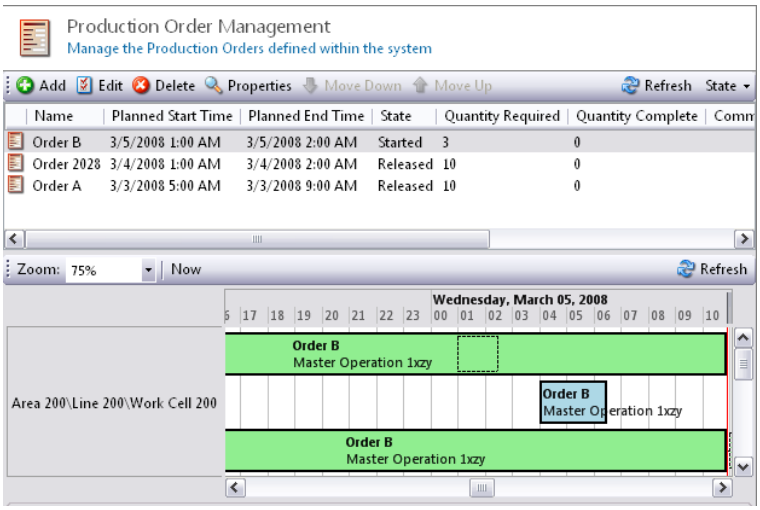
16.9.5 Fields & Buttons:

Name & Type	Description & Use
File → Exit Type: Menu Item	Description: Closes the window. Use: Click **File** and select **Exit**.
View → Server Explorer Type: Menu Item	Description: Displays or hides the Server Explorer section of the window (the tree on the left). Use: • Click **View** and select **Server Explorer**. • Or, press **F7** function key on the keyboard.
View → Activity Log Type: Menu Item	Description: Displays or hides the activity log at the bottom of the window. Use: • Click **View** and select **Activity Log**. • Or, press **F8** function key on the keyboard.

Name & Type	Description & Use
Tools → Options (Production Schedules & Production Orders) Type: Menu Item	Description: Displays the options window.  Use: Click **Tools** and select **Options**. Click on **Production Schedules** or **Production Orders** and the **State Filters** will display to the right for the production schedules or orders. • Click the box before each state to add or remove a check mark from the box. When a check is displayed in the box, each schedule or order in that state will be displayed on the Production Schedule Management or Production Order Management screen. • Under "**Automatically refresh when"** click the box before **Filter is changed** to add or remove a check mark from the box. When a check is displayed in the box, the list of production schedules or orders will be updated each time the filter is modified. If the box is not checked the **Refresh** button must be clicked before the list is updated. • Under "**Automatically refresh when"** click the box before **Message is received** to add or remove a check mark from the box. When a check is displayed in the box, the list of production schedules or orders will be updated each time a message is received.

Name & Type	Description & Use
Tools → Options (Dispatch Area) Type: Menu Item	Description: Displays the options window.  Use: Click **Tools** and select **Options**. Click on **Dispatch Area** and the following options will display to the right. • When "**Show relative path to work center**" is checked the specific paths will be displayed in the grid to the left. • When checked this will display: Area 200\Line 201\Work Cell 201 • When it is not checked this would display: Work Cell 201 • To customize the timeline, check "**Snap dropped location to ## minutes**". • Set "Look Backward" and "Look Forward" the number of day the grid is to scroll back and forward.
Help → About Manufacturing Director Dispatch Client Type: Menu Item	Description: Displays the **About Manufacturing Director** box which lists the software version number. Use: Click **Help** and select **About Manufacturing Director Dispatch Client**.

Name & Type	Description & Use
Server Explorer Tree Type: Tree	Description: Displayed on the left side of the Dispatching Client is the Server Explorer Tree.  **Note:** Use **View →Server Explorer** or the **F7** function key to display or hide the Server Explorer Tree. Use: • Click the + (plus signs) to expand the levels of the tree. Click the – (minus signs) to collapse the levels of the tree. • In the tree, under sites click on Production Schedules and the **Production Schedule Management screen** displays to the right. Click on a specific production schedule and the **Production Order Management screen** displays to the right. • Right click on any level and different options will be displayed. • **Refresh** – all levels under the selected level will be refreshed. The levels will collapse. • **Modify Server** – See the section **Add a New Server**. • **Remove Server** – the selected server will be removed from the tree. This does not affect the state of the database.

Name & Type	Description & Use
Production Schedule Management Type: Area	Description: Manage the production schedules defined within the system.  Use: To display, click on a **Production Schedules** folder in the Server Explorer Tree. See also, **Production Schedule Management**.
Production Order Management Type: Area	Description: Manage the production orders defined within the system.  Use: To display, click on a specific production schedule in the Server Explorer Tree. See also, **Production Order Management** and **Work Center Grid**.

16.9.6 Do Next:

See the Dispatching Client Overview section.

16.9.7 See Also:

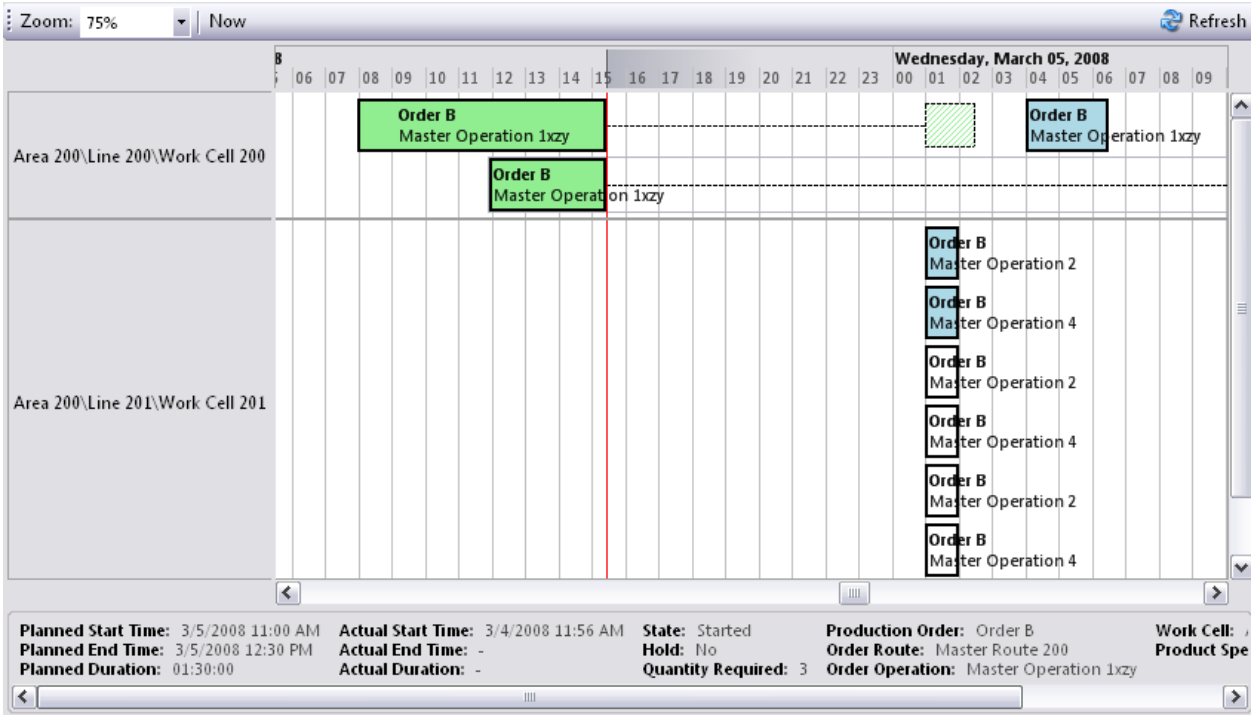
Parent Window:
None

Other Tabs:
None

Child Windows:
Production Order Management
Production Schedule Management
Work Center Grid

Other:
Dispatching Client Overview
Add a New Server

16.10 Work Center Grid



16.10.1 How to Access:

The work Center grid is displayed when the Production Order Management screen is displayed.

As production orders are dispatched they are graphically displayed on the grid.

16.10.2 Overview:

The work center grid displays work orders on a timeline sorted by location.

16.10.3 Details:

Production orders are dispatched and displayed on the grid either with the Production Order Dispatch window or the Work Center Dispatch window.

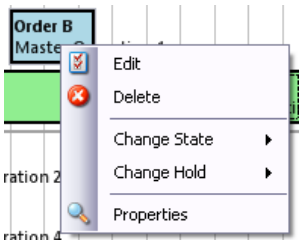
- **Production Order Dispatch**: On the Production Order Management screen, right click a production order and select **Dispatch**. On the Production Order Dispatch window work orders are dispatched and displayed on the gird.
- **Work Center Dispatch**: On the Production Order Management screen, drag a production order to the grid and drop it in the appropriate work cell at the time the order is to start. On the Work Center Dispatch window work orders are dispatched and displayed on the grid.

Details Displayed:

When work orders are selected on the grid the details about that work order are listed below the grid.

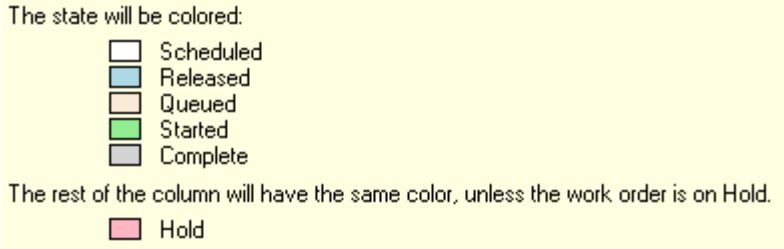
Right Click Pop-up:

Right click on a work order in the grid and the following pop-up displays. See the Fields & Button section for details on these options. Note that depending on the current state of the work order different options may be available in the pop-up list.



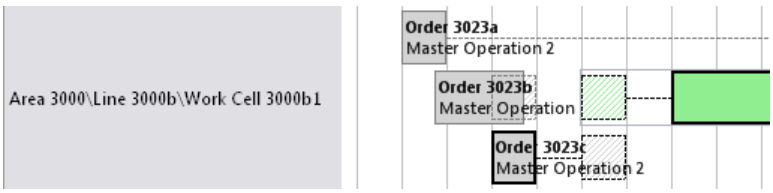
Color Status:

The work orders are colors according to their current state.



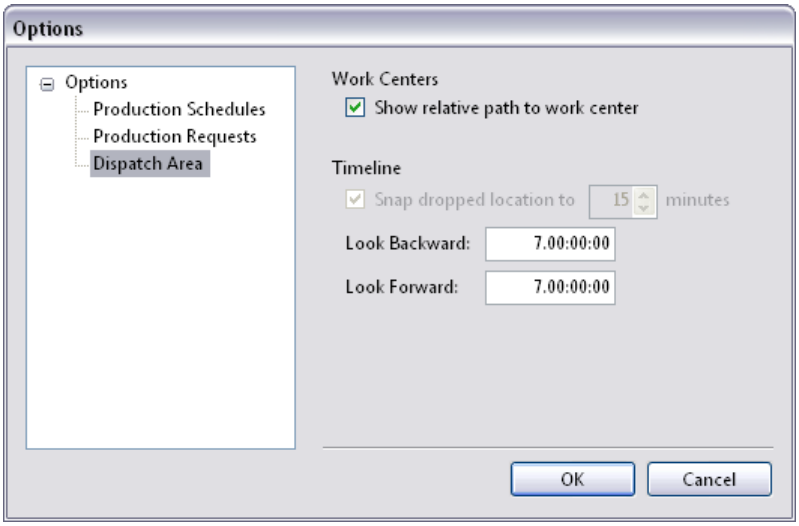
Grid Formats:

Besides color, work orders display the planned start and end times verses the actual start and end times. The solid boxes represent the actual start and end times while the dashed lined boxes represent the planned start and end times. The dashed lines between the two boxes connect boxes that represent the same work order.



Configuration Options:

See the **Tools→Option**s, **Options→Dispatch Area** configuration options.



Work Centers:

When "**Show relative path to work center**" is checked the specific paths will be displayed in the grid to the left.

When checked this will display: Area 200\Line 201\Work Cell 201

When it is not checked this would display: Work Cell 201

Timeline:

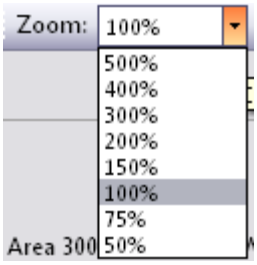
To customize the timeline, check "**Snap dropped location to ## minutes**".

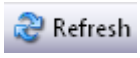
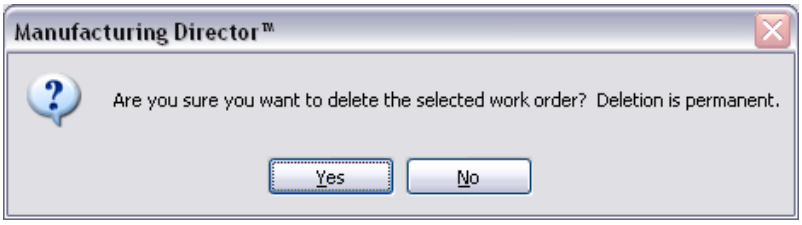
The date/time range of the grid can be defined. Set **Look Backward** and **Look Forward** the number of day the grid is to scroll back and forward.

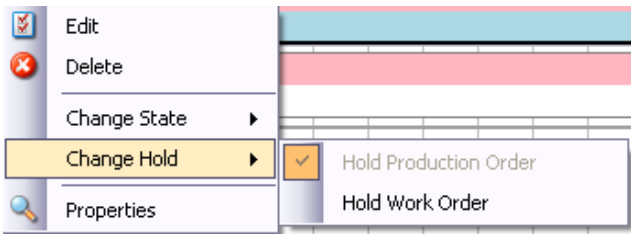
16.10.4 Prerequisites:

See the Dispatching Client Overview.

16.10.5 Fields & Buttons:

Name & Type	Description & Use
Zoom: 100% **Zoom** Type: Drop-Down	 Description: Rescales the grid. When 100% is selected the range of one day will be displayed in the part of the grid that is currently displayed. Use the scroll bar to display additional time forward and backward. The higher the percentage selected the more the graph will zoom in and display less time span at in a current view. Choose a lower percentage to zoom out and see a larger range of time. Use the Configuration Options to define the range of the entire scrollable window. Use: Click the drop-down and select the percentage to view.
Now **Now** Type: Button	Description: Resets the grid to display the current time. "Now" is represented on the grid with a red vertical line. Use: Click **Now**.

Name & Type	Description & Use
 **Refresh** Type: Button	Description: Refresh the grid. Use: Click **Refresh**.
Edit Type: Drop-Down	Description: Displays the Work Order window for the selected work order. Use: 1. Right click on a work order in the grid. 2. Select **Edit**.
Delete Type: Drop-Down	Description: Delete a work order. **NOTE:** Once it is deleted there is no getting it back (unless you go to a database backup). Use: 1. Right click on a work order in the grid. 2. Select **Delete**. Confirmation is required via a pop-up.  3. Click **Yes** to delete the work order or click **No** to cancel.
Change State Type: Drop-Down	Description: Changes the state of the selected work order. Use: 1. Right click on a work order in the grid. 2. Select **Change State**. 3. Select an available state.

Name & Type	Description & Use
Change Hold Type: Drop-Down	Description: Change the hold status of the selected work order. The Hold Work Order option toggles the on-hold status. In the grid a production order or a work order with an on-hold status will be pink. The **Change Hold→Hold Production Order** indicates if the production order is on-hold or is not on-hold. A check mark indicates the production order is on-hold.  Use: 1. Right click on a work order in the grid. 2. Select **Change Hold→Hold Work Order**.
Properties Type: Drop-Down	Description: Displays the Work Order window pre-populated with the selected work order in read only mode. Use: 1. Right click on a work order in the grid. 2. Select **Properties**.

16.10.6 Do Next:

Use the Dispatching Client grid or Shop Floor Client to view and manage the work orders.

16.10.7 See Also:

Parent Window:
Dispatching Client

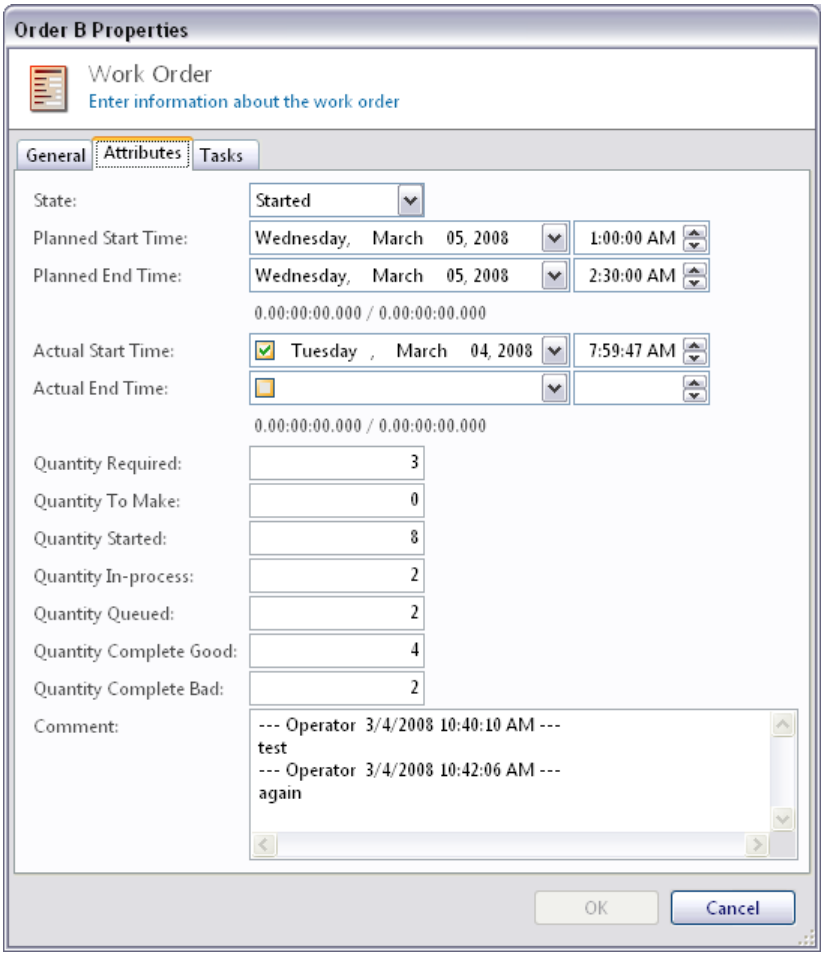
Other Tabs:
None

Child Windows:

Dispatch
Work Order (Attribute)
Work Order (General)
Work Order (Tasks)

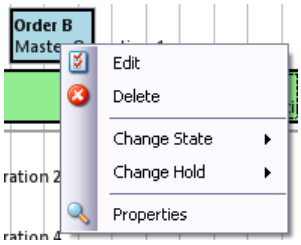
Other:
Dispatching Client Overview
Production Order Definition (Attributes)
Production Order Definition (Comment)
Production Order Definition (General)
Production Order Definition (Product Specification)
Production Order Definition (Route)
Production Order Management
Production Schedule Definition (General)
Production Schedule Management

16.11 Work Order (Attribute)



16.11.1 How to Access:

1. Right click on a work order in the Work Center grid and the following pop-up displays.



2. Select **Edit** and the Work Order window will display.
3. Click the **Attributes** tab.

Select **Properties** in the pop-up and a read only version of the Work Order window will display.

16.11.2 Overview:

The Work Order window displays information about the work order. Dates, states, comments, and quantity counts can be updated here.

16.11.3 Details:

See the Dispatching Client Overview.

16.11.4 Prerequisites:

See the Dispatching Client Overview.

16.11.5 Fields & Buttons:

Name & Type	Description & Use
State Type: Drop-Down	Description: Set the state of the work order. States may be changed automatically by the Shop Floor Client. For rules around the setting of states see the **Dispatching Client Overview** section. Use: Select a state from the drop-down list.

Name & Type	Description & Use
Planned Start Time Type: Date	Description: The date and time when processing of the work order is planned to start. If this date is changed the work order on the grid will move accordingly. Use: • Click the drop-down, and use the calendar to select a month, day, and year. • Or, click on each field and type to make a change. • For the month type the number that represents the month (i.e., 1 for January, 12 for December) • For the year just type the last digit or two of the year. It will assume 2000.
Planned End Time Type: Date	Description: The date and time when processing of the work order is planned to end. If this date is changed the work order on the grid will move accordingly. Use: • Click the drop-down, and use the calendar to select a month, day, and year. • Or, click on each field and type to make a change. • For the month type the number that represents the month (i.e., 1 for January, 12 for December) • For the year just type the last digit or two of the year. It will assume 2000.

Name & Type	Description & Use
Actual Start Time Type: Date	Description: The date and time when processing of the work order actually started. The Shop Floor Client may populate this field automatically. To set this field for the first time click the checkmark box and the current date and time will display in the field. If this date is changed the work order on the grid will move accordingly. Use: • Click the drop-down, and use the calendar to select a month, day, and year. • Or, click on each field and type to make a change. • For the month type the number that represents the month (i.e., 1 for January, 12 for December) • For the year just type the last digit or two of the year. It will assume 2000.
Actual End Time Type: Date	Description: The date and time when processing of the work order actually ended. The Shop Floor Client may populate this field automatically. To set this field for the first time click the checkmark box and the current date and time will display in the field. If this date is changed the work order on the grid will move accordingly. Use: • Click the drop-down, and use the calendar to select a month, day, and year. • Or, click on each field and type to make a change. • For the month type the number that represents the month (i.e., 1 for January, 12 for December) • For the year just type the last digit or two of the year. It will assume 2000.

Name & Type	Description & Use
Quantity Required Type: Numeric Filed	Description: Defines the quantity required for this work order. This is defined in the production order but can be changed on the Dispatch window. Use: Type in a value.
Quantity To Make Type: Numeric Filed	Description: Defines the quantity yet to make for this work order. Use: Type in a value.
Quantity Started Type: Numeric Filed	Description: Defines the quantity started for this work order. Use: Type in a value.
Quantity In-process Type: Numeric Filed	Description: Defines the quantity in-process for this work order. Use: Type in a value.
Quantity Queued Type: Numeric Filed	Description: Defines the quantity queued for this work order. Use: Type in a value.
Quantity Complete Good Type: Numeric Filed	Description: Defines the quantity that is complete and good for this work order. Use: Type in a value.

Name & Type	Description & Use
Quantity Complete Bad Type: Numeric Filed	Description: Defines the quantity that is complete and bad for this work order. Use: Type in a value.
Comment Type: Text	Description: Free form comment. If comments were entered on the Production Order Dispatch or Work Order Dispatch windows before dispatching, they will display here. Use: Type text.
OK Type: Button	Description: Closes the window and saves any configuration changes. Use: Click **OK**. The window will close and changes will be saved.
Cancel Type: Button	Description: Closes the window and does not save any configuration changes. Use: Click **Cancel**. The window will close and changes will not be saved.

16.11.6 Do Next:

Define the configuration on each tab.

Use the Dispatching Client grid or Shop Floor Client to view and manage the work orders.

16.11.7 See Also:

Parent Window:
Work Center Grid

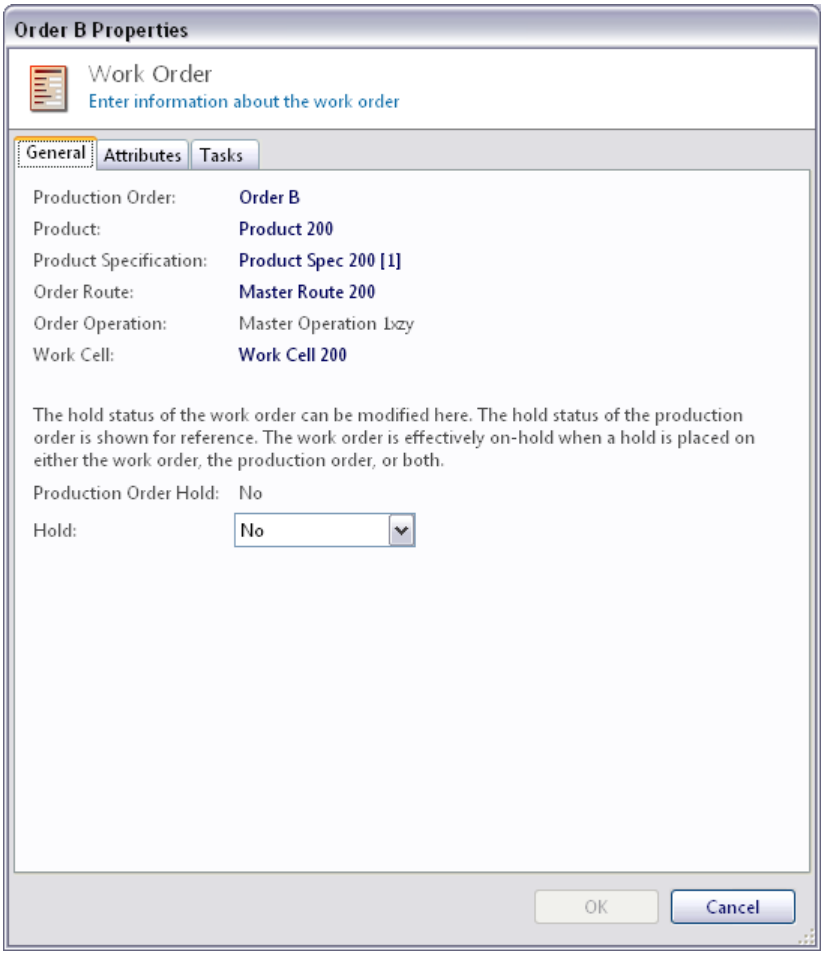
Other Tabs:
Work Order (General)

Work Order (Tasks)

Child Windows:
None

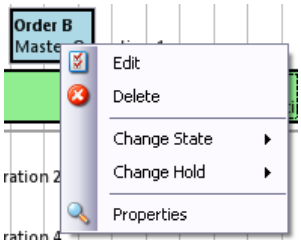
Other:
Dispatching Client Overview

16.12 Work Order (General)



16.12.1 How to Access:

1. Right click on a work order in the Work Center grid and the following pop-up displays.



2. Select **Edit** and the Work Order window will display.

Select **Properties** in the pop-up and a read only version of the Work Order window will display.

16.12.2 Overview:

The Work Order window displays information about the work order.

The "Hold" state can be changed on this tab.

Access to a read only version of the following windows can be accessed by clicking on the corresponding links:

- Production Order Definition (Route)
- Product Definition (General)
- Product Specification Definition (General)
- Order Route Definition
- Work Cell Definition (General)

16.12.3 Details:

See the Dispatching Client Overview.

16.12.4 Prerequisites:

See the Dispatching Client Overview.

16.12.5 Fields & Buttons:

Name & Type	Description & Use

Name & Type	Description & Use
Production Order Type: Link	Description: The production order from which this work order was created. Use: Click the production order and the Production Order Definition will display in read only mode.
Product Type: Link	Description: The product from which this work order was created. Use: Click the product and the Product Definition will display in read only mode.
Product Specification Type: Link	Description: The product specification from which this work order was created. Use: Click the product specification and the Product Specification Definition will display in read only mode.
Order Route Type: Link	Description: The route from which this work order was created. Use: Click the route and the Order Route Definition will display in read only mode.
Order Operation Type: Text Only	Description: Display the order operation. Use: This is informational only.
Work Cell Type: Text Only	Description: The route from which this work order was created. Use: Click the work cell and the Work Cell Definition will display in read only mode.

Name & Type	Description & Use
Production Order Hold Type: Text Only	Description: Indicates if the parent production order is on hold. In the grid a work order with a on-hold state will be pink. Use: This is informational only.
Hold Type: Drop-Down	Description: Change the hold status of the selected work order. The hold status can also be set from the right click drop-down on the grid. In the grid a work order with an on-hold state will be pink. Use: From the drop-down select **Yes** or **No**.
OK Type: Button	Description: Closes the window and saves any configuration changes. Use: Click **OK**. The window will close and changes will be saved.
Cancel Type: Button	Description: Closes the window and does not save any configuration changes. Use: Click **Cancel**. The window will close and changes will not be saved.

16.12.6 Do Next:

Define the configuration on each tab.

Use the Dispatching Client grid or Shop Floor Client to view and manage the work orders.

16.12.7 See Also:

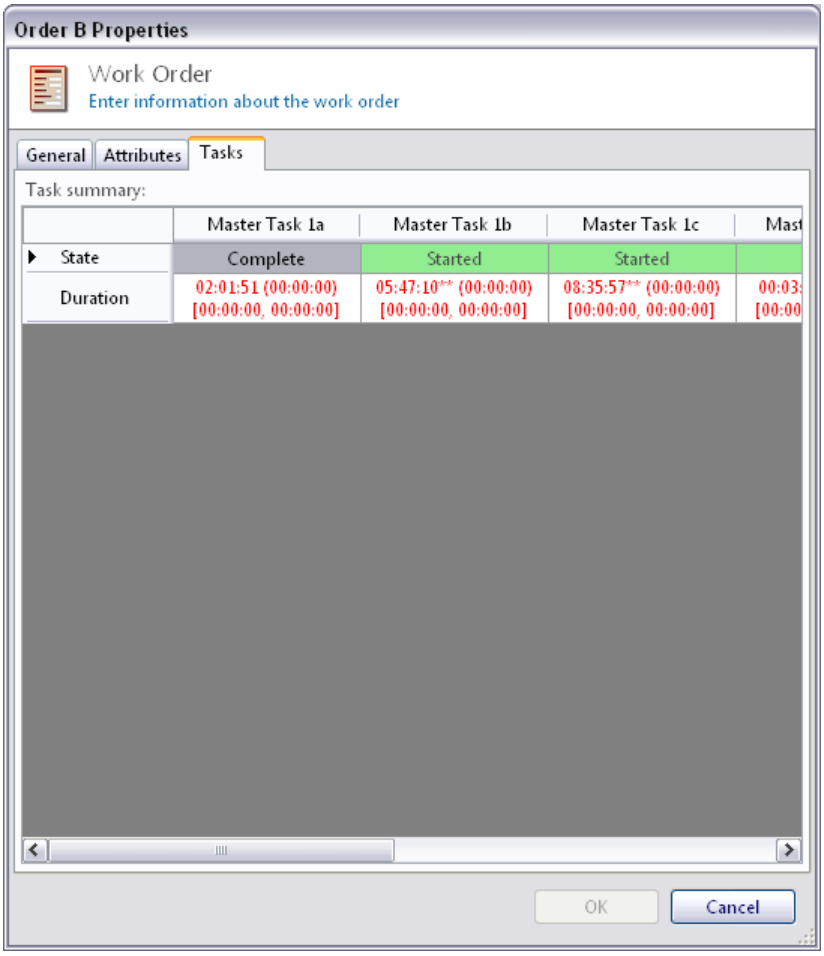
Parent Window:
Work Center Grid
Production Order Management

Other Tabs:
Work Order (Attribute)
Work Order (Tasks)

Child Windows:
Production Order Definition (Route)
Product Definition (General)
Product Specification Definition (General)
Order Route Definition
Work Cell Definition (General)

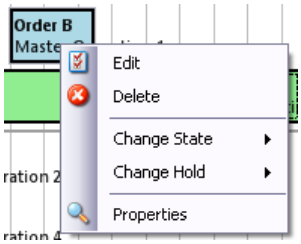
Other:
Dispatching Client Overview

16.13 Work Order (Tasks)



16.13.1 How to Access:

1. Right click on a work order in the Work Center grid and the following pop-up displays.



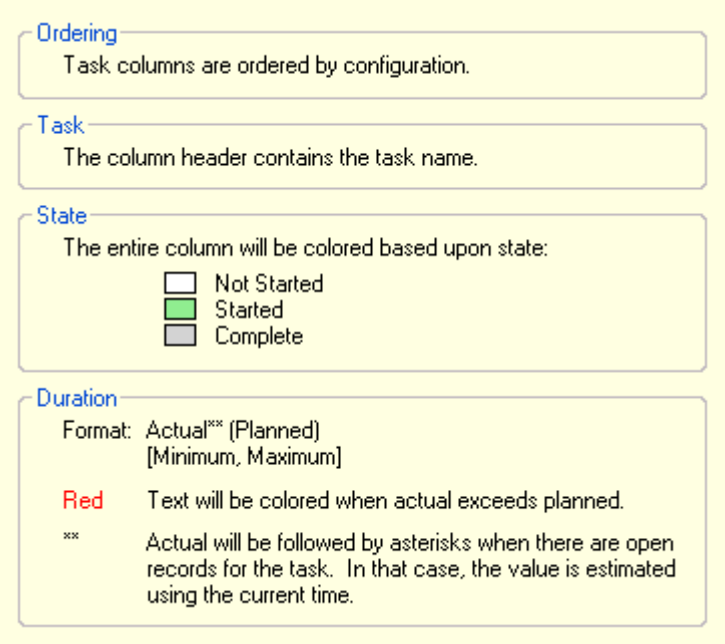
2. Select **Edit** and the Work Order window will display.

3. Click the **Tasks** tab.

Select **Properties** in the pop-up and a read only version of the Work Order window will display.

16.13.2 Overview:

The Work Order window displays information about the work order. Dates, states, comments, and quantity counts can be updated here.



16.13.3 Details:

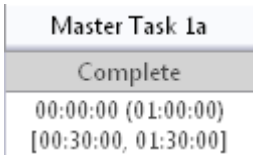
The **Task** tab is just informational. Nothing can be modified.

Across the top of the **Task summary** are the tasks defined for the work order, followed by the status of each task, and finally the duration of each task.

The durations will be in red text if they do not fall within the planned time for the task.

The durations are listed in the format dd:hh:mm:ss (days:hours:minutes:seconds).

The following are examples from the Tasks tab:



There are four durations listed for each task:

- **Actual Time** – The first time listed is the time from Actual Start Time to Actual End Time or to now if the task does not have an end time. The time is will be followed with two asterisks (**) if current time is used in place of Actual End Time.
- **Plan Time** – The time in the parentheses () is the Plan Time as specified in the production order route for the order task. If the Plan Time field is blank the value is be all zeros.
- **Minimum Time & Maximum Time** – The time in the square brackets [] is the Minimum Time and Maximum Time as specified in the production order route for the order task. If these fields are blank the values will be all zeros.

16.13.4 Prerequisites:

See the Dispatching Client Overview.

16.13.5 Fields & Buttons:

Name & Type	Description & Use
OK Type: Button	Description: Closes the window and saves any configuration changes. Use: Click **OK**. The window will close and changes will be saved.
Cancel Type: Button	Description: Closes the window and does not save any configuration changes. Use: Click **Cancel**. The window will close and changes will not be saved.

16.13.6 Do Next:

Define the configuration on each tab.

Use the Dispatching Client grid or Shop Floor Client to view and manage the work orders.

16.13.7 See Also:

Parent Window:
Work Center Grid

Other Tabs:
Work Order (Attribute)
Work Order (General)

Child Windows:
None

Other:
Dispatching Client Overview

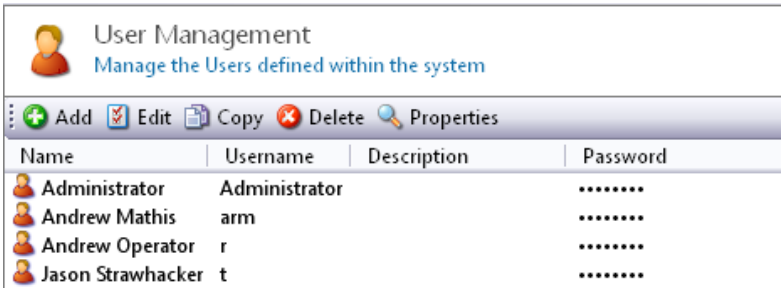
17.0 Configuring Users

Users are defined on the User Definition screen. Names, passwords, and their rolls are defined here.

The following sections describe in more details each of the screen mentioned above:

- User Management
- User Definition (General)
- User Definition (Member Of)

17.1 User Management



17.1.1 How to Access:

From Configuration Client:

- Select **Security** in the explorer view.
- Select **Users** in the explorer view. The User Management screen displays on the right.

17.1.2 Overview:

The User Management window is where users are defined and managed.

17.1.3 Details:

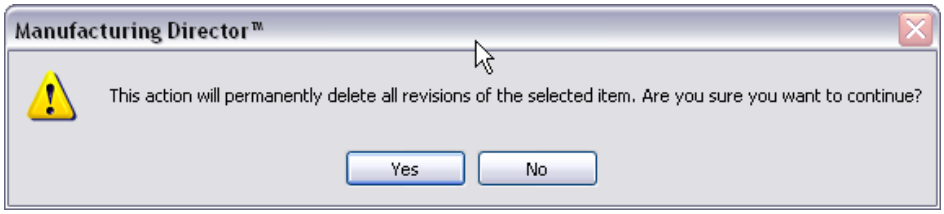
Users provide access to Manufacturing Director Clients. Users are assigned Roles to set their authorization level.

17.1.4 Prerequisites:

None

17.1.5 Fields & Buttons:

Name & Type	Description & Use

Name & Type	Description & Use
Add Type: Button	Description: Add user. The Users Definition window will display. Use: Click **Add**.
Edit Type: Button	Description: Edit a user. The Users Definition window will display. Use: 1. Select the user to be edited. 2. Click **Edit**.
Copy Type: Button	Description: Copy a user. The Copy window will display. Use: 1. Select the unit of measure to be copied. 2. Click **Copy**. The Copy window will display.
Delete Type: Button	Description: Delete a user. Use: 1. Select a user from the list. 2. Click **Delete**. A Confirmation pop-up will displayed.  3. Click **Yes** to delete the entity or click **No** to cancel to deletion.

Name & Type	Description & Use
Properties Type: Button	Description: Displays the User Definition window pre-populated with the selected unit of measure. The values in the window will be read only. Use: 1. Select a unit of measure from the list. 2. Click **Properties**. The User Definition window will display with the configuration for the unit of measure which was selected.
Selection List Type: List	Description: The list of all the users that are configured. The columns which display are defined in the Modeling Client. Use: • Right-click in the list area and select **Add** to display the Shift Definition window. • Select an entity and right-click. The **Edit, Copy, Delete** and **Properties** options will display. The Edit and Properties option will display the User Definition window. The Delete option will remove the entity after Yes is selected on the Confirmation pop-up. The Copy option will display the copy window.

17.1.6 Do Next:

Define and modify users on the User Definition window.

17.1.7 See Also:

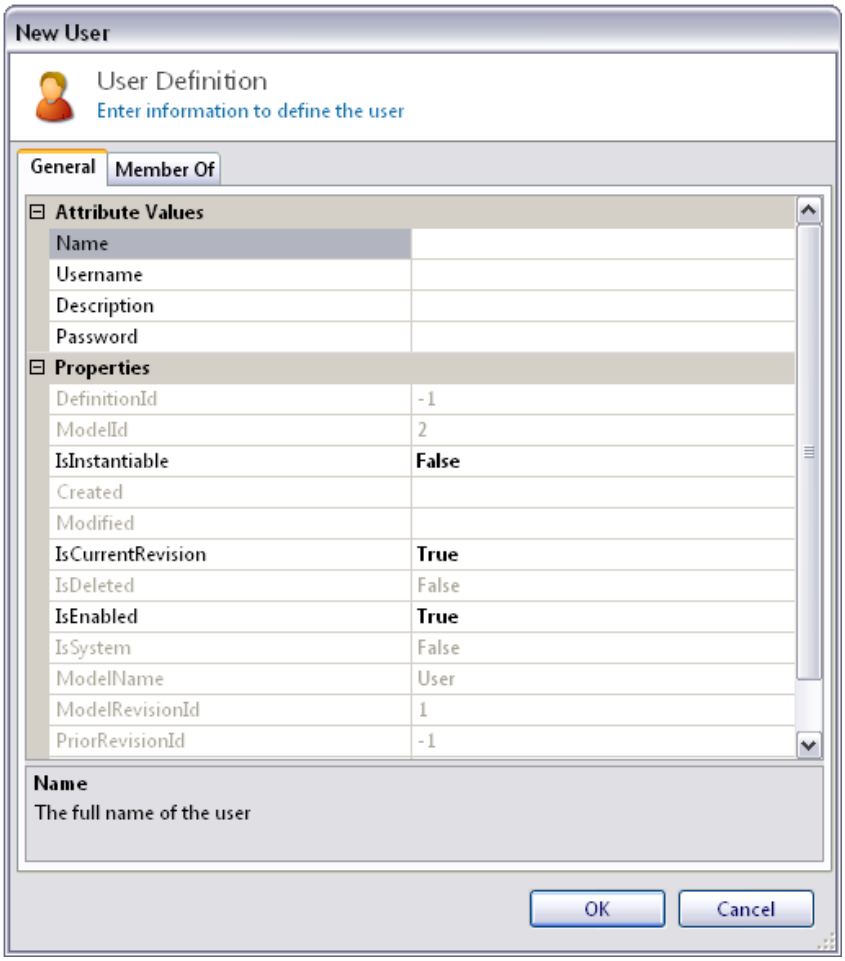
Parent Windows:
None

Other Tabs:
None

Child Windows:
User Definition (General)
User Definition (Member Of)

Other:
None

17.2 User Definition (General)



17.2.1 How to Access:

From Configuration Client:

- Select **Security** in the explorer view.
- Select **Users** in the explorer view. The User Management screen displays on the right.
    - Click Add or select a user and click Edit.

> To see this window in Read Only view:
>
> - Double click on a user.
> - Or, select a user and click **Properties**.

17.2.2 Overview:

The User Management window is where users are defined and managed.

17.2.3 Details:

Users provide access to Manufacturing Director Clients. Users are assigned Roles to set their authorization level.

17.2.4 Prerequisites:

None

17.2.5 Fields & Buttons:

Name & Type	Description & Use
Name Type: Text Field	Description: A name of the user. Use: Click in the field and type.
Username Type: Text Field	Description: Login username. Use: Click in the field and type.
Description Type: Text Field	Description: The description of the user. This is optional. Use: Click in the field and type.
Password Type: Text Field	Description: Password of the user. Use: Click in the field and then click the ellipse. The Password Entry form will display.

Name & Type	Description & Use
Is Instantiable Type: True/False	Description: Not applicable. This field should always be set to **False**. Use: • In the field, type an "F" and **False** will automatically fill in. OR • Click the drop-down list and select **False**.
Is Current Revision Type: True/False	Description: If **True**, this revision will be the current one, the one used by the program. Use: • In the field, type either a "T" or an "F" and **True** or **False** will automatically fill in. OR • Click the drop-down list and select either **True** or **False**.
Is Enabled Type: True/False	Description: If **True**, the revision will be enabled for use. Use: • In the field, type either a "T" or an "F" and **True** or **False** will automatically fill in. OR • Click the drop-down list and select either **True** or **False**.
OK Type: Button	Description: Closes the window and saves any configuration changes. Use: Click **OK**. The window will close and changes will be saved.

Name & Type	Description & Use
Cancel Type: Button	Description: Closes the window and does not save any configuration changes. Use: Click **Cancel**. The window will close and changes will not be saved.

17.2.6 Do Next:

Define the configuration on each tab.

17.2.7 See Also:

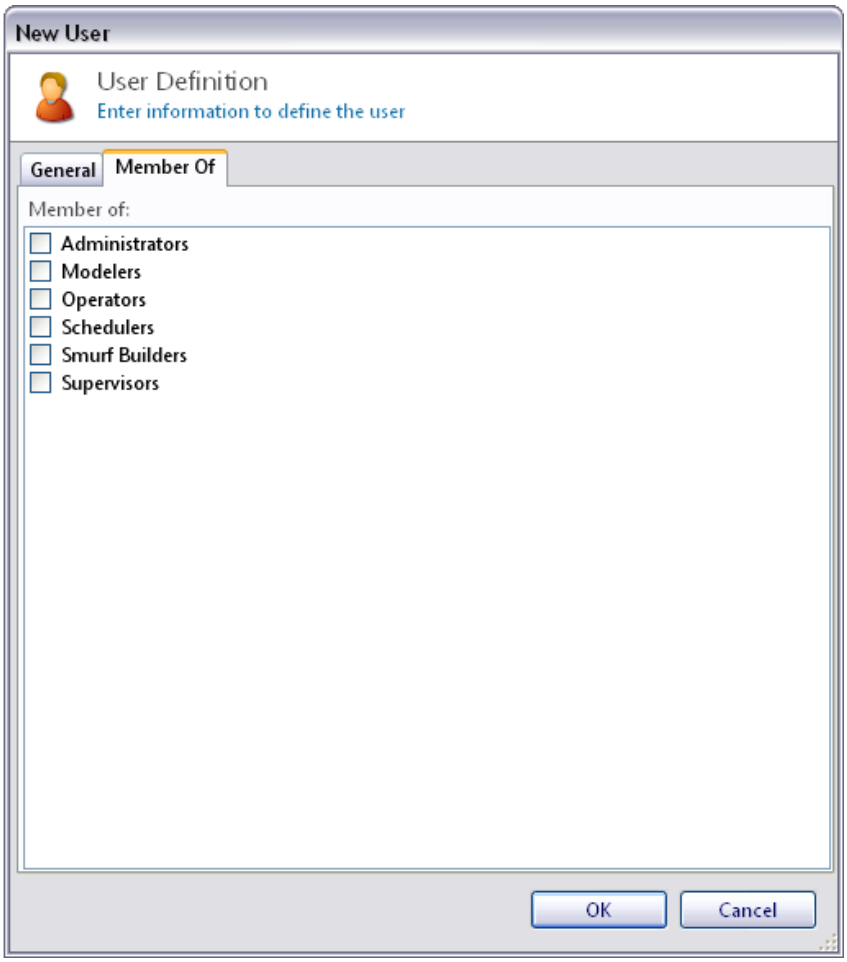
Parent Window:
Users Management

Other Tabs:
Users Definition (Member Of)

Child Window:
None

Other:
None

17.3 User Definition (Member Of)



17.3.1 How to Access:

From Configuration Client:

- Select **Security** in the explorer view.
- Select **Users** in the explorer view. The User Management screen displays on the right.
    - Click Add or select a user and click Edit.
        1. Set Is Instantiable to true.
        2. Click the Attributes Tab.

To see this window in Read Only view:

- Double click on a user.
- Or, select a user and click **Properties**.

17.3.2 Overview:

The User Management window is where users are defined and managed.

17.3.3 Details:

Users provide access to Manufacturing Director Clients. Users are assigned Roles to set their authorization level.

17.3.4 Prerequisites:

Define roles on the Roles Management screen.

17.3.5 Fields & Buttons:

Name & Type	Description & Use
Member Of Type: Checkbox list	Description: Lists all defined roles. Use: Select checkbox next to roles user is associated with.

17.3.6 Do Next:

Define the configuration on each tab.

17.3.7 See Also:

Parent Window:
Users Management

Other Tabs:
Users Definition (General)

Child Window:
None

Other:
None

18.0 Appendix F – Bill of Materials

Bill of Materials	
Vendor	**Part Description**
Kirby Risk	Cognex - Dataman – DMR-100X-00, PN: 808-0009-1 Rev A
	Cognex - DataMan Basic Accessory Kit - DM100-BAK-000
	Cognex - I/O Module - DM100-IOBOX-000
	Cognex - DM100-UBRK-000 - Universal Mounting Bracket
	Cognex - Dataman Model: DM7500 – Dataman hand held scanner
	Cognex - Dataman RS232 Cable: DM42206139-04 (For Dataman DM7500 hand held scanner
	Cognex - DataMan – DM42206416-01 – DataMan 7500 USB Cable
	CGNX CCB-84901-1004-10 COGCBL 10M ENET
	Pand C1WH6 - 6ft
	Pand G1x2WH6 - 6ft
	AB 1492J3 IEC 1-Circuit Feed-Through Block
	AB 1492ERL35 Screwless End Retainers
	AB 1492EBJ3 End barrier Gray
	AB 1492JG3 IEC 1-CKT Feed-Through Grd Blk
	IS5110-00 In-Sight® 5110 ID Reader Vision Sensor
	CCB-84901-0901-02 In-Sight Std. I/O Module Cable, 2M (6')
	LFC-25F1 Fujinon 2/3 " 25mm Lens F/1.4
	LTC-PL Polarizing Lens Filter
	LNS-CVR38-2 In-Sight® Lens Cover Kit 5000 Series 38mm
	CIO-1350-XX Breakout Module
	NTRO 104TX - 4 Port 10/100BaseTX Industrial Ethernet Switch
	Sola SDP1-24-100T 30W 24V DIN Plastic 115/230VIN
	LFC-25F1 Fujinon 2/3" 25mm lens
	CGNX DMPS5U-41 COG DATAMAN 7500 PWR Supply 5V US
Allied Automation	
	Mitsubishi FX3U-16MT/DSS 8 in/8 source out PLC
	Mitsubishi FX3U-ENET high speed ethernet module
	Mitsubishi FX3U-232-BD RS232 Adapter module
	GT1020-LBD 3.7inch monochrome touchscreen
	GT10-C30R4-8P 3 meter RS422 cable
	SC-Q programming cable for GT1020
	SC-09 programming cable for FX
	GX-DEV-C1 PLC programming software
	GT-WORKS2-C1 HMI development software
	GX-CONFIG-C1 software
	USB Bulkhead Interfaces – M22-USB-SA Moeller
	RJ45 Socket – M22-RJ45-SA Moeller
	MEAU QJ71E71-000
	MEAU QJ71E71-000 Returned
	MEAU GT10-C100R4-8P
	Mitsubishi FX3U-16MT/DSS
	MEAU FX3U-ENET

	Mitsubishi FX3U-232-BD
	MEAU MX-OPC-C1
ScanSource	
	Zebra 284Z-10400-0001
Vorne	
	Display Board XL800-32080T
L-comm.com	
	DGB9FT DB9 Female Conector for Field termination
	SDRA9AG Right Angle Assembled D-sub Hood kit, D89/HD15 metal
	SDC9M Assembled D-sub Hood Kit, D89/HD15 metalized chrome
	DGB9F Slimline Gender Changer, D89 Female/Female
	DGB9M Slimline Gender Changer, D89 Male/Male
	DS4-100 4 Conductor 24 AWG Bulk Cable, 100 ft Coil
	SD9P Solder Cup D-Sub Connector, DB9 Male
	SD9S Solder cup D-Sub Connector, DB9 Female
CDW	
	MS MBL SQL SRV STD 2005 X64 (MFG #:MBL-228-04103
	MS MBL SQL USER CAL 2005 (MFG#:MBL-359-01822)